AI & COMPUTING

Plane Geometry

G. A. Wentworth

FAHAD.COM — Free Book Library

Public Domain Edition

The Project Gutenberg EBook of Plane Geometry, by George Albert Wentworth This eBook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org Title: Plane Geometry Author: George Albert Wentworth Release Date: July 3, 2010 [EBook #33063] Language: English Character set encoding: ISO-8859-1 *** START OF THIS PROJECT GUTENBERG EBOOK PLANE GEOMETRY ***

Produced by Jeremy Weatherford, Stan Goodman, Kevin Handy and the Online Distributed Proofreading Team at http://www.pgdp.net transcriber's note—Minor typographical corrections and presentational changes have been made without comment. This PDF file is optimized for screen viewing, but may easily be recompiled for printing. Please see the preamble of the L ATEX source file for instructions.

PLANE GEOMETRY BY G.A. WENTWORTH Author of a Series of Text-Books in Mathematics REVISED EDITION GINN & COMPANY BOSTON· NEW YORK· CHICAGO· LONDON

Entered, according to Act of Congress, in the year 1888, by G.A. WENTWORTH in the Office Of the Librarian of Congress, at Washington Copyright, 1899 By G.A. WENTWORTH ALL RIGHTS RESERVED 67 10 The Athenæum Press GINN & COMPANY · PRO- PRIETORS· BOSTON· U.S.A.

iii PREFACE. Most persons do not possess, and do not easily acquire, the power of ab- straction requisite for apprehending geometrical conceptions, and for keeping in mind the successive steps of a continuous argument. Hence, with a very large proportion of beginners in Geometry, it depends mainly upon theform in which the subject is presented whether they pursue the study with indifference, not to say aversion, or with increasing interest and pleasure. Great care, therefore, has been taken to make the pages attractive. The figures have been carefully drawn and placed in the middle of the page, so that they fall directly under the eye in immediate connection with the text; and in no case is it necessary to turn the page in reading a demonstration. Full, long-dashed, and short-dashed lines of the figures indicate given, resulting, and auxiliary lines, respectively. Bold-faced, italic, and roman type has been skilfully used to distinguish the hypothesis, the conclusion to be proved, and the proof. As a further concession to the beginner, the reason for each statement in the early proofs is printed in small italics, immediately following the statement. This prevents the necessity of interrupting the logical train of thought by turning to a previous section, and compels the learner to become familiar with a large number of geometrical truths by constantly seeing and repeating them. This help is gradually discarded, and the pupil is left to depend upon the knowledge already acquired, or to find the reason for a step by turning to the given reference. It must not be inferred, because this is not a geometry of interrogation points, that the author has lost sight of the real object of the study. The training to be obtained from carefully following the logical steps of a complete proof has been provided for by the Propositions of the Geometry, and the development of the power to grasp and prove new truths has been provided for by original exercises. The chief value of any Geometry consists in the happy combination of these two kinds of training. The exercises have been arranged

according to the test of experience, and are so abundant that it is not expected that any one class will work them all out. The methods of attacking and proving original theorems are fully explained in the first Book, and illustrated by sufficient examples; and the methods of attacking and solving original problems are explained in the second Book, and illustrated

iv by examples worked out in full. None but the very simplest exercises are inserted until the student has become familiar with geometrical methods, and is furnished with elementary but much needed instruction in the art of handling original propositions; and he is assisted by diagrams and hints as long as these helps are necessary to develop his mental powers sufficiently to enable him to carry on the work by himself. The law of converse theorems, the distinction between positive and negative quantities, and the principles of reciprocity and continuity have been briefly explained; but the application of these principles is left mainly to the discretion of teachers. The author desires to express his appreciation of the valuable suggestions and assistance which he has received from distinguished educators in all parts of the country. He also desires to acknowledge his obligation to Mr. Charles Hamilton, the Superintendent of the composition room of the Athenæum Press, and to Mr. I. F. White, the compositor, for the excellent typography of the book. Criticisms and corrections will be thankfully received. G. A. WENTWORTH. Exeter, N.H., June, 1899.

v NOTE TO TEACHERS. It is intended to have the first sixteen pages of this book simply read in the class, with such running comment and discussion as may be useful to help the beginner catch the spirit of the subject-matter, and not leave him to the mere letter of dry definitions. In like manner, the definitions at the beginning of each Book should be read and discussed in the recitation room. There is a decided advantage in having the definitions for each Book in a single group so that they can be included in one survey and discussion. For a similar reason the theorems of limits are considered together. The subject of limits is exceedingly interesting in itself, and it was thought best to include in the theory of limits in the second Book every principle required for Plane and Solid Geometry. When the pupil is reading each Book for the first time, it will be well to let him write his proofs on the blackboard in his own language, care being taken that his language be the simplest possible, that the arrangement of work be vertical, and that the figures be accurately constructed. This method will furnish a valuable exercise as a language lesson, will cultivate the habit of neat and orderly arrangement of work, and will allow a brief interval for deliberating on each step. After a Book has been read in this way, the pupil should review the Book, and should be required to draw the figures free-hand. He should state and prove the propositions orally, using a pointer to indicate on the figure every line and angle named. He should be encouraged in reviewing each Book, to do the original exercises; to state the converse propositions, and determine whether they are true or false; and also to give well-considered answers to questions which may be asked him on many propositions. The Teacher is strongly advised to illustrate, geometrically and arithmeti- cally, the principles of limits. Thus, a rectangle with a constant base b, and a variable altitude x, will afford an obvious illustration of the truth that the product of a constant and a variable is also a variable; and that the limit of the product of a constant and a variable is the product of the constant by the limit

of the variable. If x increases and approaches the altitude a as a limit, the area of the rectangle increases and approaches the area of the rectangle ab as a limit; if, however, x decreases and approaches zero as a limit, the area of the rectangle decreases and approaches zero as a limit.

vi An arithmetical illustration of this truth may be given by multiplying the approximate values of any repetend by a constant. If, for example, we take the repetend 0.3333 etc., the approximate values of the repetend will be 3 10, 33 100, 333 1000, 3333 10000, etc., and these values multiplied by 60 give the series 18, 19 .8, 19.98, 19.998, etc., which evidently approaches 20 as a limit; but the product of 60 into 1 3 (the limit of the repetend 0.333 etc.) is also 20. Again, if we multiply 60 into the different values of the decreasing series 1 30, 1 300, 1 3000, 1 30000, etc., which approaches zero as a limit, we shall get the decreasing series 2, 1 5, 1 50, 1 500, etc.; and this series evidently approaches zero as a limit. The Teacher is likewise advised to give frequent written examinations. These should not be too difficult, and sufficient time should be allowed for accurately constructing the figures, for choosing the best language, and for determining the best arrangement. The time necessary for the reading of examination books will be diminished by more than one half, if the use of symbols is allowed. Exeter, N.H., 1899.

CONTENTS vii Contents GEOMETRY. 1 INTRODUCTION. . . . . 1 GENERAL TERMS. . . . . 3 GENERAL AXIOMS. . . . . 6 SYMBOLS AND ABBREVIATIONS. . . . . 6 PLANE GEOMETRY. 7 BOOK I. RECTILINEAR FIGURES. 7 DEFINITIONS. . . . . 7 THE STRAIGHT LINE. . . . . 8 THE PLANE ANGLE. . . . . 10 PERPENDICULAR AND OBLIQUE LINES. . . . . 17 PARALLEL LINES. . . . . 26 TRIANGLES. . . . . 33 LOCI OF POINTS. . . . . 48 QUADRILATERALS. . . . . 51 POLYGONS IN GENERAL. . . . . 61 SYMMETRY. . . . . 65 EXERCISES. 72

CONTENTS viii BOOK II. THE CIRCLE. 89 DEFINITIONS. . . . . 89 ARCS, CHORDS, AND TANGENTS. . . . . 91 MEASUREMENT. . . . . 109 THEORY OF LIMITS. . . . . 111 MEASURE OF ANGLES. . . . . 119 PROBLEMS OF CONSTRUCTION. . . . . 135 EXERCISES. . . . . 158 BOOK III. PROPORTION. SIMILAR POLYGONS. 168 THEORY OF PROPORTION. . . . . 168 SIMILAR POLYGONS. . . . . 183 EXERCISES. . . . . 195 NUMERICAL PROPERTIES OF FIGURES. . . . . 197 EXERCISES. . . . . 207 PROBLEMS OF CONSTRUCTION. . . . . 210 EXERCISES. . . . . 216 BOOK IV. AREAS OF POLYGONS. 226 COMPARISON OF POLYGONS. . . . . 235 EXERCISES. . . . . 239 PROBLEMS OF CONSTRUCTION. . . . . 242 EXERCISES. . . . . 252 BOOK V. REGULAR POLYGONS AND CIRCLES. 258 PROBLEMS OF CONSTRUCTION. . . . . 274 MAXIMA AND MINIMA. . . . .

. . . . . 282 EXERCISES. . . . . 289 TABLE OF FORMULAS. 302 INDEX. 305

GEOMETRY. INTRODUCTION. 1. If a block of wood or stone is cut in the shape represented in Fig. 1, it will have six flat faces . Each face of the block is called a surface; and if the faces are made smooth by polishing, so that, when a straight edge is applied to any one of them, the straight edge in every part will touch the surface, the faces are called plane surfaces, or planes. Fig. 1. 2. The intersection of any two of these surfaces is called a line. 3. The intersection of any three of these lines is called a point. 4. The block extends in three principal directions: From left to right, A to B. From front to back, A to C. From top to bottom, A to D. These are called the dimensions of the block, and are named in the order given, length, breadth (or width), and thickness (height or depth).

GEOMETRY. 2 5. A solid, in common language, is a limited portion of space filled with matter; but in Geometry we have nothing to do with the matter of which a body is composed; we study simply its shape and size; that is, we regard a solid as a limited portion of space which may be occupied by a physical body, or marked out in some other way. Hence, A geometrical solid is a limited portion of space. 6. The surface of a solid is simply the boundary of the solid, that which separates it from surrounding space. The surface is no part of a solid and has no thickness. Hence, A surface has only two dimensions, length and breadth. 7. A line is simply a boundary of a surface, or the intersection of two sur- faces. Since the surfaces have no thickness, a line has no thickness. Moreover, a line is no part of a surface and has no width. Hence, A line has only one dimension, length. 8. A point is simply the extremity of a line, or the intersection of two lines. A point, therefore, has no thickness, width, or length; therefore, no magnitude. Hence, A point has no dimension, but denotes position simply. 9. It must be distinctly understood at the outset that the points, lines, surfaces, and solids of Geometry are purely ideal, though they are represented to the eye in a material way. Lines, for example, drawn on paper or on the blackboard, will have some width and some thickness, and will so far fail of being true lines ; yet, when they are used to help the mind in reasoning, it is assumed that they represent true lines, without breadth and without thickness. A B C D F Fig. 2. 10. A point is represented to the eye by a fine dot, and named by a letter, as A (Fig. 2). A line is named by two letters, placed one at each end, as BF . A surface is represented and named by the lines which bound it, as BCDF . A solid is represented by the faces which bound it.

GENERAL TERMS. 3 11. A point in space may be considered by itself, without reference to a line. 12. If a point moves in space, its path is a line. This line may be considered apart from the idea of a surface. 13. If a line moves in space, it generates, in general, a surface. A surface can then be considered apart from the idea of a solid. 14. If a surface moves in space, it generates, in general, a solid. A B C D E F G H Fig. 3. Thus, let the upright surface ABCD (Fig. 3) move to the right to the position EFGH , the points A, B, C, and D generating the lines AE, BF , CG, and DH, respectively. The lines AB, BC, CD, and DA will generate the surfaces AF . BG, CH , and DE, respectively. The surface ABCD will generate the solid AG. 15. Geometry is the

science which treats of position, form, and magni- tude. 16. A geometrical figure is a combination of points, lines, surfaces, or solids. 17. Plane Geometry treats of figures all points of which are in the same plane. Solid Geometry treats of figures all points of which are not in the same plane. GENERAL TERMS. 18. A proof is a course of reasoning by which the truth or falsity of any statement is logically established.

GEOMETRY. 4 19. An axiom is a statement admitted to be true without proof. 20. A theorem is a statement to be proved. 21. A construction is the representation of a required figure by means of points and lines. 22. A postulate is a construction admitted to be possible. 23. A problem is a construction to be made so that it shall satisfy certain given conditions. 24. A proposition is an axiom, a theorem, a postulate, or a problem. 25. A corollary is a truth that is easily deduced from known truths. 26. A scholium is a remark upon some particular feature of a proposition. 27. The solution of a problem consists of four parts: 1. The analysis, or course of thought by which the construction of the required figure is discovered. 2. The construction of the figure with the aid of ruler and compasses. 3. The proof that the figure satisfies all the conditions. 4. The discussion of the limitations, if any, within which the solution is possible. 28. A theorem consists of two parts: the hypothesis, or that which is assumed; and the conclusion, or that which is asserted to follow from the hypothesis. 29. The contradictory of a theorem is a theorem which must be true if the given theorem is false, and must be false if the given theorem is true. Thus, A theorem: If A is B, then C is D. Its contradictory: If A is B, then C is not D.

GENERAL TERMS. 5 30. The opposite of a theorem is obtained by making both the hypothesis and the conclusion negative . Thus, A theorem: If A is B, then C is D. Its opposite: If A is not B, then C is not D. 31. The converse of a theorem is obtained by interchanging the hypothesis and conclusion. Thus, A theorem: If A is B, then C is D. Its converse: If C is D, then A is B. 32. The converse of a truth is not necessarily true. Thus, Every horse is a quadruped is true, but the converse, Every quadru- ped is a horse, is not true. 33. If a direct proposition and its opposite are true, the converse proposition is true; and if a direct proposition and its converse are true, the opposite proposition is true. Thus, if it were true that 1. If an animal is a horse, the animal is a quadruped; 2. If an animal is not a horse, the animal is not a quadruped; it would follow that 3. If an animal is a quadruped, the animal is a horse. Moreover, if 1 and 3 were true, then 2 would be true.

GEOMETRY. 6 34. GENERAL AXIOMS. 1. Magnitudes which are equal to the same magnitude, or equal magni- tudes, are equal to each other. 2. If equals are added to equals, the sums are equal. 3. If equals are taken from equals, the remainders are equal. 4. If equals are added to unequals, the sums are unequal in the same order; if unequals are added to unequals in the same order, the sums are unequal in that order. 5. If equals are taken from unequals, the remainders are unequal in the same order; if unequals are taken from equals, the remainders are unequal in the reverse order. 6. The doubles of the same magnitude, or of equal magnitudes are equal; and the doubles of unequals are unequal. 7. The halves of the same magnitude, or of equal magnitudes are equal; and the halves of unequals are unequal. 8. The whole is greater than any of its parts. 9. The whole is equal to the sum of all its parts. 35.

SYMBOLS AND ABBREVIATIONS. > is (or are) greater than. Def. . . . definition. < is (or are) less than. Ax. . . . axiom. □ is (or are) equivalent to. Hyp. . . . hypothesis. □ therefore. Cor. . . . corollary. ⊥ perpendicular. Scho. . . . scholium. ⊥s perpendiculars. Ex. . . . exercise. ∥ parallel. ∥s parallels. Adj. . . . adjacent. ∠ angle. ∠s angles. Iden. . . . identical. △ triangle. △s triangles. Const.. . . . construction. / / parallelogram. Sup. . . . supplementary. / / s parallelograms. Ext. . . . exterior. ⊙ circle. ⊙s circles. Int. . . . interior. rt. right. st. straight. Alt. . . . alternate. q.e.d. stands for quod erat demonstrandum, which was to be proved . q.e.f. stands for quod erat faciendum, which was to be done. The signs +,−,×,÷, =, have the same meaning as in Algebra.

PLANE GEOMETRY. BOOK I. RECTILINEAR FIGURES. DEFINITIONS. A B C D E F Fig. 4. 36. A straight line is a line such that any part of it, however placed on any other part, will lie wholly in that part if its extremities lie in that part, as AB. 37. A curved line is a line no part of which is straight, as CD. 38. A broken line is made up of different straight lines, as EF . Note. A straight line is often called simply a line. 39. A plane surface, or a plane, is a surface in which, if any two points are taken, the straight line joining these points lies wholly in the surface. 40. A curved surface is a surface no part of which is plane. 41. A plane figure is a figure all points of which are in the same plane. 42. Plane figures which are bounded by straight lines are calledrectilinear figures; by curved lines, curvilinear figures. 43. Figures that have the same shape are called similar. Figures that have the same size but not the same shape are called equivalent. Figures that have the same shape and the same size are called equal or congruent.

BOOK I. PLANE GEOMETRY. 8 THE STRAIGHT LINE. 44. Postulate. A straight line can be drawn from one point to another. 45. Postulate. A straight line can be produced indefinitely. 46. Axiom.∗ Only one straight line can be drawn from one point to an- other. Hence, two points determine a straight line. 47. Cor. 1. Two straight lines which have two points in common coincide and form but one line. 48. Cor. 2. Two straight lines can intersect in only one point. For if they had two points common, they would coincide and not intersect. Hence, two intersecting lines determine a point. 49. Axiom. A straight line is the shortest line that can be drawn from one point to another. 50. Def. The distance between two points is the length of the straight line that joins them. 51. A straight line determined by two points may be considered as pro- longed indefinitely. 52. If only the part of the line between two fixed points is considered, this part is called a segment of the line. 53. For brevity, we say “the line AB,” to designate a segment of a line limited by the points A and B. 54. If a line is considered as extending from a fixed point, this point is called the origin of the line. ∗The general axioms on page 6 apply to all magnitudes. Special geometrical axioms will be given when required.

THE STRAIGHT LINE. 9 A B C Fig. 5. 55. If any point,C, is taken in a given straight line,AB, the two partsCA and CB are said to have opposite directions from the point C (Fig. 5). Every straight line, asAB, may be considered as extending in either of two opposite directions, namely, fromA towardsB, which is expressed byAB, and read segmentAB; and fromB towardsA, which is expressed byBA, and read segmentBA. 56.

If the magnitude of a given line is changed, it becomes longer or shorter. Thus (Fig. 5), by prolongingAC toB we addCB toAC, andAB =AC + CB. By diminishing AB toC, we subtractCB fromAB, andAC =AB−CB. If a given line increases so that it is prolonged by its own magnitude several times in succession, the line is multiplied, and the resulting line is called a multiple of the given line. A B C D E Fig. 6. Thus (Fig. 6), if AB = BC = CD = DE, then AC = 2AB, AD = 3AB, and AE = 4AB. Hence, Lines of given length may be added and subtracted; they may also be mul- tiplied by a number.

BOOK I. PLANE GEOMETRY. 10 THE PLANE ANGLE. DE F Fig. 7. 57. The opening between two straight lines drawn from the same point is called a plane angle. The two lines, ED and EF , are called the sides, and E, the point of meeting, is called the vertex of the angle. The size of an angle depends upon the extent of opening of its sides, and not upon the length of its sides. 58. If there is but one angle at a given vertex, the angle is designated by a capital letter placed at the vertex, and is read by simply naming the letter. A BC D E F Fig. 8. A B abcd Fig. 9. If two or more angles have the same vertex, each angle is designated by three letters, and is read by naming the three letters, the one at the vertex between the others. Thus, DAC (Fig. 8) is the angle formed by the sides AD and AC. An angle is often designated by placing a small italic letter between the sides and near the vertex, as in Fig. 9. 59. Postulate of Superposition. Any figure may be moved from one place to another without altering its size or shape. 60. The test of equality of two geometrical magnitudes is that they may be made to coincide throughout their whole extent. Thus, Two straight lines are equal, if they can be placed one upon the other so that the points at their extremities coincide. Two angles are equal, if they can be placed one upon the other so that their vertices coincide and their sides coincide, each with each.

THE PLANE ANGLE. 11 61. A line or plane that divides a geometric magnitude into two equal parts is called the bisector of the magnitude. If the anglesBAD andCAD (Fig. 8) are equal,AD bisects the angleBAC. 62. Two angles are calledadjacent angles when they have the same vertex and a common side between them; as the angles BOD and AOD (Fig. 10). A B D O Fig. 10. A B C D Fig. 11. 63. When one straight line meets another straight line and makes the adjacent angles equal , each of these angles is called a right angle; as angles DCA and DCB (Fig. 11). 64. A perpendicular to a straight line is a straight line that makes a right angle with it. Thus, if the angle DCA (Fig. 11) is a right angle, DC is perpendicular to AB, and AB is perpendicular to DC. 65. The point (as C, Fig. 11) where a perpendicular meets another line is called the foot of the perpendicular. 66. When the sides of an angle extend in opposite directions, so as to be in the same straight line, the angle is called a straight angle. A B C Fig. 12. Thus, the angle formed atC (Fig. 12) with its sidesCA andCB extending in opposite directions from C is a straight angle. 67. Cor. A right angle is half a straight angle.

BOOK I. PLANE GEOMETRY. 12 A Fig. 13. A D O Fig. 14. 68. An angle less than a right angle is called an acute angle; as, angle A (Fig. 13). 69. An angle greater than a right angle and less than a straight angle is called an obtuse angle; as, angle AOD (Fig. 14). 70. An angle greater than a straight angle and less than two straight angles

is called a reflex angle; as, angleDOA, indicated by the dotted line (Fig. 14). 71. Angles that are neither right nor straight angles are called oblique angles; and intersecting lines that are not perpendicular to each other are called oblique lines.

THE PLANE ANGLE. 13 EXTENSION OF THE MEANING OF ANGLES. A B C D O A′ B′ Fig. 15. 72. Suppose the straight lineOC (Fig. 15) to move in the plane of the paper from coincidence with OA, about the point O as a pivot, to the position OC; then the line OC describes or generates the angle AOC, and the magnitude of the angle AOC depends upon the amount of rotation of the line from the position OA to the position OC. If the rotating line moves from the position OA to the position OB, per- pendicular toOA, it generates the right angleAOB; if it moves to the position OD, it generates the obtuse angle AOD; if it moves to the position OA′, it generates the straight angleAOA′; if it moves to the positionOB′ it generates the reflex angle AOB′, indicated by the dotted line; and if it moves to the position OA again, it generates two straight angles. Hence, 73. The angular magnitude about a point in a plane is equal to two straight angles, or four right angles; and the angular magnitude about a point on one side of a straight line drawn through the point is equal to a straight angle, or two right angles. 74. The whole angular magnitude about a point in a plane is called a perigon; and two angles whose sum is a perigon are calledconjugate angles. Note. This extension of the meaning of angles is necessary in the applications of Geometry, as in Trigonometry, Mechanics, etc.

BOOK I. PLANE GEOMETRY. 14 a bc d Fig. 16. B C D O Fig. 17. A B D O Fig. 18. 75. When two angles have the same vertex, and the sides of the one are prolongations of the sides of the other, they are called vertical angles; as, angles a and b, c and d (Fig. 16). 76. Two angles are called complementary when their sum is equal to a right angle; and each is called the complement of the other; as, angles DOB and DOC (Fig. 17). 77. Two angles are called supplementary when their sum is equal to a straight angle; and each is called the supplement of the other; as, angles DOB and DOA (Fig. 18). UNIT OF ANGLES. 78. By adopting a suitable unit for measuring angles we are able to express the magnitudes of angles by numbers. If we suppose OC (Fig. 15) to turn about O from coincidence with OA until it makes one three hundred sixtieth of a revolution, it generates an angle at O, which is taken as the unit for measuring angles. This unit is called a degree. The degree is subdivided into sixty equal parts, called minutes, and the minute into sixty equal parts, called seconds. Degrees, minutes, and seconds are denoted by symbols. Thus, 5 degrees 13 minutes 12 seconds is written 5◦ 13′ 12″. A right angle is generated when OC has made one fourth of a revolution and contains 90◦; a straight angle, when OC has made half of a revolution and contains 180◦; and a perigon, when OC has made a complete revolution and contains 360◦.

THE PLANE ANGLE. 15 Note. The natural angular unit is one complete revolution. But this unit would require us to express the values of most angles by fractions. The advantage of using the degree as the unit consists in its convenient size, and in the fact that 360 is divisible by so many different integral numbers. AB C DE F G H Fig. 19. 79. By the method of superposition we are able to compare magnitudes of the same kind. Suppose we have two angles, ABC and DEF (Fig. 19). Let the side ED be

placed on the side BA, so that the vertex E shall fall on B; then, if the side EF falls on BC, the angle DEF equals the angle ABC; if the side EF falls between BC and BA in the position shown by the dotted line BG, the angle DEF is less than the angle ABC; but if the side EF falls in the position shown by the dotted line BH, the angle DEF is greater than the angle ABC.

BOOK I. PLANE GEOMETRY. 16 AB C DE F F H Fig. 20. AB C D M P Fig. 21. 80. If we have the angles ABC and DEF (Fig. 20), and place the vertex E on B and the side ED on BC, so that the angle DEF takes the position CBH , the angles DEF and ABC will together be equal to the angle ABH. If the vertex E is placed on B, and the side ED on BA, so that the angle DEF takes the position ABF , the angle FBC will be the difference between the angles ABC and DEF . If an angle is increased by its own magnitude two or more times in succes- sion, the angle is multiplied by a number. Thus, if the angles ABM , MBC , CBP , PBD (Fig. 21) are all equal, the angle ABD is 4 times the angle ABM . Therefore, Angles may be added and subtracted; they may also be multiplied by a num- ber.

PERPENDICULAR AND OBLIQUE LINES. 17 PERPENDICULAR AND OBLIQUE LINES. Proposition I. Theorem. 81. All straight angles are equal. A BC D E F Let the angles ACB and DEF be any two straight angles. To prove that ∠ACB = ∠DEF . Proof. Place the ∠ACB on the ∠DEF , so that the vertex C shall fall on the vertexE, and the side CB on the side EF . Then CA will fall on ED, § 47 (because ACB and DEF are straight lines). □ ∠ACB = ∠DEF . § 60 q.e.d. 82. Cor. 1. All right angles are equal. Ax. 7 83. Cor. 2. At a given point in a given line there can be but one perpen- dicular to the line. For, if there could be two⊥s, we should have rt.∠s of different magnitudes; but this is impossible, § 82. 84. Cor. 3. The complements of the same angle or of equal angles are equal. Ax. 3 85. Cor. 4. The supplements of the same angle or of equal angles are equal. Ax. 3

BOOK I. PLANE GEOMETRY. 18 Note. The beginner must not forget that in Plane Geometry all the points of a figure are in the same plane. Without this restriction in Cor. 2, an indefinite number of perpendiculars can be erected at a given point in a given line. Proposition II. Theorem. 86. If two adjacent angles have their exterior sides in a straight line, these angles are supplementary. A B D O Let the exterior sides OA and OB of the adjacent angles AOD and BOD be in the straight line AB. To prove that ∠sAOD and BOD are supplementary. Proof. AOB is a straight line. Hyp. □ ∠AOB is a st. ∠. § 66 But ∠AOD + ∠BOD = the st. ∠AOB. Ax. 9 □ the ∠sAOD and BOD are supplementary. § 77 q.e.d. 87. Def. Adjacent angles that are supplements of each other are called supplementary-adjacent angles. Since the angular magnitude about a point is neither increased nor dimin- ished by the number of lines which radiate from the point, it follows that, 88. Cor. 1. The sum of all the angles about a point in a plane is equal to a perigon, or two straight angles. 89. Cor. 2. The sum of all the angles about a point in a plane, on the same side of a straight line passing through the point, is equal to a straight angle, or two right angles.

PERPENDICULAR AND OBLIQUE LINES. 19 Proposition III. Theorem. 90. Conversely: If two adjacent angles are supplementary, their exterior sides are in the same straight line. A BC F O Let the adjacent angles OCA and OCB be

supplementary. To prove that AC and CB are in the same straight line. Proof. Suppose CF to be in the same line with AC. Then ∠sOCA and OCF are supplementary, § 86 (if two adjacent angles have their exterior sides in a straight line, these angles are supplementary). But ∠sOCA and OCB are supplementary. Hyp. ∠ sOCF and OCB have the same supplement. ∠OCF = ∠OCB . § 85 CB and CF coincide. § 60 AC and CB are in the same straight line. q.e.d. Since Propositions II. and III. are true, their opposites are true. Hence, § 33 91. Cor. 1. If the exterior sides of two adjacent angles are not in a straight line, these angles are not supplementary. 92. Cor. 2. If two adjacent angles are not supplementary, their exterior sides are not in the same straight line.

BOOK I. PLANE GEOMETRY. 20 Proposition IV. Theorem. 93. If one straight line intersects another straight line, the vertical angles are equal. A B C P O Let the lines OP and AB intersect at C. To prove that ∠OCB = ∠ACP . Proof. ∠OCA and ∠OCB are supplementary. § 86 ∠OCA and ∠ACP are supplementary, § 86 (if two adjacent angles have their exterior sides in a straight line, these angles are supplementary). ∠ sOCB and ACP have the same supplement. ∠OCB = ∠ACP . § 85 In like manner, ∠ACO = ∠PCB . q.e.d. 94. Cor. If one of the four angles formed by the intersection of two straight lines is a right angle, the other three angles are right angles. Ex. 1. Find the complement and the supplement of an angle of 49 ◦. Ex. 2. Find the number of degrees in an angle if it is double its comple- ment; if it is one fourth of its complement. Ex. 3. Find the number of degrees in an angle if it is double its supplement; if it is one third of its supplement.

PERPENDICULAR AND OBLIQUE LINES. 21 Proposition V. Theorem 95. Two straight lines drawn from a point in a perpendicular to a given line, cutting off on the given line equal segments from the foot of the perpendicular, are equal and make equal angles with the perpendicular. A B C E F K Let CF be a perpendicular to the line AB, and CE and CK two straight lines cutting off on AB equal segments F E and F K from F . To prove that CE =CK ; and ∠FCE = ∠FCK . Proof. Fold overCFA , onCF as an axis, until it falls on the plane at the right of CF . FA will fall along FB , (since ∠CFA = ∠CFB , each being a rt. ∠, by hyp.). PointE will fall on point K, (since FE =FK , by hyp.). CE =CK , § 60 (their extremities being the same points); and ∠FCE = ∠FCK , § 60 (since their vertices coincide, and their sides coincide, each with each). q.e.d. Ex. 4. Find the number of degrees in the angle included by the hands of a clock at 1 o’clock. 3 o’clock. 4 o’clock. 6 o’clock.

BOOK I. PLANE GEOMETRY. 22 Proposition VI. Theorem. 96. Only one perpendicular can be drawn to a given line from a given external point. A BCD P P ′ Let AB be the given line, P the given external point, P C a perpen- dicular to AB from P , and P D any other line from P to AB. To prove that PD is not⊥ to AB. Proof. Produce PC to P′, making CP′ equal to PC . DrawDP′. By construction, PCP′ is a straight line. PDP′ is not a straight line, § 46 (only one straight line can be drawn from one point to another). Hence, ∠PDP′ is not a straight angle. Since PC is⊥ to DC, and PC =CP′, AC is⊥ to PP′ at its middle point. ∠PDC = ∠P′DC, § 95 (two straight lines from a point in a ⊥ to a line, cutting off on the line equal segments from the foot of the ⊥, make equal ∠s with the ⊥) Since ∠PDP′ is not a straight angle, ∠PDC , the

half of ∠PDP′, is not a right angle. □PD is not⊥ to AB. q.e.d.

PERPENDICULAR AND OBLIQUE LINES. 23 Proposition VII. Theorem. 97. The perpendicular is the shortest line that can be drawn to a straight line from an external point. A BCD P P ′ Let AB be the given straight line, P the given point, P C the perpen- dicular, and P D any other line drawn from P to AB. To prove that PC <PD . Proof. Produce PC to P′, making CP′ =PC . DrawDP′. Then PD =DP′, § 95 (two straight lines drawn from a point in a ⊥ to a line, cutting off on the line equal segments from the foot of the ⊥, are equal). □PD +DP′ = 2PD , and PC +CP′ = 2PC . Const. But PC +CP′ <PD +DP′. § 49 □ 2PC < 2PD . □PC <PD . Ax. 7 q.e.d. 98. Cor. The shortest line that can be drawn from a point to a given line is perpendicular to the given line.

BOOK I. PLANE GEOMETRY. 24 99. Def. The distance of a point from a line is the length of the perpen- dicular from the point to the line. Proposition VIII. Theorem. 100. The sum of two lines drawn from a point to the extremities of a straight line is greater than the sum of two other lines similarly drawn, but included by them. A B C E O Let CA and CB be two lines drawn from the point C to the extrem- ities of the straight line AB. Let OA and OB be two lines similarly drawn, but included by CA and CB. To prove that CA +CB >OA +OB. Proof. Produce AO to meet the line CB at E. Then CA +CE >OA +OE, and BE +OE >OB, § 49 (a straight line is the shortest line from one point to another). Add these inequalities, and we have CA +CE +BE +OE >OA +OE +OB. Ax. 4 Substitute for CE +BE its equal CB, then CA +CB +OE >OA +OE +OB. Take awayOE from each side of the inequality. CA +CB >OA +OB. Ax. 5 q.e.d.

PERPENDICULAR AND OBLIQUE LINES. 25 Proposition IX. Theorem. 101. Of two straight lines drawn from the same point in a perpendicular to a given line, cutting off on the line unequal segments from the foot of the perpendicular, the more remote is the greater. A BC D E F G O Let OC be perpendicular to AB, OG and OE two straight lines to AB, and CE greater than CG. To prove that OE >OG. Proof. TakeCF equal to CG, and draw OF . Then OF =OG, § 95 (two straight lines drawn from a point in a ⊥ to a line, cutting off on the line equal segments from the foot of the ⊥, are equal). Produce OC to D, making CD =OC. DrawED and FD . Then OE =ED, and OF =FD . § 95 But OE +ED >OF +FD , § 100 □ 2OE >2OF , OE >OF , and OE >OG. q.e.d. 102. Cor. Only two equal straight lines can be drawn from a point to a straight line; and of two unequal lines, the greater cuts off on the line the greater segment from the foot of the perpendicular.

BOOK I. PLANE GEOMETRY. 26 PARALLEL LINES. 103. Def. Two parallel lines are lines that lie in the same plane and cannot meet however far they are produced. Proposition X. Theorem. 104. Two straight lines in the same plane perpendicular to the same straight line are parallel. A B C D Let AB and CD be perpendicular to AC. To prove that AB and CD are parallel. Proof. If AB and CD are not parallel, they will meet if sufficiently prolonged; and we shall have two perpendicular lines from their point of meeting to the same straight line; but this is impossible, § 96 (only one perpendicular can be drawn to a given line from a given external point). □AB and CD are parallel. q.e.d. 105. Axiom. Through a given point only one straight line can be drawn parallel

to a given straight line. 106. Cor. Two straight lines in the same plane parallel to a third straight line are parallel to each other. For if they could meet, we should have two straight lines from the point of meeting parallel to a straight line; but this is impossible. § 105

PARALLEL LINES. 27 Proposition XI. Theorem. 107. If a straight line is perpendicular to one of two parallel lines, it is perpendicular to the other also. A B CE F H K M N O Let AB and EF be two parallel lines, and let HK be perpendicular to AB, and cut EF at C. To prove that HK is⊥ to EF . Proof. Suppose MN drawn through C⊥ to HK . Then MN is‖ to AB. § 104 But EF is‖ to AB. Hyp. ∴EF coincides with MN . § 105 But MN is⊥ to HK . Const. ∴EF is⊥ to HK , that is, HK is⊥ to EF . q.e.d. 108. Def. A straight line that cuts two or more straight lines is called a transversal of those lines.

BOOK I. PLANE GEOMETRY. 28 A B C D E F a b c d e f g h 109. If the transversal EF cuts AB and CD, the angles a, d, g, f are called interior angles; b, c, h, e are called exterior angles. The angles d and f, and a and g, are called alternate-interior angles; the angles b and h, and c and e, are called alternate-exterior angles. The angles b and f, c and g, e and a, h and d, are called exterior-interior angles.

PARALLEL LINES. 29 Proposition XII. Theorem. 110. If two parallel lines are cut by a transversal, the alternate-interior angles are equal. A B C D E F G H O Let EF and GH be two parallel lines cut by the transversal BC. To prove that ∠EBC = ∠BCH . Proof. Through O, the middle point of BC, suppose AD drawn⊥ to GH. Then AD is likewise⊥ to EF , § 107 (a straight line ⊥ to one of two ‖s is ⊥ to the other), that is, CD and BA are both⊥ to AD. Apply the figure COD to the figure BOA, so that OD shall fall along OA. Then OC will fall along OB, § 93 (since ∠COD = ∠BOA, being vertical ∠s); and C will fall on B, (since OC =OB, by construction). Then the⊥CD will fall along the ⊥BA, § 96 (only one ⊥ can be drawn to a given line from a given external point). ∴ ∠OCD coincides with ∠OBA, and is equal to it, § 60 (two angles are equal, if their vertices coincide and their sides coincide, each with each). q.e.d.

BOOK I. PLANE GEOMETRY. 30 Proposition XIII. Theorem. 111. Conversely: When two straight lines in the same plane are cut by a transversal, if the alternate-interior angles are equal, the two straight lines are parallel. A B C D E F H K M N Let EF cut the straight lines AB and CD in the points H and K, and let the angles AHK and HKD be equal. To prove that AB is‖ to CD. Proof. Suppose MN drawn through H‖ to CD. Then ∠MHK = ∠HKD , § 110 (being alt.-int. ∠s of ‖ lines). But ∠AHK = ∠HKD . Hyp. ∴ ∠MHK = ∠AHK . Ax. 1 ∴MN and AB coincide. § 60 But MN is‖ to CD. Const. ∴AB, which coincides with MN , is‖ to CD. q.e.d. Ex. 5. Find the complement and the supplement of an angle that contains 37◦ 53′ 49′′. Ex. 6. If the complement of an angle is one third of its supplement, how many degrees does the angle contain?

PARALLEL LINES. 31 Proposition XIV. Theorem. 112. If two parallel lines are cut by a transversal, the exterior-interior angles are equal. A B C D E F H K Let AB and CD be two parallel lines cut by the transversal EF , in the points H and K. To prove that ∠EHB = ∠HKD . Proof. ∠EHB = ∠AHK , § 93 (being vertical ∠s). ∠AHK = ∠HKD , § 110 (being alt.-int. ∠s of ‖ lines). ∴ ∠EHB = ∠HKD . Ax. 1 In like manner ∠EHA =

∠HKC . q.e.d. 113. Cor. The alternate-exterior angles EHB and CKF , and also AHE and DKF , are equal.

BOOK I. PLANE GEOMETRY. 32 Proposition XV. Theorem. 114. Conversely: When two straight lines in a plane are cut by a transversal, if the exterior-interior angles are equal, these two straight lines are parallel. (Proof similar to that in § 111.) Proposition XVI. Theorem. 115. If two parallel lines are cut by a transversal, the two interior angles on the same side of the transversal are supplementary. A B C D E F H K Let AB and CD be two parallel lines cut by the transversal EF in the points H and K. To prove that ∠sBHK and HKD are supplementary. Proof. ∠EHB + ∠BHK = a st. ∠, § 86 (being sup.-adj. ∠s). But ∠EHB = ∠HKD , § 112 (being ext.-int. ∠s of ∥ lines). ∠BHK + ∠HKD = a st. ∠. ∠ sBHK and HKD are supplementary. § 77 q.e.d.

TRIANGLES. 33 Proposition XVII. Theorem. 116. Conversely: When two straight lines in a plane are cut by a transversal, if two interior angles on the same side of the transversal are sup- plementary, the two straight lines are parallel. (Proof similar to that in § 111.) TRIANGLES. 117. A triangle is a portion of a plane bounded by three straight lines; as, ABC (Fig. 1). A B C D Fig. 1. A B C D Fig. 2. The bounding lines are called the sides of the triangle, and their sum is called its perimeter; the angles included by the sides are called the angles of the triangle, and the vertices of these angles, the vertices of the triangle. 118. Adjacent angles of a rectilinear figure are two angles that have one side of the figure common; as, angles A and B (Fig. 2). 119. An exterior angle of a triangle is an angle included by one side and another side produced; as, ACD (Fig. 2). The interior angle ACB is adja- cent to the exterior angle; the interior angles, A and B, are called opposite interior angles.

BOOK I. PLANE GEOMETRY. 34 Scalene. Isosceles. Equilateral. 120. A triangle is called a scalene triangle when no two of its sides are equal; an isosceles triangle, when two of its sides are equal; an equilateral triangle, when its three sides are equal. Right. Obtuse. Acute. Equiangular. 121. A triangle is called a right triangle , when one of its angles is a right angle; an obtuse triangle, when one of its angles is an obtuse angle; an acute triangle, when all three of its angles are acute angles; an equiangular triangle, when its three angles are equal. 122. In a right triangle, the side opposite the right angle is called the hypotenuse, and the other two sides the legs. 123. The side on which a triangle is supposed to stand is called the base of the triangle. In the isosceles triangle, the equal sides are called the legs, and the other side, the base; in other triangles, any one of the sides may be taken as the base. 124. The angle opposite the base of a triangle is called the vertical angle, and its vertex, the vertex of the triangle. 125. The altitude of a triangle is the perpendicular from the vertex to the base, or to the base produced; as, AD (Fig. 1).

TRIANGLES. 35 126. The three perpendiculars from the vertices of a triangle to the opposite sides (produced if necessary) are called the altitudes of the triangle; the three bisectors of the angles are called the bisectors of the triangle; and the three lines from the vertices to the middle points of the opposite sides are called the medians of the triangle. 127. If two triangles have the angles of the one equal, respectively, to the angles of the other, the equal angles are called homologous angles, and the sides opposite the equal angles are called homologous sides. 128.

Two triangles are equal in all respects if they can be made to coincide (§ 60). The homologous sides of equal triangles are equal, and the homologous angles are equal.

BOOK I. PLANE GEOMETRY. 36 Proposition XVIII. Theorem. 129. The sum of the three angles of a triangle is equal to two right angles. A B C E F Let A, B, and BCA be the angles of the triangle ABC. To prove that ∠A + ∠B + ∠BCA = 2 rt. ∠s. Proof. Suppose CE drawn‖ to AS, and prolong AC to F . Then ∠ECF + ∠ECB + ∠BCA = 2 rt. ∠s, § 89 (the sum of all the ∠s about a point on the same side of a straight line passing through the point is equal to 2 rt. ∠s). But ∠A = ∠ECF , § 112 (being ext.-int. ∠s of the ‖ lines AB and CE), and ∠B = ∠BCE , § 110 (being alt.-int. ∠s of the ‖ lines AB and CE). Put for the ∠sECF and BCE their equals, the ∠sA and B. Then ∠A + ∠B + ∠BCA = 2 rt. ∠s. q.e.d. 130. Cor. 1. The sum of two angles of a triangle is less than two right angles. 131. Cor. 2. If the sum of two angles of a triangle is taken from two right angles, the remainder is equal to the third angle. 132. Cor. 3. If two triangles have two angles of the one equal to two angles of the other, the third angles are equal. 133. Cor. 4. If two right triangles have an acute angle of the one equal to an acute angle of the other, the other acute angles are equal.

TRIANGLES. 37 134. Cor. 5. In a triangle there can be but one right angle, or one obtuse angle. 135. Cor. 6. In a right triangle the two acute angles are together equal to one right angle, or 90◦. 136. Cor. 7. In an equiangular triangle, each angle is one third of two right angles, or 60◦. 137. Cor. 8. An exterior angle of a triangle is equal to the sum of the two opposite interior angles, and therefore greater than either of them. Proposition XIX. Theorem. 138. The sum of two sides of a triangle is greater than the third side, and their difference is less than the third side. A B C In the triangle ABC, let AC be the longest side. To prove that AB +BC >AC, and AC−BC <AB. Proof. AB +BC >AC, § 49 (a straight line is the shortest line from one point to another). Take awayBC from both sides. Then AB >AC−BC, Ax. 5 or AC−BC <AB. q.e.d.

BOOK I. PLANE GEOMETRY. 38 Proposition XX. Theorem. 139. Two triangles are equal if two angles and the included side of the one are equal, respectively, to two angles and the included side of the other. A B C D E F In the triangles ABC, DEF , let the angle A be equal to the angle D, B to E, and the side AB to DE. To prove that △ABC =△DEF . Proof. Apply the △ABC to the △DEF so that AB shall coincide with its equal, DE. Then AC will fall along DF , and BC along EF , (for ∠A = ∠D, and ∠B = ∠E, by hyp.). □C will fall on F , § 48 (two straight lines can intersect in only one point). □ the two△s coincide, and are equal. § 60 q.e.d. 140. Cor. 1. Two triangles are equal if a side and any two angles of the one are equal to the homologous side and two angles of the other. § 132 141. Cor. 2. Two right triangles are equal if the hypotenuse and an acute angle of the one are equal, respectively, to the hypotenuse and an acute angle of the other. § 133 142. Cor. 3. Two right triangles are equal if a leg and an acute angle of the one are equal, respectively, to a leg and the homologous acute angle of the other. § 133

TRIANGLES. 39 Proposition XXI. Theorem. 143. Two triangles are equal if two sides and the included angle of the one are equal, respectively, to two sides and the included angle of the other. A B C D E F In the triangles ABC and DEF , let AB be

equal to DE, AC to DF , and the angle A to the angle D. To prove that △ABC =△DEF . Proof. Apply the △ABC to the △DEF so that AB shall coincide with its equal, DE. Then AC will fall along DF , (for ∠A = ∠D, by hyp.); and C will fall on F , (for AC =DF , by hyp.). □CB =FE , (their extremities being the same points). □ the two△s coincide, and are equal. q.e.d. 144. Cor. Two right triangles are equal if their legs are equal, each to each. Note. In§ 139 we have given two angles and the included side, in § 143 two sides and the included angle; hence, by interchanging the words sides and angles, either theorem is changed to the other. This is called the Principle of Duality , or the Principle of Reciprocity. The reciprocal of a theorem is not always true, just as the converse of a theorem is not always true.

BOOK I. PLANE GEOMETRY. 40 Proposition XXII. Theorem. 145. In an isosceles triangle the angles opposite the equal sides are equal. A B CD Let ABC be an isosceles triangle, having AB and AC equal. To prove that ∠B = ∠C. Proof. Suppose AD drawn so as to bisect the ∠BAC. In the△sADB and ADC, AB =AC, Hyp. AD =AD, Iden. and ∠BAD = ∠CAD. Const. □△ADB =△ADC, § 143 (two △s are equal if two sides and the included ∠ of the one are equal, respectively, to two sides and the included ∠ of the other). □ ∠B = ∠C, § 128 (being homologous angles of equal triangles). q.e.d. 146. Cor. An equilateral triangle is equiangular, and each angle is two thirds of a right angle. Ex. 7. If the equal sides of an isosceles triangle are produced, the angles on the other side of the base are equal.

TRIANGLES. 41 Proposition XXIII. Theorem. 147. If two angles of a triangle are equal, the sides opposite the equal angles are equal, and the triangle is isosceles. A B CD In the triangle ABC, let the angle B be equal to the angle C. To prove that AB =AC. Proof. Suppose AD drawn⊥ to BC. In the rt.△sADB and ADC, AD =AD, Iden. and ∠B = ∠C. Hyp. □ rt.△ADB = rt.△ADC, § 142 (having a leg and an acute ∠ of the one equal, respectively, to a leg and the homologous acute ∠ of the other). □AB =AC, § 128 (being homologous sides of equal △s). q.e.d. 148. Cor. 1. An equiangular triangle is also equilateral. 149. Cor. 2. The perpendicular from the vertex to the base of an isosceles triangle bisects the base, and bisects the vertical angle of the triangle.

BOOK I. PLANE GEOMETRY. 42 Proposition XXIV. Theorem. 150. Two triangles are equal if the three sides of the one are equal, respec- tively, to the three sides of the other. A B C B′ A′ B′ C′ In the triangles ABC and A′B′C′, let AB be equal to A′B′, AC to A′C′, BC to B′C′. To prove that △ABC =△A′B′C′. Proof. Place△A′B′C′ in the position △AB′C having its greatest side △A′C′ in coincidence with its equal△AC, and its vertex atB′, opposite B; and drawBB′. Since AB =AB′ Hyp. ∠ABB′ = ∠AB′B § 145 (in an isosceles △ the ∠s opposite the equal sides are equal). Since CB =CB′, Hyp. ∠CBB′ = ∠CB′B. § 145 □ ∠ABB′ + ∠CBB′ = ∠AB′B + ∠CB′B. Ax. 2 Hence, ∠ABC = ∠AB′C. □△ABC =△AB′C, § 143 (two △s are equal if two sides and the included ∠ of the one are equal, respectively, to two sides and the included ∠ of the other). □△ABC =△A′B′C′. q.e.d.

TRIANGLES. 43 Proposition XXV. Theorem. 151. Two right triangles are equal if a leg and the hypotenuse of the one are equal, respectively, to a leg and the hypotenuse of the other. A BC C A′ B′ C′ A′ B′ C′ In the right triangles ABC and A′B′C′, let AB be equal to A′B′, and AC to A′C′. To prove that △ABC =△A′B′C′. Proof. Apply the△ABC to

the△A′B′C′, so thatAB shall coincide withA′B′, A falling on A′, B on B′, and C and C′ on opposite sides of A′B′. Then BC will fall along C′B′ produced, (for ∠ABC = ∠A′B′C′, each being a rt. ∠.). Since AC =A′C′, Hyp. the△A′CC′ is an isosceles triangle. § 120 ∠C = ∠C′, § 145 (∠s opposite the equal sides of an isosceles △ are equal). △sABC and A′B′C′ are equal, § 141 (two right △s are equal if they have the hypotenuse and an acute ∠ of, the one equal to the hypotenuse and an acute ∠ of the other). q.e.d. Ex. 8. How many degrees are there in each of the acute angles of an isosceles right triangle?

BOOK I. PLANE GEOMETRY. 44 Proposition XXVI. Theorem. 152. If two sides of a triangle are unequal, the angles opposite are unequal, and the greater angle is opposite the greater side. A BC E In the triangle ACB, let AB be greater than AC. To prove that ∠ACB is greater than ∠B. Proof. On AB takeAE equal to AC. DrawEC. ∠AEC = ∠ACE § 145 (being ∠s opposite equal sides). But ∠AEC is greater than ∠B § 137 (an exterior ∠ of a △ is greater than either opposite interior ∠), and ∠ACB is greater than ∠ACE. Ax. 8 Substitute for ∠ACE its equal ∠AEC, then ∠ACB is greater than ∠AEC. Since ∠AEC is greater than ∠B, and ∠ACB is greater than ∠AEC, ∠ACB is greater than ∠B. q.e.d. Ex. 9. If any angle of an isosceles triangle is equal to two thirds of a right angle (60◦), what is the value of each of the two remaining angles? Ex. 10. One angle of a triangle is 34 ◦. Find the other angles, if one of them is twice the other.

TRIANGLES. 45 Proposition XXVII. Theorem. 153. Reciprocally: If two angles of a triangle are unequal, the sides opposite are unequal, and the greater side is opposite the greater angle. A BC In the triangle ACB, let the angle C be greater than the angle B. To prove that AB >AC. Proof. NowAB =AC, or <AC , or >AC . But AB is not equal to AC; for then the ∠C would be equal to the ∠B, § 145 (being ∠s opposite equal sides). And AB is not less than AC; for then the ∠C would be less than the ∠B. § 152 Both these conclusions are contrary to the hypothesis that the ∠C is greater than the ∠B. Hence, AB cannot be equal to AC or less than AC. ∴AB >AC. q.e.d. Ex. 11. If the vertical angle of an isosceles triangle is equal to 30 ◦, find the exterior angle included by a side and the base produced. Ex. 12. If the vertical angle of an isosceles triangle is equal to 36 ◦, find the angle included by the bisectors of the base angles.

BOOK I. PLANE GEOMETRY. 46 Proposition XXVIII. Theorem. 154. If two triangles have two sides of the one equal, respectively, to two sides of the other, but the included angle of the first triangle greater than the included angle of the second, then the third side of the first is greater than the third side of the second. A B E A B C E F A B C In the triangles ABC and ABE, let AB be equal to AB, BC to BE; but let the angle ABC be greater than the angle ABE. To prove that AC >AE. Proof. Place the△s so that AB of the one shall fall on AB of the other, and BE within the ∠ABC. Suppose BF drawn to bisect the ∠EBC , and draw EF . The△sEBF and CBF are equal. § 143 For BF =BF , Iden. BE =BC, Hyp. and ∠EBF = ∠CBF . Const. ∴EF =FC . § 128 Now AF +FE >AE . § 138 ∴AF +FC >AE . ∴AC >AE. q.e.d.

TRIANGLES. 47 Proposition XXIX. Theorem. 155. Conversely: If two sides of a triangle are equal, respectively, to two sides of another, but the third side of the first triangle is greater than the third side of the second, then the angle opposite the third

side of the first triangle is greater than the angle opposite the third side of the second. A B C D E F In the triangles ABC and DEF , let AB be equal to DE, AC to DF ; but let BC be greater than EF. To prove that the ∠A is greater than the ∠D. Proof. Now the ∠A is equal to the ∠D, or less than the ∠D, or greater than the ∠D. But the ∠A is not equal to the ∠D; for then the△ABC would be equal to the △DEF , § 143 (having two sides and the included ∠ of the one equal, respectively, to two sides and the included ∠ of the other), and BC would be equal to EF . And the ∠A is not less than the ∠D, for then BC would be less than EF .§ 154 Both these conclusions are contrary to the hypothesis that BC is greater than EF . Since the ∠A is not equal to the ∠D or less than the ∠D, the ∠A is greater than the ∠D. q.e.d.

BOOK I. PLANE GEOMETRY. 48 LOCI OF POINTS. 156. If it is required to find a point which shall fulfil a single geometric condition, the point may have an unlimited number of positions . If, however, all the points are in the same plane, the required point will be confined to a particular line, or group of lines . A point in a plane at a given distance from a fixed straight line of indefinite length in that plane, is evidently in one of two straight lines, so drawn as to be everywhere at the given distance from the fixed line, one on one side of the fixed line, and the other on the other side of it. A point in a plane equidistant from two parallel lines in that plane is evidently in a straight line drawn between the two given parallel lines and everywhere equidistant from them. 157. All points in a plane that satisfy a single geometrical condition lie, in general, in a line or group of lines; and this line or group of lines is called the locus of the points that satisfy the given condition. 158. To provecompletely that a certain line is the locus of points that fulfil a given condition, it is necessary to prove 1. Any point in the line satisfies the given condition; and any point not in the line does not satisfy the given condition. Or, to prove 2. Any point that satisfies the given condition lies in the line; and any point in the line satisfies the given condition . Note. The word locus (pronounced lo´kus) is a Latin word that signifies place. The plural of locus is loci (pronounced lo´si). 159. Def. A line which bisects a given line and is perpendicular to it is called the perpendicular bisector of the line.

LOCI OF POINTS. 49 Proposition XXX. Theorem. 160. The perpendicular bisector of a given line is the locus of points equidis- tant from the extremities of the line. A B C D O P R Let P R be the perpendicular bisector of the line AB, O any point in P R, and C any point not in P R. DrawOA and OB, CA and CB. To prove OA and OB equal,CA and CB unequal. Proof. 1. △OPA =△OPB , § 144 for PA =PB by hypothesis, and OP is common, (two right △s are equal if their legs are equal, each to each). □OA =OB. § 128 2. Since C is not in the ⊥, CA or CB will cut the⊥. Let CA cut the⊥ at D, and draw DB. Then, by the first part of the proof DA =DB. But CB <CD +DB. § 138 □CB <CD +DA. That is, CB <CA. □PR is the locus of points equidistant from A and B. § 158,1 q.e.d. 161. Cor. Two points each equidistant from the extremities of a line de- termine the perpendicular bisector of the line.

BOOK I. PLANE GEOMETRY. 50 Proposition XXXI. Theorem. 162. The bisector of a given angle is the locus of points equidistant from the sides of the angle. A B C D F G O P Q Let O be any point equidistant from the sides of the angle P AQ. To prove that O is in the bisector of the ∠PAQ . Proof. DrawAO. Suppose OF drawn⊥ to AP and

OG⊥ to AQ. In the rt.△sAFO and AGO, AO =AO, Iden. OF =OG, Hyp. ∴△AFO =△AGO. § 151 ∴ ∠FAO = ∠GAO. § 128 ∴O is in the bisector of the ∠PAQ . Let D be any point in the bisector of the angle PAQ. To prove that D is equidistant from AP and AQ. Proof. Suppose DB drawn⊥ to AP and DC⊥ to AQ. In the rt.△sABD and ACD, AD =AD, Iden. ∠DAB = ∠DAC, Hyp. ∴△ABD =△ACD. § 141 ∴DB =DC. § 128 ∴D is equidistant from AP and AQ.

QUADRILATERALS. 51 ∴ the bisector of the ∠PAQ is the locus of points that are equidistant from its sides. § 158, 2 QUADRILATERALS. 163. A quadrilateral is a portion of a plane bounded by four straight lines. The bounding lines are the sides, the angles formed by these sides are the angles, and the vertices of these angles are the vertices, of the quadrilateral. 164. A trapezium is a quadrilateral which has no two sides parallel. 165. A trapezoid is a quadrilateral which has two sides, and only two sides, parallel. 166. A parallelogram is a quadrilateral which has its opposite sides par- allel. Trapezium. Trapezoid. Parallelogram. 167. A rectangle is a parallelogram which has its angles right angles. 168. A square is a rectangle which has its sides equal. 169. A rhomboid is a parallelogram which has its angles oblique angles. 170. A rhombus is a rhomboid which has its sides equal. Square. Rectangle. Rhombus. Rhomboid. 171. The side upon which a parallelogram stands, and the opposite side, are called its lower and upper bases.

BOOK I. PLANE GEOMETRY. 52 172. Two parallel sides of a trapezoid are called its bases, the other two sides its legs, and the line joining the middle points of the legs is called the median of the trapezoid. A B C E P Q 173. A trapezoid is called an isosceles trapezoid if its legs are equal. 174. The altitude of a parallelogram or trapezoid is the perpendicular distance between its bases, as PQ . 175. A diagonal of a quadrilateral is a straight line joining two opposite vertices, as AC.

QUADRILATERALS. 53 Proposition XXXII. Theorem. 176. Two angles whose sides are parallel, each to each, are either equal or supplementary. A B C D H M N a c x a′c′ Let BA be parallel to HD, and BC be parallel to M N. To prove ∠sa, a′ and c equal;a and c′ supplementary. Proof. Let HD and BC prolonged intersect at x. Then ∠a = ∠x, and ∠a′ = ∠x. § 112 ∴ ∠a = ∠a′. Ax. 1 Also ∠c = ∠a′ (§ 93). ∴ ∠c = ∠a. Ax. 1 Now ∠a′ and ∠c′ are supplementary. § 89 Put ∠a for its equal, ∠a′. Then ∠a and ∠c′ are supplementary. q.e.d. 177. Cor. The opposite angles of a parallelogram are equal, and the adja- cent angles are supplementary.

BOOK I. PLANE GEOMETRY. 54 Proposition XXXIII. Theorem. 178. The opposite sides of a parallelogram are equal. A B C E Let the figure ABCE be a parallelogram. To prove BC =AE, and AB =EC . Proof. Draw the diagonal AC. △ABC =△CEA. § 139 ForAC is common, ∠BAC = ∠ACE, and ∠ACB = ∠CAE, § 110 (being alt-int. ∠s of ∥ lines). ∴BC =AE, and AB =CE, § 128 (being homologous sides of equal △s). q.e.d. 179. Cor. 1. A diagonal divides a parallelogram into two equal triangles. 180. Cor. 2. Parallel lines comprehended between parallel lines are equal. A B CD 181. Cor. 3. Two parallel lines are everywhere equally distant. For ifAB andDC are parallel,⊥s dropped from any points inAB toDC, are equal,§ 180. Hence, all points in AB are equidistant from DC.

QUADRILATERALS. 55 Proposition XXXIV. Theorem. 182. If the opposite sides of a quadrilateral are equal, the figure is a paral- lelogram. A B C E Let the figure ABCE be a quadrilateral, having BC equal to AE and AB to EC. To prove that the figure ABCE is a / / . Proof. Draw the diagonal AC. In the△sABC and CEA, BC =AE, Hyp. AB =CE, Hyp. and AC =AC, Iden. □△ABC =△CEA, § 150 (having three sides of the one equal, respectively, to the three sides of the other). □ ∠ACB = ∠CAE, § 128 and ∠BAC = ∠ACE, (being homologous ∠s of equal △s). □BC is‖ to AE, and AB is‖ to EC, § 111 (two lines in the same plane cut by a transversal are parallel, if the alt.-int. ∠s are equal). □ the figure ACBE is a / / , § 166 (having its opposite sides parallel). q.e.d.

BOOK I. PLANE GEOMETRY. 56 Proposition XXXV. Theorem. 183. If two sides of a quadrilateral are equal and parallel, then the other two sides are equal and parallel, and the figure is a parallelogram. A B C E Let the figure ABCE be a quadrilateral, having the side AE equal and parallel to BC. To prove that AB is equal and parallel to EC . Proof. DrawAC. The△sABC and CEA are equal, § 143 (having two sides and the included ∠ of each equal, respectively). For AC is common, BC =AE Hyp. and ∠BCA = ∠CAE, § 110 (being alt.-int. ∠s of ‖ lines). □AB =EC, and ∠BAC = ∠ACE, § 128 (being homologous parts of equal △s). □AB is‖ to EC, § 111 (two lines are ‖, if the alt.-int. ∠s are equal). □ the figure ABCE is a / / , § 166 (the opposite sides being parallel). q.e.d.

QUADRILATERALS. 57 Proposition XXXVI. Theorem. 184. The diagonals of a parallelogram bisect each other. A B C E O Let the figure ABCE be a parallelogram, and let the diagonals AC and BE cut each other at O. To prove that AO =OC, and BO =OE. Proof. In the△sAOE and COB , AE =BC, § 178 (being opposite sides of a / /). ∠OAE = ∠OCB , § 110 and ∠OEA = ∠OBC , (being alt.-int. ∠s of ‖ lines). □△AOE =△COB , § 139 (having two ∠s and the included side of the one equal, respectively, to two ∠s and the included side of the other). □AO =OC, and BO =OE, § 128 (being homologous sides of equal △s). q.e.d. Ex. 13. The median from the vertex to the base of an isosceles triangle is perpendicular to the base, and bisects the vertical angle. Ex. 14. If two straight lines are cut by a transversal so that the alternate- exterior angles are equal, the two straight lines are parallel. Ex. 15. If two parallel lines are cut by a transversal, the two exterior angles on the same side of the transversal are supplementary.

BOOK I. PLANE GEOMETRY. 58 Ex. 16. If two straight lines are cut by a transversal so as to make the exterior angles on the same side of the transversal supplementary, the two lines are parallel. Proposition XXXVII. Theorem. 185. Two parallelograms are equal, if two sides and the included angle of the one are equal, respectively, to two sides and the included angle of the other. A B C D A′ B′ C′ D′ In the parallelograms ABCD and A′B′C′D′, let AB be equal to A′B′, AD to A′D′, and angle A to A′. To prove that the / / s are equal. Proof. Place the / / ABCD on the / / A′B′C′D′, so that AD will fall on and coincide with its equal, A′D′. Then AB will fall on A′B′, and B on B′; (for ∠A = ∠A′, and AB =A′B′, by hyp.) Now,BC and B′C′ are both‖ to A′D′ and drawn through B′. □BC and B′C′ coincide, § 105 (through a given point only one line can be drawn ‖ to a given line). Also DC and D′C′ are‖ to A′B′ and drawn through D′. □DC and D′C′ coincide. § 105 □C falls on C′, § 48 (two lines can intersect in only one

point), □ the two / / s coincide, and are equal. q.e.d. 186. Cor. Two rectangles having equal bases and altitudes are equal.

QUADRILATERALS. 59 Proposition XXXVIII. Theorem. 187. If three or more parallels intercept equal parts on one transversal, they intercept equal parts on every transversal. A B C D E F G H K M P Let the parallels AH, BK, CM , DP intercept equal parts HK, KM , M P on the transversal HP . To prove that they intercept equal parts AB, BC, CD on the transversal AD. Proof. Suppose AH, BF , and CG drawn∥ to HP . ∠s AEB, BFC , etc. = ∠s HKE , KMF , etc., respectively. § 112 But ∠s HKE , KMF , etc. are equal. § 112 □ ∠ s AEB, BFC , etc. are equal. Ax. 1 Also ∠s BAE, CBF , etc. are equal. § 112 NowAE =HK , BF =KM , CG =MP , § 180 (parallels comprehended between parallels are equal). □AE =BF =CG. Ax. 1 □△ABE =△BCF =△CDG, § 139 (having two ∠s and the included side of each respectively equal). □AB =BC =CD. § 128 q.e.d.

BOOK I. PLANE GEOMETRY. 60 A B C D E F 188. Cor. 1. If a line is parallel to the base of a triangle and bisects one side, it bisects the other side also. Let DE be∥ to BC and bisect AB. Suppose a line is drawn through A∥ toBC. Then this line is ∥ toDE, by§ 106. The three parallels by hypothesis intercept equal parts on the transversal AB, and therefore, by § 187, they intercept equal parts on the transversal AC; that is, the line DE bisects AC. 189. Cor. 2. The line which joins the middle points of two sides of a triangle is parallel to the third side, and is equal to half the third side. A line drawn throughD, the middle point of AB,∥ toBC, passes through E, the middle point of AC, by § 188. Therefore the line joining D and E coincides with this parallel and is ∥ to BC. Also, since EF drawn∥ to AB bisects AC, it bisects BC, by§ 188; that is, BF =FC = 1 2BC. But BDEF is a / / by§ 166, and therefore DE =BF = 1 2BC. A B CD E F G 190. Cor. 3. The median of a trapezoid is parallel to the bases, and is equal to half the sum of the bases. Draw the diagonal DB. In the△ADB joinE, the middle point of AD, to F , the middle point of DB. Then, by § 189, EF is∥ to AB and = 1 2AB. In the△DBC join F to G, the middle point of BC. Then FG is∥ to DC and = 1 2DC. AB and FG , being∥ to DC, are∥ to each other. But only one line can be drawn through F∥ to AB (§ 105). Therefore FG is the prolongation of EF . Hence, EFG is parallel to AB and DC, and equal to 1 2(AB +DC).

POLYGONS IN GENERAL. 61 POLYGONS IN GENERAL. 191. A polygon is a portion of a plane bounded by straight lines. The bounding lines are the sides, and their sum, the perimeter of the polygon. The angles included by the adjacent sides are the angles of the polygon, and the vertices of these angles are the vertices of the polygon. The number of sides of a polygon is evidently equal to the number of its angles. 192. A diagonal of a polygon is a line joining the vertices of two angles not adjacent; as, AC (Fig. 1). A B C D E Fig. 1. A B C D E Fig. 2. D E F Fig. 3. 193. An equilateral polygon is a polygon which has all its sides equal. 194. An equiangular polygon is a polygon which has all its angles equal. 195. A convex polygon is a polygon of which no side, when produced, will enter the polygon. 196. A concave polygon is a polygon of which two or more sides, if produced, will enter the polygon. 197. Each angle of a convex polygon (Fig. 2) is called a salient angle, and is less than a straight angle. 198. The angleEDF of the concave polygon (Fig. 3) is called a

re-entrant angle, and is greater than a straight angle. When the term polygon is used, a convex polygon is meant.

BOOK I. PLANE GEOMETRY. 62 199. Two polygons are equal when they can be divided by diagonals into the same number of triangles, equal each to each, and similarly placed; for if the polygons are applied to each other, the corresponding triangles will coincide, and hence the polygons will coincide and be equal. 200. Two polygons are mutually equiangular , if the angles of the one are equal to the angles of the other, each to each, when taken in the same order. Figs. 1 and 2. 201. The equal angles in mutually equiangular polygons are called homolo- gous angles; and the sides which are included by homologous angles are called homologous sides. 202. Two polygons are mutually equilateral , if the sides of the one are equal to the sides of the other, each to each, when taken in the same order. Figs. 1 and 2. Fig. 4. Fig. 5. Fig. 6. Fig. 7. 203. Two polygons may be mutually equiangular without being mutually equilateral; as, Figs. 4 and 5. And, except in the case of triangles , two polygons may be mutually equi- lateral without being mutually equiangular; as, Figs. 6 and 7. If two polygons are mutually equilateral and mutually equiangular they are equal, for they can be made to coincide. 204. A polygon of three sides is called a triangle; one of four sides, a quadrilateral; one of five sides, a pentagon; one of six sides, a hexagon; one of seven sides, a heptagon; one of eight sides, an octagon; one of ten sides, a decagon; one of twelve sides, a dodecagon.

POLYGONS IN GENERAL. 63 Proposition XXXIX. Theorem. 205. The sum of the interior angles of a polygon is equal to two right angles, taken as many times less two as the figure has sides. A B C DE F Let the figure ABCDEF be a polygon, having n sides. To prove that ∠A + ∠B + ∠C, etc. = (n− 2)2 rt. ∠s. Proof. FromA draw the diagonals AC, AD, and AE. The sum of the ∠s of the△s is equal to the sum of the ∠s of the polygon. Now, there are (n− 2)△s, and the sum of the ∠s of each△ = 2 rt. ∠s. § 129 ∴ the sum of the ∠s of the△s, that is, the sum of the ∠s of the polygon is equal to (n− 2)2 rt.△s. q.e.d. 206. Cor. The sum of the angles of a quadrilateral equals 4 right angles; and if the angles are all equal, each is a right angle. In general, each angle of an equiangular polygon of n sides is equal to 2(n− 2) n right angles. Ex. 17. How many diagonals can be drawn in a polygon of n sides?

BOOK I. PLANE GEOMETRY. 64 Proposition XL. Theorem. 207. The exterior angles of a polygon, made by producing each of its sides in succession, are together equal to four right angles. A B C D E a b c d e Let the figure ABCDE be a polygon, having its sides produced in succession. To prove the sum of the ext. ∠s = 4 rt. ∠s. Proof. Denote the int. ∠s of the polygon by A, B, C, D, E, and the corre- sponding ext. ∠s bya, b, c, d, e. ∠A + ∠a = 2 rt. ∠s, § 89 and ∠B + ∠b = 2 rt. ∠s, (being sup.-adj. ∠s). In like manner each pair of adj. ∠s = 2 rt. ∠s. ∴ the sum of the interior and exterior ∠s of a polygon of n sides is equal to 2n rt. ∠s. But the sum of the interior ∠s = (n− 2)2 rt. ∠s § 205 = 2n rt. ∠s− 4 rt. ∠s. ∴ the sum of the exterior ∠s = 4 rt. ∠s. q.e.d. Ex. 18. How many sides has a polygon if the sum of its interior ∠s is twice the sum of its exterior ∠s? ten times the sum of its exterior ∠s?

SYMMETRY. 65 SYMMETRY. 208. Two points are said to be symmetrical with respect to a third point, called the centre of symmetry , if this third point bisects the

straight line which joins them. O P P ′ XX′ A B C A′ B′C′ O A B C D B′ C′ XX′ Two points are said to be symmetrical with respect to a straight line, called the axis of symmetry, if this straight line bisects at right angles the straight line which joins them. Thus, P and P′ are symmetrical with respect to O as a centre, and XX′ as an axis, if O bisects the line PP′, and if XX′ bisects PP′ at right angles. 209. A figure is symmetrical with respect to a point as a centre of symme- try, if the point bisects every straight line drawn through it and terminated by the boundary of the figure. 210. A figure is symmetrical with respect to a line as an axis of symmetry if one of the parts of the figure coincides, point for point, with the other part when it is folded over on that line as an axis.

BOOK I. PLANE GEOMETRY. 66 A B C A′ B′ C′ XX′ 211. Two figures are said to be symmetrical with respect to an axis if every point of one has a corresponding symmetrical point in the other. Thus, if every point in the figure A′B′C′ has a symmetrical point in ABC, with respect toXX′ as an axis, the figureA′B′C′ is symmetrical toABC with respect to XX′ as an axis.

SYMMETRY. 67 Proposition XLI. Theorem. 212. A quadrilateral which has two adjacent sides equal, and the other two sides equal, is symmetrical with respect to the diagonal joining the vertices of the angles formed by the equal sides, and the diagonals are perpendicular to each other. A B C D O Let ABCD be a quadrilateral, having AB equal to AD, and CB equal to CD, and having the diagonals AC and BD. To prove that the diagonal AC is an axis of symmetry, and that it is ⊥ to the diagonal BD. Proof. In the△sABC and ADC, AB =AD, and BC =DC, Hyp. and AC =AC. Iden. □△ABC =△ADC. § 150 □ ∠BAC = ∠DAC, and ∠BCA = ∠DCA. Hence, if ABC is turned on AC as an axis until it falls on ADC, AB will fall upon AD, CB on CD, and OB on OD. □ the△ABC will coincide with the △ADC. □AC is an axis of symmetry (§ 210) and is⊥ to BD. § 208 q.e.d.

BOOK I. PLANE GEOMETRY. 68 Proposition XLII. Theorem. 213. If a figure is symmetrical with respect to two axes perpendicular to each other, it is symmetrical with respect to their intersection as a centre. A B C D E FG H I K L M N O XX′ Y Y ′ Let the figure ABCDEF GH be symmetrical with respect to the two perpendicular axes XX ′, Y Y ′, which intersect at O. To prove that O is the centre of symmetry of the figure. Proof. Let N be any point in the perimeter. Suppose NMI drawn⊥ to YY ′, IKL⊥ to XX′. Then NI is‖ to XX′ and IL is‖ to YY ′. § 104 DrawLO, ON, and KM . Now KI =KL, § 208 (the figure being symmetrical with respect to XX′). But KI =OM. § 180 □KL =OM, and KLOM is a / / . § 183 □LO is equal and parallel to KM . § 183 In like manner ON is equal and parallel to KM . □LON is a straight line. § 105 □O bisectsLN, any straight line and thereforeevery straight line drawn through O and terminated by the perimeter. □O is the centre of symmetry of the figure. q.e.d.

SYMMETRY. 69 REVIEW QUESTIONS ON BOOK I. 1. What is the subject-matter of Geometry? 2. What is a geometric magnitude? 3. What is an axiom? a theorem? a converse theorem? an opposite theorem? a contradictory theorem? 4. Define a straight line; a curved line; a broken line; a plane surface; a curved surface. 5. How many points are necessary to determine a straight line? 6. How many straight lines are necessary to determine a point? 7. On what does the magnitude of an angle

depend? 8. Define a straight angle; a right angle; an oblique angle. 9. Define adjacent angles; complementary angles; supplementary angles; conjugate angles. 10. Define parallel lines and give the axiom of parallels. 11. If two lines in the same plane are parallel and cut by a transversal, what pairs of angles are equal? what pairs are supplementary? 12. Define a right triangle; an isosceles triangle; a scalene triangle. 13. To how many right angles is the sum of the angles of a triangle equal? the sum of the acute angles of a right triangle? 14. To what angles is the exterior angle of a triangle equal? 15. What is the test of equality of two geometric magnitudes? 16. How does a reciprocal theorem differ from a converse theorem? 17. State the three cases in which two triangles are equal. 18. State the cases in which two right triangles are equal. 19. What is meant by a locus of points? 20. Where are the points located in a plane that are each equidistant from two given points? from two intersecting lines? 21. Define a parallelogram; a trapezoid; an isosceles trapezoid. 22. When is a figure symmetrical with respect to a centre?

BOOK I. PLANE GEOMETRY. 70 23. When is a figure symmetrical with respect to an axis? 24. Must a triangle be equiangular if equilateral? must a triangle be equilateral if equiangular? 25. When are two polygons said to be mutually equiangular? 26. When are two polygons said to be mutually equilateral? 27. Can two polygons of more than three sides be mutually equiangular without being mutually equilateral? mutually equilateral without being mutually equiangular? 28. What line do two points each equidistant from the extremities of a given straight line determine? METHODS OF PROVING THEOREMS. 214. There are three general methods of proving theorems, the synthetic, the analytic, and the indirect methods. The synthetic method is the method employed in most of the theorems already given, and consists in putting together known truths in order to obtain a new truth. The analytic method is the reverse of the synthetic method. It asserts that the conclusion is true if another proposition is true, and so on step by step, until a known truth is reached. Thus, proposition A is true if proposition B is true, and B is true if C is true; but C is true, hence A and B are true. If a known truth suggests the required proof, it is best to use the synthetic form at once. If no proof occurs to the mind, it is necessary to use the analytic method to discover the proof, and then the synthetic proof may be given. The indirect method, or the method of reductio ad absurdum, is illustrated on page 45. It consists in proving a theorem to be true by proving its contra- dictory to be false.

SYMMETRY. 71 215. Generally auxiliary lines are required, as a line connecting two points; a line parallel to or perpendicular to a given line ; a line produced by its own length; a line making with another line an angle equal to a given angle. Two lines are proved equal by proving them homologous sides of equal triangles; or legs of an isosceles triangle ; or opposite sides of a parallelogram. Two angles are proved equal by proving them alternate-interior an- gles or exterior-interior angles of parallel lines ; or homologous angles of equal triangles; or base angles of an isosceles triangle ; or opposite angles of a paral- lelogram. Two suggestions are of special importance to the beginner: 1. Draw as accurate figures as possible. 2. Draw as general figures as possible.

BOOK I. PLANE GEOMETRY. 72 EXERCISES. Prove by the analytic method: A B CD E Ex. 19. A median of a triangle is less than half the sum of the two adjacent sides. To prove the median AD <1 2(AB +AC). Now AD <1 2(AB +AC), if 2AD <AB +AC. This suggests producing AD by its own length to E, and joining BE. Then AE = 2AD, and 2AD <AB +AC if AE <AB +AC. But AE <AB +BE. § 138 ∴AE <AB +AC if AC =BE. And AC =BE if△ACD =△EBD . § 128 But △ACD =△EBD . § 143 For CD =DB, Hyp. AD =DE, Const. and ∠ADC = ∠BDE . § 93 ∴AE <AB +AC.

EXERCISES. 73 ∴AD <1 2(AB +AC). A B C D E G Ex. 20. A straight line which bisects two sides of a triangle is parallel to the third side. If AD =DB and AE =EC, to prove DE∥ to BC. DrawCG∥ to BA, and produce DE to meet it at G. DE is∥ to BC if BCGD is a / / . § 166 BCGD is a / / if CG =BD. § 183 CG =BD if each is equal to AD. Ax. 1 Now BD =AD. Hyp. And CG =AD if△CGE =△ADE. § 128 But △CGE =△ADE. § 139 For EC =AE. Hyp. ∠GEC = ∠AED. § 93 ∠ECG = ∠DAE. § 110 ∴DE is∥ to BC. Prove by the synthetic method: Ex. 21. The middle point of the hypotenuse of a right triangle is equidistant from the three vertices. FromD, the middle point, draw DE⊥ to CB. DE is∥ to AC (why?), and DE bisects CB (why?). ∴D is equidistant from B, A, and C. (Why?)

BOOK I. PLANE GEOMETRY. 74 A BC D E A BC D E abc 2a Ex. 22. If one acute angle of a right triangle is double the other, the hypotenuse is double the shorter leg. The median CD =BD =AD (Ex. 21). Then ∠b = ∠a; and ∠c = ∠2a. (Why?) Nowa + 2a = 90◦. (Why?) ∴ ∠a = 30◦; ∠2a = 60◦; ∠c = 60◦. ∴△ACD is equilateral (why?), andAD, half ofAB =AC. ∴AB = 2AC.

EXERCISES. 75 Ex. 23. If two triangles have two sides of the one equal, respectively, to two sides of the other, and the angles opposite two equal sides equal, the angles opposite the other two equal sides are equal or supplementary, and if equal the triangles are equal. Let AC =A′C′, BC =B′C′, and ∠B = ∠B′. Place△A′B′C′ on△ABC so that B′C′ shall coincide with BC, and ∠A′ and ∠A shall be on the same side of BC. A B C D A′ B′ C′ A′ B′ C′ Since ∠B′ = ∠B, B′A′ will fall along BA, and A′ will fall at A or at some other point in BA, as D. If A′ falls at A, the△sA′B′C′ and ABC coincide and are equal. If A′ falls at D, the△sA′B′C′ and DBC coincide and are equal. Since CD =C′A′ =CA, ∠A = ∠CDA. (Why?) But ∠sCDA and CDB are supplements. (Why?) ∴ ∠ sA and CDB are supplements. (Why?) Draw figures and show that the triangles are equal: 1. If the given angles B and B′ are both right or both obtuse angles. 2. If the required angles A and A′ are both acute, both right, or both obtuse. 3. If AC and A′C′ are not less than BC and B′C′, respectively.

BOOK I. PLANE GEOMETRY. 76 A B C DE FO A B C D E F D′ E′ F ′ O Ex. 24. The bisectors of the angles of a triangle meet in a point which is equidistant from the sides of the triangle. Let the bisectors AD and BE intersect at O. Then O being in AD is equidistant from AC and AB. (Why?) And O being in BE is equidistant from BC and AB. Hence, O is equidistant from AC and BC, and therefore in the bisector CF . (Why?) Ex. 25. The perpendicular bisectors of the sides of a triangle meet in a point which is equidistant from the vertices of the triangle. Let the⊥ bisectors EE′ and DD′ intersect at O. Then O being in EE′ is equidistant from A and C. (Why?) And O being in DD′ is equidistant from A and B. Hence, O is equidistant from B and C, and therefore is in the ⊥ bisector FF′. (Why?) Ex. 26. The perpendiculars from the

vertices of a triangle to the opposite sides meet in a point. Let the⊥s be AH, BP , and CK . Through A, B, C suppose B′C′, A′C′, A′B′, drawn∥ toBC,AC,AB, respectively. Then AH is⊥ toB′C′. (Why?) Now ABCB′ and ACBC′ are / / s (why?) and AB′ = BC, and AC′ = BC. (Why?) That is, A is the middle point of B′C′. In the same way, B andC are the middle points of A′C′ and A′B′, respectively. Therefore, AH, BP , and CK are the⊥ bisectors of the sides of the △A′B′C′. Hence, they meet in a point. (Why?)

EXERCISES. 77 A B CH K O P A′ B′C′ A B C D E F G H O Ex. 27. The medians of a triangle meet in a point which is two thirds of the distance from each vertex to the middle of the opposite side. Let the two medians AD and CE meet in O. Take F the middle point of OA, and G of OC. Join GF , FE , ED, and DG. In △AOC, GF is∥ to AC and equal to 1 2AC. (Why?) DE is∥ toAC and equal to 1 2AC. (Why?) Hence, DGFE is a / / . (Why?) Hence, AF = FO = OD, and CG = GO = OE. (Why?) Hence, any median cuts off on any other median two thirds of the distance from the vertex to the middle of the opposite side. Therefore, the median from B will cut off AO, two thirds of AD; that is, will pass through O. Note. If three or more lines pass through the same point, they are called con- current lines. Ex. 28. If an angle is bisected, and if a line is drawn through the vertex perpendicular to the bisector, this line forms equal angles with the sides of the given angle. A BC D E F A B C D EF A B C D E F G A B C D E F G Ex. 29. The bisectors of two supplementary adjacent angles are perpen- dicular to each other. Ex. 30. If the bisectors of two adjacent angles are perpendicular to each other, the adjacent angles are supplementary. Ex. 31. The bisector of one of two vertical angles bisects the other. Ex. 32. The bisectors of two vertical angles form one line.

BOOK I. PLANE GEOMETRY. 78 Ex. 33. The bisectors of the two pairs of vertical angles formed by two intersecting lines are perpendicular to each other. Ex. 34. The bisector of the vertical angle of an isosceles triangle bisects the base, and is perpendicular to the base. △ADC =△BDC (§ 143) A B C D A B C D E Ex. 35. The perpendicular bisector of the base of an isosceles triangle passes through the vertex and bisects the angle at the vertex (§ 160). Ex. 36. If the perpendicular bisector of the base of a triangle passes through the vertex, the triangle is isosceles. Ex. 37. Any point in the bisector of the vertical angle of an isosceles triangle is equidistant from the extremities of the base (Ex. 34, § 160). Ex. 38. If the bisector of an angle of a triangle is perpendicular to the opposite side, the triangle is isosceles. Ex. 39. If two isosceles triangles are on the same base, a straight line passing through their vertices is perpendicular to the base, and bisects the base (§ 161). Ex. 40. Two isosceles triangles are equal when a side and an angle of the one are equal, respectively, to the homologous side and angle of the other.

EXERCISES. 79 A B C D E Ex. 41. The bisector of an exterior angle of an isosceles triangle, formed by producing one of the legs through the vertex, is parallel to the base. Why does ∠DAC = ∠B + ∠C? Why is ∠DAE = ∠ABC? Why is AE∥ to BC? Ex. 42. If the bisector of an exterior angle of a triangle is parallel to one side, the triangle is isosceles. A B C D Ex. 43. If one of the legs of an isosceles triangle is produced through the vertex by its own length, the line joining the end of the leg produced to the nearer end of the base is perpendicular to the base. ∠CBA = ∠A, and ∠CBD = ∠D. (Why?) □ ∠ABD = ∠A + ∠D. Ex. 44. A line drawn from the vertex of the right

angle of a right triangle to the middle point of the hypotenuse divides the triangle into two isosceles triangles. Ex. 45. If the equal sides of an isosceles triangle are produced through the vertex so that the external segments are equal, the extremities of these segments will be equally distant from the extremities of the base, respectively.

BOOK I. PLANE GEOMETRY. 80 A B C D H KP Ex. 46. If through any point in the bisector of an angle a line is drawn parallel to either of the sides of the angle, the triangle thus formed is isosceles. Ex. 47. Through any pointC in the lineAB an intersecting line is drawn, and from any two points in this line equidistant from C perpendiculars are dropped on AB or AB produced. Prove that these perpendiculars are equal. Ex. 48. If the median drawn from the vertex of a triangle to the base is equal to half the base, the vertical angle is a right angle. A B C D EF Ex. 49. The lines joining the middle points of the sides of a triangle divide the triangle into four equal triangles. Ex. 50. The altitudes upon the legs of an isosceles triangle are equal. Rt.△BEC = rt.△CDB (§ 141). Ex. 51. If the altitudes upon two sides of a triangle are equal, the triangle is isosceles. Rt.△BEC = rt.△CDB (§ 151).

EXERCISES. 81 A B C DE A B C DE O A B C DE A B C D E F G H K Ex. 52. The medians drawn to the legs of an isosceles triangle are equal. △BEC =△CDB (§ 143). Ex. 53. If the medians to two sides of a triangle are equal, the triangle is isosceles. BO =CO, and OE =OD (Ex. 27). ∠BOE = ∠COD. □△BOE =△COD (§ 143). Ex. 54. The bisectors of the base angles of an isosceles triangle are equal. △BEC =△CDB (§ 139). Ex. 55. Opposite Theorem. If a triangle is not isosceles, the bisectors of the base angles are not equal. Let ∠ABC be greater than ∠ACB; then KC >KB . (Why?) NowCD >BE, if KD is greater than or equal to KE. But suppose KD < KE. Lay off KH = KD and KG = KB, join HG, and draw GF∥ to BE. △KDB =△KHG . (Why?) □ ∠KHG = ∠KDB . (Why?) □ ∠KEC is greater than ∠KHG . (Why?) □GF >HE . (Why?) ∠GFC is greater than ∠FCG (1 2ACB). □CG>GF , and >HE . □KC−KG>KE −KH , or KC +KD >KB +KE, or CD >BE. Ex. 56. State the converse theorem of Ex. 54. Is the converse theorem true?

BOOK I. PLANE GEOMETRY. 82 A B CD E F Ex. 57. The perpendiculars dropped from the middle point of the base upon the legs of an isosceles triangle are equal. △BED =△CFD (§ 141). Ex. 58. State and prove the converse. △BED =△CFD (§ 151). Ex. 59. The difference of the distances from any point in the base produced of an isosceles triangle to the equal sides of the triangle is constant. Rt.△DGC = rt.△DFC . (Why?) □DF =DG. □DE−DF =DE−DG =EG, the⊥ distance between the two∥s, BA and CH . A B C D E F G H A B C DE F G P Ex. 60. The sum of the perpendiculars dropped from any point in the base of an isosceles triangle to the legs is constant, and equal to the altitude upon one of the legs. Let PE and PD be the⊥s and BF the altitude. DrawPG⊥ to BF . EPGF is a parallelogram. (Why?) □ GF = PE . It remains to prove GB =PD . The rt.△PGB = the rt.△BDP . (Why?)

EXERCISES. 83 Ex. 61. The sum of the perpendiculars dropped from any point within an equilateral triangle to the three sides is constant, and equal to the altitude. AD is the altitude, PE , PG , and PF the three perpendiculars. Through P drawHK∥ to BC, meeting AD at M. Then MD =PE . (Why?) PG +PF =AM (Ex. 60). A B C DE F G H

K MP A B C D Ex. 62. ABC and ABD are two triangles on the same base AB, and on the same side of it, the vertex of each triangle being without the other. If AC equals AD, show that BC cannot equal BD (§ 154). Ex. 63. The sum of the lines which join a point within a triangle to the three vertices is less than the perimeter, but greater than half the perimeter. A B C O A B CD E F Ex. 64. If from any point in the base of an isosceles triangle parallels to the legs are drawn, a parallelogram is formed whose perimeter is constant, being equal to the sum of the legs of the triangle. Ex. 65. The bisector of the vertical angle A of a triangle ABC, and the bisectors of the exterior angles at the base formed by producing the sides AB and AC, meet in a point which is equidistant from the base and the sides produced (§ 162).

BOOK I. PLANE GEOMETRY. 84 A B C D EFM N A B C DE O Ex. 66. If the bisectors of the base angles of a triangle are drawn, and through their point of intersection a line is drawn parallel to the base, the length of this parallel between the sides is equal to the sum of the segments of the sides between the parallel and the base. ∠EOB = ∠OBC = ∠OBE . □BE =EO. Ex. 67. The bisector of the vertical angle of a triangle makes with the perpendicular from the vertex to the base an angle equal to half the difference of the base angles. Let ∠B be greater than ∠A. ∠DCE = 90◦− ∠A− ∠ACD. ∠ACD = 90◦− 1 2 ∠A− 1 2 ∠B. □ ∠DCE = 90◦− ∠A− (90◦− 1 2 ∠A− 1 2 ∠B) = 1 2 ∠B− 1 2 ∠A. A B C D E A B CD O Ex. 68. If the diagonals of a quadrilateral bisect each other, the figure is a parallelogram. Prove△AOB =△COD.

EXERCISES. 85 A B CD Ex. 69. The diagonals of a rectangle are equal. Prove△ABC =△BAD. Ex. 70. If the diagonals of a parallelogram are equal, the figure is a rectangle. Ex. 71. The diagonals of a rhombus are perpendicular to each other, and bisect the angles of the rhombus. Ex. 72. The diagonals of a square are perpendicular to each other, and bisect the angles of the square. A B CD E F M N Ex. 73. Lines from two opposite vertices of a parallelogram to the middle points of the opposite sides trisect the diagonal. EBFD is a / / (why?), and DF is∥ to EB. AM =MN , and MN =CN (§ 188). Ex. 74. The lines joining the middle points of the sides of any quadrilateral, taken in order, enclose a parallelogram. ProveHG and EF∥ to AC; and FG and EH∥ to BD (§ 189). Then HG and EF are each equal to 1 2AC.

BOOK I. PLANE GEOMETRY. 86 A B C D E F G H A B C D E FG H O A B CD E F G H A B CD E F G H Ex. 75. The lines joining the middle points of the sides of a rhombus, taken in order, enclose a rectangle. (Proof similar to that of Ex. 74.) Ex. 76. The lines joining the middle points of the sides of a rectangle (not a square), taken in order, enclose a rhombus. Ex. 77. The lines joining the middle points of the sides of a square, taken in order, enclose a square. Ex. 78. The lines joining the middle points of the sides of an isosceles trapezoid, taken in order, enclose a rhombus or a square. SHR and QFP drawn⊥ to AB are parallel. □ PQSR is a / / , and by Const. is a rectangle or a square. □EFGH is a rhombus or a square (Exs. 76, 77). A B CD E F G H P Q R S A B CD E F L K A B CD E F L K A B C D Ex. 79. The bisectors of the angles of a rhomboid enclose a rectangle. Ex. 80. The bisectors of the angles of a rectangle enclose a square. Ex. 81. If two parallel lines are cut by a transversal, the bisectors of the interior angles form a rectangle.

EXERCISES. 87 A B CD E F G H Ex. 82. The median of a trapezoid passes through the middle points of the two diagonals. The median EF is‖ to AB and bisects AD (§ 190). □ it bisects DB. LikewiseEF bisects BC and BD. Ex. 83. The lines joining the middle points of the diagonals of a trapezoid is equal to half the difference of the bases. △BFG =△DFC . (Why?) □EF = 1 2AG (§ 180). CF =FG , DC =BG. □AG =AB−DC. □EF = 1 2(AB−DC) A B CD E F G A B C D E A B CD A B CD Ex. 84. In an isosceles trapezoid each base makes equal angles with the legs. Draw CE ‖ to DB. CE = DB. (Why?) ∠A = ∠CEA, ∠B = ∠CEA, ∠sC and D have equal supplements. Ex. 85. If the angles at the base of a trapezoid are equal, the other angles are equal, and the trapezoid is isosceles. Ex. 86. In an isosceles trapezoid the opposite angles are supplementary: ∠C = ∠D (Ex. 84)

BOOK I. PLANE GEOMETRY. 88 Ex. 87. The diagonals on an isosceles trapezoidal are equal. Prove△ACD =△BDC . Ex. 88. If the diagonals of a trapezoid are equal, the trapezoid is isosceles. DrawCE and DF⊥ to AB. △ADF =△BCE . (Why?) □ ∠ADF = ∠CBA. △ABC =△BAD. A B C D E F O A B CD E F Ex. 89. If from the diagonal DB, of a square ABCD ,BE is cut off equal toBC, and EF is drawn perpendicular to BD meetingDC atF , then DE is equal to EF and also to FC . ∠EDF = 45◦, and ∠DFE = 45◦; and DE = DF . Rt. △BEF = rt.△BCF (§ 151); and EF =FC . Ex. 90. Two angles whose sides are so perpendicular, each to each, are either equal or supplementary.

BOOK II. THE CIRCLE. DEFINITIONS. 216. A circle is a portion of a plane bounded by a curved line, all points of which are equally distant from a point within called thecentre. The bounding line is called the circumference of the circle. 217. A radius is a straight line from the centre to the circumference; and a diameter is a straight line through the centre, with its ends in the circumference. By the definition of a circle, all its radii are equal . All its diameters are equal, since a diameter is equal to two radii. 218. Postulate. A circumference can be described from any point as a centre, with any given radius. 219. A secant is a straight line of unlimited length which intersects the circumference in two points; as, AD (Fig. 1). A B C D Fig. 1. 220. A tangent is a straight line of unlimited length which has one point, and only one, in common with the circumference; as, BC (Fig. 1). In this case the circle is said to be tangent to the straight line. The common point is called the point of contact, or point of tangency . 221. Two circles are tangent to each other, if both are tangent to a straight line at the same point; and are said to be tangent internally or externally, according as one circle lies wholly within or without the other.

BOOK II. PLANE GEOMETRY. 90 222. An arc is any part of the circumference; as, BC (Fig. 3). Half a circumference is called a semicircumference. Two arcs are called conjugate arcs, if their sum is a circumference. 223. A chord is a straight line that has its extremities in the circumference; as, the straight line BC (Fig. 3). 224. A chord subtends two conjugate arcs. If the arcs are unequal, the less is called the minor arc, and the greater the major arc. A minor arc is generally called simply an arc. A B CD N O SEMICIRCLE SEGMENT SE CTOR QUADRANT Fig. 2. A B C D E Fig. 3. A B C D E F G H Fig. 4. 225. A segment of a circle is a portion of the circle bounded by an arc and its chord (Fig. 2). 226. A semicircle is a segment equal to half the circle (Fig. 2). 227. A sector of a circle is a portion of the circle bounded by two radii and the arc

which they intercept. The angle included by the radii is called the angle of the sector (Fig. 2). 228. A quadrant is a sector equal to a quarter of the circle (Fig. 2). 229. An angle is called a central angle, if its vertex is at the centre and its sides are radii of the circle; as, ∠AOD (Fig. 2). 230. An angle is called an inscribed angle, if its vertex is in the circum- ference and its sides are chords; as, ∠ABC (Fig. 3). An angle is inscribed in a segment , if its vertex is in the arc of the segment and its sides pass through the extremities of the arc.

ARCS, CHORDS, AND TANGENTS. 91 231. A polygon is inscribed in a circle , if its sides are chords; and a circle is circumscribed about a polygon , if all the vertices of the polygon are in the circumference (Fig. 3). 232. A circle is inscribed in a polygon , if the sides of the polygon are tangent to the circle; and a polygon is circumscribed about a circle if its sides are tangents (Fig. 4). 233. Two circles are equal, if they have equal radii. For they will coincide, if their centres are made to coincide. Conversely: Two equal circles have equal radii. 234. Two circles are concentric , if they have the same centre. ARCS, CHORDS, AND TANGENTS. Proposition I. Theorem. 235. A straight line cannot meet the circumference of a circle in more than two points. H K M O P Let HK be any line meeting the circumference HKM in H and K. To prove that HK cannot meet the circumference in any other point . Proof. If possible, let HK meet the circumference in P . Then the radii OH, OP , and OK are equal. § 217 □P does not lie in the straight line HK . § 102 □HK meets the circumference in only two points. q.e.d.

BOOK II. PLANE GEOMETRY. 92 Proposition II. Theorem. 236. In the same circle or in equal circles, equal central angles intercept equal arcs; and of two unequal central angles the greater intercepts the greater arc. A B C O P A′ B′ O′ P ′ In the equal circles whose centres are O and O′, let the angles AOB and A′O′B′ be equal, and angle AOC be greater than angle A′O′C′. To prove that 1. arcAB = arcA′B′; 2. arcAC >arcA′B′. Proof. 1. Place the⊙A′B′P′ on the⊙ABP so that the ∠A′O′B′ shall coincide with its equal, the ∠AOB. Then A′ falls on A, and B′ on B. § 233 □ arcA′B′ coincides with arcAB. § 216 2. Since the ∠AOC is greater than the ∠A′O′B′, it is greater than the ∠AOB, the equal of the ∠A′O′B′. Therefore, OC falls without the ∠AOB. □ arcAC >arcAB. Ax. 8 □ arcAC >arcA′B′, the equal of arc AB. q.e.d.

ARCS, CHORDS, AND TANGENTS. 93 237. Conversely: In the same circle or in equal circles, equal arcs sub- tend equal central angles; and of two unequal arcs the greater subtends the greater central angle. To prove that 1. ∠AOB = ∠A′O′B′; 2. ∠AOC is greater than ∠A′O′B′. Proof. 1. Place the⊙A′B′P′ on the⊙ABP so that O′A shall fall on its equal OA, and the arc A′B′ on its equal AB. Then O′B′ will coincide with OB. § 47 □ ∠A′O′B′ = ∠AOB. § 60 2. Since arcAC > A′B′, it is greater than arc AB, the equal of A′B′, and OB will fall within the ∠AOC. □ ∠AOC is greater than ∠AOB. Ax. 8 □ ∠AOC is greater than ∠A′O′B′. q.e.d. 238. Cor. 1. In the same circle or in equal circles, two sectors that have equal angles are equal; two sectors that have unequal angles are unequal, and the greater sector has the greater angle. 239. Cor. 2. In the same circle or in equal circles, equal sectors have equal angles; and of two unequal sectors the greater has the greater angle.

BOOK II. PLANE GEOMETRY. 94 240. Law of Converse Theorems.It was stated in§ 32 that the converse of a theorem is not necessarily true. If, however, a theorem is in fact a group of three theorems, and if one of the hypotheses of the group must be true, and no two of the conclusions can be true at the same time , then the converse of the theorem is necessarily true. Proposition II. is a group of three theorems. It asserts that the arc AB is equal to the arc A′B′, if the angle AOB is equal to the angle A′O′B′; that the arc AB is greater than the arc A′B′, if the angle AOB is greater than the angle A′O′B′; that the arc AB is less than the arc A′B′, if the angle AOB is less than the angle A′O′B′. One of these hypotheses must be true; for the angle AOB must be equal to, greater than, or less than, the angle A′O′B′. No two of the conclusions can be true at the same time, for the arc AB cannot be both equal to and greater than the arc A′B′; nor can it be both equal to and less than the arc A′B′; nor both greater than and less than the arc A′B′. In such a case, the converse theorem is necessarily true, and no proof like that given in the text is required to establish it.

ARCS, CHORDS, AND TANGENTS. 95 Proposition III. Theorem. 241. In the same circle or in equal circles, equal arcs are subtended by equal chords; and of two unequal arcs the greater is subtended by the greater chord. A BF O A′ B′ O′ In the equal circles whose centres are O and O′, let the arcs AB and A′B′ be equal, and the arc AF greater than arc A′B′. To prove that 1. chord AB = chord A′B′; 2. chord AF > chordA′B′. Proof. Draw the radii OA, OB, OF , O′A′, O′B′. 1. The△sAOB and A′O′B′ are equal. § 143 ForOA =O′A′, and OB =O′B′, § 233 (radii of equal circles), and ∠AOB = ∠A′O′B′, § 237 (in equal ⊙s equal arcs subtend equal central ∠s). ∴ chordAB = chord A′B′. § 128 2. In the△sAOF and A′O′B′, OA =O′A′, and OF =O′B′. § 233 But the ∠AOF is greater than the ∠A′O′B′, § 237 (in equal ⊙s, the greater of two unequal arcs subtends the greater ∠). ∴ chordAF >chordA′B′. § 154 q.e.d.

BOOK II. PLANE GEOMETRY. 96 242. Cor. In the same circle or in equal circles, the greater of two unequal major arcs is subtended by the less chord.

ARCS, CHORDS, AND TANGENTS. 97 Proposition IV. Theorem. 243. Conversely: In the same circle or in equal circles, equal chords subtend equal arcs; and of two unequal chords the greater subtends the greater arc. A BF O A′ B′ O′ In the equal circles whose centres are O and O′, let the chords AB and A′B′ be equal, and the chord AF greater than A′B′. To prove that 1. arc AB = arcA′B′; 2. arc AF >arcA′B′. Proof. Draw the radii OA, OB, OF , O′A′, O′B′. 1. The△sOAB and O′A′B′ are equal. § 150 ForOA =O′A′, and OB =O′B′, § 233 and chord AB = chord A′B′. Hyp. ∴ ∠AOB = ∠A′O′B′. § 128 ∴ arcAB = arcA′B′, § 236 (in equal ⊙s equal central ∠s intercept equal arcs). 2. In the△sOAFandO′A′B′, OA =O′A′ and OF =O′B′. § 233 But chord AF >chordA′B′. Hyp. ∴ the ∠AOF is greater than the ∠A′O′B′. § 155 ∴ arcAF >arcA′B′, § 236 (in equal ⊙s the greater central ∠ intercepts the greater arc). q.e.d.

BOOK II. PLANE GEOMETRY. 98 244. Cor. In the same circle or in equal circles, the greater of two unequal chords subtends the less major arc. Proposition V. Theorem. 245. A diameter perpendicular to a chord bisects the chord and the arcs subtended by it. A B E M O S Let ES be a diameter perpendicular to the chord AB at M. To prove that AM =BM , AS =BS, and AE =BE. Proof. DrawOA and OB from O, the centre of the circle. The rt.△sOAM and OBM are equal. § 151 For OM =OM, Iden. and

OA =OB. § 217 □AM =BM , and ∠AOS = ∠BOS. § 128 Likewise ∠AOE = ∠BOE. § 85 □AS =BS, and AE =BE. § 236 q.e.d. 246. Cor. 1. A diameter bisects the circumference and the circle. 247. Cor. 2. A diameter which bisects a chord is perpendicular to it. 248. Cor. 3. The perpendicular bisector of a chord passes through the centre of the circle, and bisects the arcs of the chord.

ARCS, CHORDS, AND TANGENTS. 99 Proposition VI. Theorem. 249. In the same circle or in equal circles, equal chords are equally distant from the centre. Conversely: Chords equally distant from the centre are equal. A B C F H O P Let AB and CF be equal chords of the circle ABF C. To prove that AB and CF are equidistant from the centre O. Proof. DrawOP⊥ to AB, OH⊥ to CF , and join OA and OC. OP bisects AB, and OH bisects CF . § 245 The rt.△sOPA and OHC are equal. § 151 AP =CH , Ax. 7 and OA =OC, § 217 Hence, OP =OH. § 128 □AB and CF are equidistant from O. Conversely: Let OP =OH. To prove AB =CF . Proof. The rt.△sOPA and OHC are equal. § 151 For OA =OC, § 217 and OP =OH, Hyp. Hence, AP =CH . § 128

BOOK II. PLANE GEOMETRY. 100 □AB =CF . Ax. 6 q.e.d. Proposition VII. Theorem. 250. In the same circle or in equal circles, if two chords are unequal, they are unequally distant from the centre; and the greater chord is at the less distance. A B C D E F G H O In the circle whose centre is O, let the chords AB and CD be unequal, and AB the greater; and let OE be perpendicular to AB and OF perpendicular to CD. To prove that OE <OF . Proof. Suppose AG drawn equal to CD, and OH⊥ to AG. DrawEH . OE bisects AB, and OH bisects AG. § 245 By hypothesis, AB >CD. □AB >AG, the equal of CD. □AE >AH. Ax. 7 □ ∠AHE is greater than ∠AEH . § 152 □ ∠OHE , the complement of ∠AHE , is less than ∠OEH , the complement of ∠AEH . Ax. 5 □OE <OH . § 153 But OH =OF . § 249 □OE <OF . q.e.d.

ARCS, CHORDS, AND TANGENTS. 101 Ex. 91. The perpendicular bisectors of the sides of an inscribed polygon are concurrent (pass through the same point). Proposition VIII. Theorem. 251. Conversely: In the same circle or in equal circles, if two chords are unequally distant from the centre, they are unequal; and the chord at the less distance is the greater. A B C D E F G H O In the circle whose centre is O, let AB and CD be unequally distant from O; and let OE, the perpendicular to AB, be less than OF , the perpendicular to CD. To prove that AB >CD. Proof. Suppose AG drawn equal to CD, and OH⊥ to AG. Then OH =OF § 249 Hence, OE <OH . DrawEH . ∠OHE is less than ∠OEH . § 152 □ ∠AHE , the complement of ∠OHE , is greater than ∠AEH , the complement of ∠OEH . Ax. 5 □AE >AH. § 153 But AE = 1 2AB, and AH = 1 2AG. § 245 □AB >AG. Ax. 6 But CD =AG. Const. □AB >CD. q.e.d.

BOOK II. PLANE GEOMETRY. 102 252. Cor. A diameter of a circle is greater than any other chord. Proposition IX. Theorem. 253. A straight line perpendicular to a radius at its extremity is a tangent to the circle. A B C HM O Let M B be perpendicular to the radius OA at A. To prove that MB is a tangent to the circle. Proof. FromO draw any other line to MB , as OH. Then OH >OA. § 97 □ the point H is without the circle. § 216 Hence, every point, exceptA, of the line MB is without the circle, and therefore MB is a tangent to the circle at A. § 220 q.e.d. 254. Cor. 1. A tangent to a circle is perpendicular to the radius drawn to the point of contact. For OA is the shortest line from O to MB , and is therefore ⊥ to MB (§ 98); that is, MB is⊥

to OA. 255. Cor. 2. A perpendicular to a tangent at the point of contact passes through the centre of the circle. For a radius is ⊥ to a tangent at the point of contact, and therefore a ⊥ erected at the point of contact coincides with this radius and passes through the centre.

ARCS, CHORDS, AND TANGENTS. 103 256. Cor. 3. A perpendicular from the centre of a circle to a tangent passes through the point of contact.

BOOK II. PLANE GEOMETRY. 104 Proposition X. Theorem. 257. Parallels intercept equal arcs on a circumference. A B C D F A′ B′ F ′ Fig. 1. A B C D E F M Fig. 2. A B C D E F G H Fig. 3. Case 1. Let AB (Fig. 1) be a tangent at F parallel to CD, a secant. To prove that arcCF = arcDF . Proof. Suppose FF′ drawn⊥ to AB. Then FF′ is a diameter of the circle. § 255 And FF′ is also⊥ to CD. § 107 □CF =DF , and CF′ =DF′. § 245 Case 2. Let AB and CD (Fig. 2) be parallel secants. To prove that arcAC = arcBD. Proof. Suppose EF‖ to CD and tangent to the circle at M. Then arcAM = arcBM , Case 1 and arcCM = arcDM. □ arcAC = arcBD. Ax. 3 Case 3. Let AB and CD (Fig. 3) be parallel tangents at E and F. To prove that arcEGF = arcEHF . Proof. Suppose GH drawn‖ to AB.

ARCS, CHORDS, AND TANGENTS. 105 Then arcEG = arcEH , Case 1 and arcGF = arcHF . □ arcEGF = arcEHF . Ax. 2 q.e.d. Proposition XI. Theorem. 258. Through three points not in a straight line one circumference, and only one, can be drawn. A B C O Let A, B, C be three points not in a straight line. To prove that one circumference, and only one, can be drawn through A, B, and C. Proof. DrawAB and BC. At the middle points of AB and BC suppose⊥s erected. These⊥s will intersect at some point O, since AB and BC are not in the same straight line. The point O is in the perpendicular bisector of AB, and is therefore equidistant from A and B; the point O is also in the perpendicular bisector of BC, and is therefore equidistant from B and C. § 160 Therefore, O is equidistant from A, B, and C; and a circumference described from O as a centre, with a radius OA, will pass through the three given points. The centre of a circumference passing through the three points must be in both perpendiculars, and hence at their intersection. As two straight lines can intersect in only one point, O is the centre of the only circumference that can pass through the three given points. q.e.d.

BOOK II. PLANE GEOMETRY. 106 259. Cor. Two circumferences can intersect in only two points. For, if two circumferences have three points common, they coincide and form one circumference. 260. Def. A tangent from an external point to a circle is the part of the tangent between the external point and the point of contact. Proposition XII. Theorem. 261. The tangents to a circle drawn from an external point are equal, and make equal angles with the line joining the point to the centre. A B C O Let AB and AC be tangents from A to the circle whose centre is O, and let AO be the line joining A to the centre O. To prove that AB =AC, and ∠BAO = ∠CAO. Proof. DrawOB and OC. AB is⊥ to OB, and AC⊥ to OC, § 254 (a tangent to a circle is ⊥ to the radius drawn to the point of contact). The rt.△sOAB and OAC are equal. § 151 ForOA is common, and the radii OB and OC are equal. § 217 □AB =AC, and ∠BAO = ∠CAO. § 128 q.e.d. 262. Def. The line joining the centres of two circles is called the line of centres. 263. Def. A tangent to two circles is called acommon external tangent

if it does not cut the line of centres, and a common internal tangent if it cuts the line of centres.

ARCS, CHORDS, AND TANGENTS. 107 Proposition XIII. Theorem. 264. If two circles intersect each other, the line of centres is perpendicular to their common chord at its middle point. A B C C′ Let C and C′ be the centres of the two circles, AB the common chord, and CC ′ the line of centres. To prove that CC′ is⊥ to AB at its middle point. Proof. DrawCA, CB, C′A, and C′B. CA =CB, and C′A =C′B. § 217 □C and C′ are two points, each equidistant from A and B. □CC′ is the perpendicular bisector of AB. § 161 q.e.d. Ex. 92. Describe the relative position of two circles if the line of centres: 1. is greater than the sum of the radii; 2. is equal to the sum of the radii; 3. is less than the sum but greater than the difference of the radii; 4. is equal to the difference of the radii; 5. is less than the difference of the radii. Illustrate each case by a figure. Ex. 93. The straight line drawn from the middle point of a chord to the middle point of its subtended arc is perpendicular to the chord. Ex. 94. The line which passes through the middle points of two parallel chords passes through the centre of the circle.

BOOK II. PLANE GEOMETRY. 108 Proposition XIV. Theorem. 265. If two circles are tangent to each other, the line of centres passes through the point of contact. A B C O C′ Let the two circles, whose centres are C and C′, be tangent to the straight line AB at Q, and CC ′ the line of centres. To prove that O is in the straight line CC′. Proof. A⊥ to AB, drawn through the point O, passes through the centres C and C′, § 255 (a⊥ to a tangent at the point of contact passes through the centre of the circle). □ the line CC′, having two points in common with this ⊥ must coincide with it. § 47 □O is in the straight line CC′. q.e.d. Ex. 95. Describe the relative position of two circles if they may have: 1. two common external and two common internal tangents; 2. two common external tangents and one common internal tangent; 3. two common external tangents and no common internal tangent; 4. one common external and no common internal tangent; 5. no common tangent. Illustrate each case by a figure. Ex. 96. The line drawn from the centre of a circle to the point of inter- section of the two tangents is the perpendicular bisector of the chord joining the points of contact.

MEASUREMENT. 109 MEASUREMENT. 266. To measure a quantity of any kind is to find the number of times it contains a known quantity of the same kind, called the unit of measure. The number which shows the number of times a quantity contains the unit of measure is called the numerical measure of that quantity. 267. No quantity is great or small except by comparison with another quantity of the same kind. This comparison is made by finding the numerical measures of the two quantities in terms of a common unit, and then dividing one of the measures by the other. The quotient is called theirratio. In other words the ratio of two quantities of the same kind is the ratio of their numerical measures expressed in terms of a common unit. The ratio of a to b is written a :b, or a b . 268. Two quantities that can be expressed inintegers in terms of a common unit are said to be commensurable, and the exact value of their ratio can be found. The common unit is called their common measure, and each quantity is called a multiple of this common measure. Thus, a common measure of 2 1 2 feet and 3 2 3 feet is 1 6 of a foot, which is contained 15 times in 2 1 2 feet, and 22

times in 3 2 3 feet. Hence, 2 1 2 feet and 3 2 3 feet are multiples of 1 6 of a foot, since 2 1 2 feet may be obtained by taking 1 6 of a foot 15 times, and 3 2 3 feet by taking 1 6 of a foot 22 times. The ratio of 2 1 2 feet to 3 2 3 feet is expressed by the fraction 15 22.

BOOK II. PLANE GEOMETRY. 110 269. Two quantities of the same kind that cannot both be expressed in integers in terms of a common unit, are said to beincommensurable, and the exact value of their ratio cannot be found. But by taking the unit sufficiently small, an approximate value can be found that shall differ from the true value of the ratio by less than any assigned value, however small. Thus, suppose the ratio, a b = √ 2. Now √ 2 = 1 .41421356··· , a value greater than 1 .414213, but less than 1.414214. If, then, a millionth part of b is taken as the unit of measure, the value of a b lies between 1.414213 and 1.414214, and therefore differs from either of these values by less than 0.000001. By carrying the decimal further, an approximate value may be found that will differ from the true value of the ratio by less than a billionth, a trillionth, or any other assigned value . In general, if a b > m n but< m + 1 n , then the error in taking either of these values for a b is less than 1 n, the difference between these two fractions. But by increasing n indefinitely, 1 n can be decreased indefinitely, and a value of the ratio can be found within any required degree of accuracy. 270. The ratio of two incommensurable quantities is called anincommen- surable ratio ; and is a fixed value which its successive approximate values constantly approach.

THEORY OF LIMITS. 111 THE THEORY OF LIMITS. 271. When a quantity is regarded as having a fixed value throughout the same discussion, it is called a constant; but when it is regarded, under the conditions imposed upon it, as having different successive values, it is called a variable. If a variable, by having different successive values, can be made to differ from a given constant by less than any assigned value, however small, but cannot be made absolutely equal to the constant, that constant is called the limit of the variable, and the variable is said to approach the constant as its limit. A BM M ′ M ″ 272. Suppose a point to move fromA towardB, under the conditions that the first second it shall move one half the distance from A to B, that is, to M; the next second, one half the remaining distance, that is, to M′; and so on indefinitely. Then it is evident that the moving point may approach as near to B as we choose, but will never arrive at B. For, however near it may be to B at any instant, the next second it will pass over half the distance still remaining; it must, therefore, approach nearer to B, since half the distance still remaining is some distance, but will not reach B, since half the distance still remaining is not the whole distance. Hence, the distance from A to the moving point is an increasing variable, which indefinitely approaches the constant AB as its limit; and the distance from the moving point to B is a decreasing variable, which indefinitely ap- proaches the constant zero as its limit.

BOOK II. PLANE GEOMETRY. 112 A B CD E F H K 273. Again, suppose a squareABCD inscribed in a circle, andE,F ,H,K the middle points of the arcs subtended by the sides of the square. If we draw the lines AE,EB,BF , etc., we shall have an inscribed polygon of double the number of sides of the square. The length of

the perimeter of this polygon, represented by the dotted lines, is greater than that of the square, since two sides replace each side of the square and form with it a triangle, and two sides of a triangle are together greater than the third side; but less than the length of the circumference, for it is made up of straight lines, each one of which is less than the part of the circumference between its extremities.

THEORY OF LIMITS. 113 By continually doubling the number of sides of each resulting inscribed figure, the length of the perimeter will increase with the increase of the number of sides, but will not become equal to the length of the circumference. The difference between the perimeter of the inscribed polygon and the circumference of the circle can be made less than any assigned value, but cannot be made equal to zero. The length of the circumference is, therefore, the limit of the length of the perimeter as the number of sides of the inscribed figure is indefinitely increased. § 271 274. Consider the decimal 0.333··· which may be written 3 10 + 3 100 + 3 1000 +··· The value of each fraction after the first is one tenth of the preceding fraction, and by continuing the series we shall reach a fraction less than any assigned value, that is, the values of the successive fractions approach zero as a limit. The sum of these fractions is less than 1 3; but the more terms we take, the nearer does the sum approach 1 3 as a limit. 275. Test for a limit. In order to prove that a variable approaches a constant as a limit, it is necessary to prove that the difference between the variable and the constant: 1. Can be made less than any assigned value, however small. 2. Cannot be made absolutely equal to zero. 276. Theorem. If the limit of a variable x is zero, then the limit of kx, the product of the variable by any finite constant k, is zero. 1. Let q be any assigned quantity, however small. Then q k is not 0. Hence x, which may differ as little as we please from 0, may be taken less than q k, and then kx will be less than q. 2. Since x cannot be 0, kx cannot be 0. Therefore, the limit of kx = 0 § 275

BOOK II. PLANE GEOMETRY. 114 277. Cor. If the limit of a variable x is zero, then the limit of the quotient of the variable by any finite constant k, is also zero. For x k = 1 k×x, which by§ 276 can be made less than any assigned value, however small, but cannot be made equal to zero. 278. Theorem. The limit of the sum of a finite number of variables x, y, z,··· is equal to the sum of their respective limits a, b, c,··· . Letd, d′, d″,··· denote the differences betweenx, y, z,··· anda, b, c,··· , respectively. Then d +d′ +d″ +··· can be made less than any assigned quan- tityq. For, ifd, d′, d″,··· are n in number and d is the largest, d +d′ +d″ +··· <nd . (1) Since d may be diminished at pleasure, we may make d so small that d< q n; and therefore nd<q . But by (1), d +d′ +d″ +··· <nd , and therefore <q . Therefore, the difference between (x +y +z +···) and (a +b +c +···) can be made less than any assigned quantity, but not zero. Therefore, the limit of (x +y +z +···) =a +b +c +··· . § 275 279. Theorem. If the limit of a variable x is not zero, and if k is any finite constant, the limit of the product kx is equal to the limit of x multiplied by k. 1. If a denotes the limit of x, then x cannot be equal to a. § 271 Therefore, kx cannot be equal to ka. 2. The limit of (a−x) = 0. Hence, the limit of ka−kx = 0. § 276 Therefore, the limit of kx =ka. § 275 280. Cor. The limit of the quotient of a variable x by any finite constant k is the limit of x divided by k. For x k = 1 k×x, and the limit of x k = 1 k× the limit of x.

THEORY OF LIMITS. 115 281. Theorem. The limit of the product of two or more variables is the product of their respective limits, provided no one of these limits is zero. Ifx andy are variables,a andb their respective limits, we may putx =a−d, y =b−d′; thend andd′ are variables which can be made less than any assigned quantity, but not zero. § 275 Now, xy = (a−d)(b−d′) =ab−ad′−bd +dd′ ∴ab−xy =ad′ +bd−dd′. Since every term on the right contains d ord′, the whole right member can be made less than any assigned quantity, but not zero. § 278 Hence,ab−xy can be made less than any assigned quantity, but not zero. Therefore, the limit of xy =ab. § 275 Similarly, for three or more variables. 282. Cor. 1. The limit of the nth power of a variable is the nth power of its limit. For the limit of the product of the variables x, y, z,⋯ to n factors is the product of their respective limits, the constantsa,b,c,⋯ ton factors (§ 281). If the n factors xyz⋯ are each equal to x, and the n factors abc⋯ are each equal to a, we have xyz⋯ =xn, and abc⋯ =an. Therefore, the limit of xn =an. 283. Cor. 2. The limit of the nth root of a variable is the nth root of its limit. For if the limit of x =a, we may put this in the following form, the limit of n√xn = n√an; that is, the limit of n√ xxx⋯ to n factors is n√ aaa⋯ to n factors. Now, xxx⋯ is a variable since each factor is a variable, and aaa⋯ is a constant since each factor is a constant. If we denotexxx⋯ ton factors by the variabley, andaaa⋯ ton factors by the constant b, we have the limit of n √ y = n √ b.

BOOK II. PLANE GEOMETRY. 116 284. Theorem. If two variables are constantly equal, and each approaches a limit, the limits are equal. Let x and y be two variables, a and b their respective limits, d and d′ the respective differences between the variables and their limits. Then, if the variables are increasing toward their limits a =x +d, and b =y +d′. Since the equation x =y is always true, we have by subtraction a−b =d−d′. Since a and b are constants, a−b is a constant; therefore, d−d′, which is equal to a−b, is a constant. But the only constant which is less than any assigned value is 0. Therefore, d−d′ = 0. Therefore, a−b = 0, and a =b. If the variables x and y are decreasing toward their limits a and b, respec- tively, then a =x−d and b =y−d′. Therefore, by subtraction a−b =d′−d. Therefore, by the same proof as for increasing variables a =b. 285. Theorem. If two variables have a constant ratio, and each approaches a limit that is not zero, the limits have the same ratio. Let x and y be two variables, a and b their respective limits. Let x y =r, a constant; then x =ry. Since x and ry are two variables that are always equal, the limit of x = the limit of ry. § 284 Now, the limit of ry =r× limit of y. § 279 But the limit of x is a, and the limit of y is b. Therefore, a =rb; that is, a b =r.

THEORY OF LIMITS. 117 Proposition XV. Problem. 286. To find the ratio of two straight lines. A B C D E F H K Let AB and CD be two straight lines. To find the ratio of AB and CD. Apply CD to AB as many times as possible. Suppose twice, with a remainder EB. Then apply EB to CD as many times as possible. Suppose three times, with a remainder FD . Then apply FD to EB as many times as possible. Suppose once, with a remainder HB. Then apply HB to FD as many times as possible. Suppose once, with a remainder KD. Then apply KD to HB as many times as possible. Suppose KD is contained just twice in HB. Then HB = 2KD; FD =HB +KD = 3KD; EB =FD +HB = 5KD; CD = 3EB +FD = 18KD; AB = 2CD +EB = 41KD; ∴ AB CD = 41KD 18KD = 41 18. q.e.f.

BOOK II. PLANE GEOMETRY. 118 Note. By the same process the ratio of two arcs of the same circle or of equal circles can be found. If the lines or arcs are incommensurable, an approximate value of the ratio can be found by the same method.

MEASURE OF ANGLES. 119 MEASURE OF ANGLES. Proposition XVI. Theorem. 287. In the same circle or in equal circles, two central angles have the same ratio as their intercepted arcs. A′ B′ C′ m Fig. 1. A B C Fig. 2. A′ B′ C′ D Fig. 3. In the equal circles whose centres are C and C′, let ACB and A′C′B′ be the angles, AB and A′B′ the intercepted arcs. To prove that ∠A′C′B′ ∠ACB = arcA′B′ arcAB . Case 1. When the arcs are commensurable (Figs. 1 and 2). Proof. Let the arc m be a common measure of A′B′ and AB. Suppose m to be contained 4 times in A′B′, and 7 times in AB. Then arcA′B′ arcAB = 4 7. At the several points of division on AB and A′B′ draw radii. These radii will divide ∠ACB into 7 parts, and ∠A′C′B′ into 4 parts, equal each to each, § 237 (in the same ⊙, or equal ⊙s, equal arcs subtend equal central ∠s). ∠A′C′B′ ∠ACB = 4 7. ∠A′C′B′ ∠ACB = arcA′B′ arcAB . Ax. 1

BOOK II. PLANE GEOMETRY. 120 Case 2. When the arcs are incommensurable (Figs. 2 and 3). Proof. DivideAB into any number of equal parts, and apply one of these parts to A′B′ as many times as A′B′ will contain it. Since AB and A′B′ are incommensurable, a certain number of these parts will extend fromA′ to some point, as D, leaving a remainder DB′ less than one of these parts. Draw C′D. By construction AB and A′D are commensurable. ∠A′C′D ∠ACB = arcA′D arcAB . Case 1 By increasing the number of equal parts into which AB is divided we can di- minish at pleasure the length of each part, and therefore make DB′ less than any assigned value, however small, since DB′ is always less than one of the equal parts into which AB is divided. We cannot make DB′ equal to zero, since, by hypothesis, AB and A′B′ are incommensurable. § 269 Hence, DB′ approaches zero as a limit, if the number of parts of AB is indefi- nitely increased. § 275 And the corresponding angle DC′B′ approaches zero as a limit. Therefore, the arc A′D approaches the arc A′B′ as a limit, § 271 and the ∠A′C′D approaches the ∠A′C′B′ as a limit. Therefore, arcA′D arcAB approaches arcA′B′ arcAB as a limit, § 280 and ∠A′C′D ∠ACB approaches ∠A′C′B′ ∠ACB as a limit. § 280 But ∠A′C′D ∠ACB is constantly equal to arcA′D arcAB , as A′D varies in value and approaches A′B′ as a limit. ∠A′C′B′ ∠ACB = arcA′B′ arcAB . § 284 q.e.d.

MEASURE OF ANGLES. 121 288. A circumference is divided into 360 equal parts, called degrees; and therefore a unit angle at the centre intercepts a unit arc on the circumference. Hence, the numerical measure of a central angle expressed in terms of the unit angle is equal to the numerical measure of its intercepted arc expressed in terms of the unit arc. This must be understood to be the meaning when it is said that A central angle is measured by its intercepted arc.

BOOK II. PLANE GEOMETRY. 122 Proposition XVII. Theorem. 289. An inscribed angle is measured by half the arc intercepted between its sides. A B C P Fig. 1. A B C E P Fig. 2. A B C F P Fig. 3. 1. Let the centre C (Fig. 1) be in one of the sides of the angle. To prove that the ∠B is measured by 1 2 the arc PA . Proof. DrawCA. CA =CB. § 217 ∠B = ∠A. § 145 But ∠PCA = ∠B + ∠A. § 137 ∠PCA = 2∠B. But ∠PCA is

measured by arcPA , § 288 (a central ∠ is measured by its intercepted arc). ∠B is measured by 1 2 arcPA . 2. Let the centre C (Fig. 2) fall within the angle PBA . To prove that the ∠PBA is measured by 1 2 the arc PA . Proof. Draw the diameter BCE . Then ∠EBA is measured by 1 2 arcAE, and ∠EBP is measured by 1 2 arcEP . Case 1

MEASURE OF ANGLES. 123 ∠EBA + ∠EBP is measured by 1 2(arcAE + arcEP), or ∠PBA is measured by 1 2 arcPA . 3. Let the centre C (Fig. 3) fall without the angle PBA . To prove that the ∠PBA is measured by 1 2 the arc PA . Proof. Draw the diameter BCF . Then ∠FBA is measured by 1 2 arcFA , and ∠FBP is measured by 1 2 arcFP . Case 1 ∠FBA− ∠FBP is measured by 1 2(arcFA− arcFP), or ∠PBA is measured by 1 2 arcPA . q.e.d. A B C Fig. 4. A B C D Fig. 5. A B C D E Fig. 6. 290. Cor. 1. An angle inscribed in a semicircle is a right angle. For it is measured by half a semicircumference (Fig. 4). 291. Cor. 2. An angle inscribed in a segment greater than a semicircle is an acute angle. For it is measured by an arc less than half a semicircumference; as, ∠CAD (Fig. 5). 292. Cor. 3. An angle inscribed in a segment less than a semicircle is an obtuse angle. For it is measured by an arc greater than half a semicircumfer- ence; as, ∠CBD (Fig. 5). 293. Cor. 4. Angles inscribed in the same segment or in equal segments are equal (Fig. 6).

BOOK II. PLANE GEOMETRY. 124 Proposition XVIII. Theorem. 294. An angle formed by two chords intersecting within the circumference is measured by half the sum of the intercepted arcs. A BC D E O Let the angle COD be formed by the chords AC and BD. To prove that the ∠COD is measured by 1 2(CD +AB). Proof. Suppose AE drawn∥ to BD. Then arcAB = arcDE, § 257 (parallels intercept equal arcs on a circumference). Also ∠COD = ∠CAE, § 112 (ext.-int. ∠s of ∥s). But ∠CAE is measured by 1 2(CD +DE), § 289 (an inscribed ∠ is measured by half its intercepted arc). Put ∠COD for its equal, the ∠CAE, and arcAB for its equal, the arcDE; then ∠COD is measured by 1 2(CD +AB). q.e.d. Ex. 97. The opposite angles of an inscribed quadrilateral are supplemen- tary. Ex. 98. If through a point within a circle two perpendicular chords are drawn, the sum of either pair of the opposite arcs which they intercept is equal to a semicircumference. Ex. 99. The line joining the centre of the square described upon the hypotenuse of a right triangle to the vertex of the right angle bisects the right angle.

MEASURE OF ANGLES. 125 Proposition XIX. Theorem. 295. An angle included by a tangent and a chord drawn from the point of contact is measured by half the intercepted arc. A E FH M O Let M AH be the angle included by the tangent M O to the circle at A and the chord AH. To prove that the ∠MAH is measured by 1 2 the arc AEH . Proof. Suppose HF drawn∥ to MO. Then arcAF = arcAEH , § 257 (parallels intercept equal arcs on a circumference). Also ∠MAH = ∠AHF , § 110 (alt.-int. ∠s of ∥s). ∠AHF is measured by 1 2AF , § 289 (an inscribed ∠ is measured by half its intercepted arc). Put ∠MAH for its equal, the ∠AHF , and arc AEH for its equal, the arc AF ; then ∠MAH is measured by 1 2 arcAEH . Likewise, the ∠OAH, the supplement of the ∠MAH , is measured by half the arc AFH , the conjugate of the arc AEH . q.e.d. Ex. 100. Two circles are tangent externally atA, and a common external tangent touches them at B and C, respectively. Show that angle BAC is a right angle.

BOOK II. PLANE GEOMETRY. 126 Proposition XX. Theorem. 296. An angle formed by two secants, two tangents, or a tangent and a secant, drawn to a circle from an external point, is measured by half the dif- ference of the intercepted arcs. A BC E O P AB E M O P A B C E O P The proof of this theorem is left as an exercise for the student.

MEASURE OF ANGLES. 127 A B C D O o / A B C B′ 297. Positive and Negative Quantities. In measurements it is con- venient to mark the distinction between two quantities that are measured in opposite directions, by calling one of them positive and the other negative. Thus, if OA is considered positive, then OC may be considered negative, and if OR is considered positive, then OD may be considered negative. When this distinction is applied to angles, an angle is considered to be positive, if the rotating line that describes it moves in the opposite direction to the hands of a clock (counter clockwise), and to be negative, if the rotating line moves in the same direction as the hands of a clock (clockwise). Arcs corresponding to positive angles are considered positive, and arcs cor- responding to negative angles are considered negative. Thus, the angleACB described by a line rotating aboutC fromCA toCB is positive, and the arc AB is positive; the angle ACB′ described by the line rotating about C from CA to CB′ is negative, and the arc AB′ is negative. 298. The Principle of Continuity. By marking the distinction between quantities measured in opposite directions, a theorem may often be so stated as to include two or more particular theorems.

BOOK II. PLANE GEOMETRY. 128 The following theorem furnishes a good illustration: 299. The angle included between two lines of unlimited length that cut or touch a circumference is measured by half the sum of the intercepted arcs. Here the word sum means the algebraic sum and includes both the arith- metical sum and the arithmetical difference of two quantities. ab c d o ab c d o ab c d o ab c dm o a b c dm n o a b c d a b c d m no 1. If the lines intersect at the centre, the two intercepted arcs are equal, and half the sum will be one of the arcs (§ 288). 2. If the lines intersect between the centre and the circumference, the angle is measured by half the sum of the arcs (§ 294). 3. If the lines intersect on the circumference, one of the arcs becomes zero and we have an inscribed angle (§ 289), or an angle formed by a tangent and a chord (§ 295). In each case the angle is measured by half the sum of the intercepted arcs. 4. If the lines intersect without the circumference, then the arc ab is nega- tive and the algebraic sum is the arithmetical difference of the included arcs. When the reasoning employed to prove a theorem is continued in the man- ner just illustrated to include two or more theorems, we are said to reason by the Principle of Continuity.

MEASURE OF ANGLES. 129 REVIEW QUESTIONS ON BOOK II. 1. What do we call the locus of points in a plane that are equidistant from a fixed point in the plane? 2. What does the chord of a segment become when the segment is a semi- circle? 3. What kind of an angle do the radii of a sector include when the sector is a semicircle? 4. What is the difference between a chord and a secant? 5. What part of a tangent is meant by a tangent to a circle from an external point? 6. Two chords are equal in equal circles under either of two conditions. What are the two conditions? 7. Points

that lie in a straight line are called collinear; points that lie in a circumference are called concyclic. How many collinear points can be concyclic? 8. What is meant by the statement that a central angle is measured by the arc intercepted between its sides? 9. What is an inscribed angle? What is its measure? 10. What kind of an angle is the angle inscribed in a semicircle? in a segment less than a semicircle? in a segment greater than a semicircle? 11. What is the measure of an angle included by two intersecting chords? by two secants intersecting without the circle? 12. What is the measure of an angle included by a tangent and a chord drawn to the point of contact? 13. When are two quantities of the same kind incommensurable? 14. When are two quantities of the same kind commensurable? 15. Define a variable and the limit of a variable. 16. Does the series 1 2 in., 3 4 in., 7 8 in., 15 16 in., etc., constitute a variable? Is the variable increasing or decreasing? 17. What is the limit of this variable? 18. What is the test of a limit?

BOOK II. PLANE GEOMETRY. 130 THEOREMS. Ex. 101. An angle formed by a tangent and a chord is equal to the angle inscribed in the opposite segment. Ex. 102. Two chords drawn perpendicular to a third chord at its extrem- ities are equal. Ex. 103. The sum of two opposite sides of a circumscribed quadrilateral is equal to the sum of the other two sides. A B CD O M A BC D O Ex. 104. If the sum of two opposite angles of a quadrilateral is equal to two right angles, a circle may be circumscribed about the quadrilateral. Let ∠A + ∠C = 2 rt. ∠s. Pass a circumference through D, A, and B, and prove that this circumference passes through C. Ex. 105. The shortest line that can be drawn from a point within a circle to the circumference is the shorter segment of the diameter through that point. Let A be the given point. Prove AB shorter than any other line AD from A to the circumference. Ex. 106. The longest line that can be drawn from a point within a circle to the circumference is the longer segment of the diameter through that point. A B C D OP Ex. 107. The shortest line that can be drawn from a point without a circle to the circumference will pass through the centre of the circle if produced. Ex. 108. The longest line that can be drawn from a point without a circle to the concave arc of the circumference passes through the centre of the circle.

MEASURE OF ANGLES. 131 A B C DO P Q A B C D E H KM N O A B C D G H K L O A B C D E F O Ex. 109. The shortest chord that can be drawn through a point within a circle is perpendicular to the diameter at that point. Ex. 110. If two intersecting chords make equal angles with the diameter drawn through the point of intersection, the two chords are equal. Rt.△COM = rt.△CON . □OM =ON. Ex. 111. The angles subtended at the centre of a circle by any two opposite sides of a circumscribed quadrilateral are supplementary. Ex. 112. The radius of a circle inscribed in an equilateral triangle is equal to one third the altitude of the triangle. △OEF is equiangular and equilateral; ∠FEA = ∠FAE . □AF =EF . □AF =FO =OD. Ex. 113. The radius of a circle circumscribed about an equilateral triangle is equal to two thirds the altitude of the triangle (Ex. 27). Ex. 114. A parallelogram inscribed in a circle is a rectangle. Ex. 115. A trapezoid inscribed in a circle is an isosceles trapezoid. Ex. 116. All chords of a circle which touch an interior concentric circle are equal, and are bisected at the point of contact. Ex. 117. If the inscribed and circumscribed circles of a triangle are con- centric, the triangle is equilateral (Ex. 116). Ex. 118. If two circles

are tangent to each other the tangents to them from any point of the common internal tangent are equal.

BOOK II. PLANE GEOMETRY. 132 Ex. 119. If two circles touch each other and a line is drawn through the point of contact terminated by the circumferences, the tangents at its ends are parallel. A B C D M N P A B C D E F Ex. 120. If two circles touch each other and two lines are drawn through the point of contact terminated by the circumferences, the chords joining the ends of these lines are parallel. ∠A = ∠MPC and ∠B = ∠NPD . ∠A = ∠B. Ex. 121. If two circles intersect and a line is drawn through each point of intersection terminated by the circumferences, the chords joining the ends of these lines are parallel. A BC D O O ′ A B C D E F O Ex. 122. Through one of the points of intersection of two circles a diameter of each circle is drawn. Prove that the line joining the ends of the diameters passes through the other point of intersection. ∠ABC = ∠ABD = 90◦ § 290 Ex. 123. If two common external tangents or two common internal tan- gents are drawn to two circles, the segments intercepted between the points of contact are equal. Ex. 124. The diameter of the circle inscribed in a right triangle is equal to the difference between the sum of the legs and the hypotenuse.

MEASURE OF ANGLES. 133 A B C D E O A B C H K M Ex. 125. If one leg of a right triangle is the diameter of a circle, the tangent at the point where the circumference cuts the hypotenuse bisects the other leg. ∠BOE = ∠DOE. ∠BOE = ∠OAD. OE and AC are∥. BE =EC (§ 188). Ex. 126. If, from any point in the circumference of a circle, a chord and a tangent are drawn, the perpendiculars dropped on them from the middle point of the subtended arc are equal. ∠BAM = ∠CAM . Ex. 127. The median of a trapezoid circumscribed about a circle is equal to one fourth the perimeter of the trapezoid (Ex. 103). Ex. 128. Two fixed circles touch each other externally and a circle of variable radius touches both externally. Show that the difference of the distances from the centre of the variable circle to the centres of the fixed circles is constant. Ex. 129. If two fixed circles intersect, and circles are drawn to touch both, show that either the sum or the difference of the distances of their centres from the centres of the fixed circles is constant, according as they touch (i) one internally and one externally, (ii) both internally or both externally.

BOOK II. PLANE GEOMETRY. 134 A B CD E F O PE′ F ′ A B C M P Ex. 130. If two straight lines are drawn through any point in a diagonal of a square parallel to the sides of the square, the points where these lines meet the sides lie on the circumference of a circle whose centre is the point of intersection of the diagonals. △POE =△POF (§ 143). OE =OF . △POE′ =△POF′. Ex. 131. IfABC is an inscribed equilateral triangle and P is any point in the arc BC, then PA =PB +PC . TakePM =PB .△ABM =△CBP (§ 143) and AM =PC . Ex. 132. The tangents drawn through the vertices of an inscribed rect- angle, which is not a square, enclose a rhombus.

PROBLEMS OF CONSTRUCTION. 135 A B CD E F G H O A B C D E F GH I M N Ex. 133. The bisectors of the angles included by the opposite sides (produced) of an inscribed quadrilateral intersect at right angles. Arc AF− arcBM = arcDF− arcCM and arcAH− arcDN = arcBH− arcCN . arcFH + arcMN = arcHM + arcFN . ∠FIH = ∠HIM . Discussion. This problem is impossible, if any two sides of the quadrilat-

eral are parallel. PROBLEMS OF CONSTRUCTION. Note. Hitherto we have supposed the figures constructed. We now proceed to explain the methods of constructing simple problems, and afterwards to apply these methods to the solution of more difficult problem.

BOOK II. PLANE GEOMETRY. 136 Proposition XXI. Problem. 300. To let fall a perpendicular upon a given line from a given external point. .... A B C H K M O Let AB be the given straight line, and C the given external point. To let fall a ⊥ to the line AB from the point C. FromC as a centre, with a radius sufficiently great, describe an arc cutting AB in two points, H and K. FromH and K as centres, with equal radii greater than 1 2HK , describe two arcs intersecting at O. DrawCO, and produce it to meet AB at M. CM is the⊥ required. Proof. Since C and O are two points each equidistant from H and K, they determine a⊥ to HK at its middle point. § 161 q.e.f. Note. Given lines of the figures are represented by full lines, resulting lines by long-dashed, and auxiliary lines by short-dashed lines.

PROBLEMS OF CONSTRUCTION. 137 Proposition XXII. Problem. 301. At a given point in a straight line, to erect a perpendicular to that line. ... ... A B CH O R Fig. 1. A B C D E Fig. 2. 1. Let O be the given point in AC. Fig. 1. TakeOH and OB equal. FromH and B as centres, with equal radii greater than OB, describe two arcs intersecting at R. Join OR. Then the line OR is the⊥ required. Proof. O and R, two points each equidistant from H and B, determine the perpendicular bisector of HB. § 161 2. Let B be the given point. Fig. 2. Take any point C without AB; and from C as a centre, with the distance CB as a radius, describe an arc intersecting AB at E. DrawEC, and prolong it to meet the arc again at D. Join BD, and BD is the⊥ required. Proof. The ∠B is a right angle. § 290 □BD is⊥ to AB. q.e.f. Discussion. The point C must be so taken that it will not be in the required perpendicular.

BOOK II. PLANE GEOMETRY. 138 Proposition XXIII. Problem. 302. To bisect a given straight line. A B C D E To bisect the given straight line AB. From A and B as centres, with equal radii greater than 1 2AB, describe arcs intersecting at C and E. Join CE. Then CE bisects AB. § 161 q.e.f.

PROBLEMS OF CONSTRUCTION. 139 Proposition XXIV. Problem. 303. To bisect a given arc. A B C D E To bisect the given arc AB. Draw the chord AB. From A and B as centres, with equal radii greater than 1 2AB, describe arcs intersecting at D and E. DrawDE. Then DE is the⊥ bisector of the chord AB. § 161 □DE bisects the arc ACB. § 248 q.e.f.

BOOK II. PLANE GEOMETRY. 140 Proposition XXV. Problem. 304. To bisect a given angle. ... . A B C D E Let AEB be the given angle. FromE as a centre, with any radius, as EA, describe an arc cutting the sides of the ∠E at A and B. FromA and B as centres, with equal radii greater than half the distance from A to B, describe two arcs intersecting at D. DrawDE. Then DE bisects the arc AB at C. § 303 □DE bisects the angle E. § 237 q.e.f. Ex. 134. To construct an angle of 45◦; of 135◦. Ex. 135. To construct an equilateral triangle, having given one side. Ex. 136. To construct an angle of 60◦; of 150◦. Ex. 137. To trisect a right angle.

PROBLEMS OF CONSTRUCTION. 141 Proposition XXVI. Problem. 305. At a given point in a given straight line, to construct an angle equal to a given angle. A E F C G H M O At C in the line CM , construct an angle equal to the given angle A. FromA as a centre, with any radius, AE, describe an arc cutting the sides of the ∠A at E and F . FromC as a centre, with a radius equal to AE, describe an arc HG cutting CM at H. FromH as a centre, with a radius equal to the chord EF , describe an arc intersecting the arc HG at O. DrawCO, and ∠ HCO is the required angle. Why? q.e.f.

BOOK II. PLANE GEOMETRY. 142 Proposition XXVII. Problem. 306. To draw a straight line parallel to a given straight line through a given external point. A B C D E FH Let AB be the given line, and C the given point. DrawECD , making any convenient ∠EDB. At the point C construct ∠ECF equal to ∠EDB. § 305 Then the line HCF is∥ to AB. Why? q.e.f.

PROBLEMS OF CONSTRUCTION. 143 Proposition XXVIII. Problem. 307. To divide a given straight line into a given number of equal parts. A B C O Let AB be the given straight line. FromA draw the line AO, making any convenient angle with AB. Take any convenient length, and apply it to AO as many times as the line AB is to be divided into parts. FromC, the last point thus found on AO, draw CB. Through the points of division on AO draw parallels to the line CB. § 306 These lines will divide AB into equal parts. § 187 q.e.f. Ex. 138. To construct an equilateral triangle, having given the perimeter. Ex. 139. To divide a line into four equal parts by two different methods. Ex. 140. Through a given point to draw a line which shall make equal angles with the two sides of a given angle. Through the given point draw a ⊥ to the bisector of the given ∠. Ex. 141. To draw a line through a given point, so that it shall form with the sides of a given angle an isosceles triangle (Ex. 140).

BOOK II. PLANE GEOMETRY. 144 Proposition XXIX. Problem. 308. To find the third angle of a triangle when two of the angles are given. A B E F H R abc Let A and B be the two given angles. At any point H in any line EF , construct ∠a equal to ∠A, and ∠b equal to ∠B. § 305 Then ∠c is the ∠ required. Why? q.e.f.

PROBLEMS OF CONSTRUCTION. 145 Proposition XXX. Problem. 309. To construct a triangle when two sides and the included angle are given. E b c A B C D b c Let b and c be the two sides of the triangle and E the included angle. TakeAB equal to the side c. AtA, construct ∠BAD equal to the given ∠E. § 305 On AD takeAC equal to b, and draw CB. Then△ACB is the△ required. q.e.f.

BOOK II. PLANE GEOMETRY. 146 Proposition XXXI. Problem. 310. To construct a triangle when a side and two angles of the triangle are given. A B c CE H K O c Let c be the given side, A and B the given angles. TakeEC equal to the side c. AtE construct the ∠CEH equal to ∠A. § 305 AtC construct the ∠ECK equal to ∠B. Produce EH and CK until they intersect at O. Then△COE is the△ required. q.e.f. Remark. If one of the given angles is opposite to the given side, find the third angle by§ 308, and proceed as above. Discussion. The problem is impossible when the two given angles are together equal to or greater than two right angles. Ex. 142. To construct an equilateral triangle, having given the altitude. To construct an isosceles triangle, having given: Ex. 143. The base and the altitude. Ex. 144. The altitude and one of the

legs. Ex. 145. The angle at the vertex and the altitude.

PROBLEMS OF CONSTRUCTION. 147 Proposition XXXII. Problem. 311. To construct a triangle when two sides and the angle opposite one of them are given. A a b A B CC′ D E aa b Let a and b be the given sides, and A the angle opposite a. Case 1. If a is less than b. Construct ∠DAE equal to the given ∠A § 305 On AD takeAB equal to b. FromB as a centre, with a radius equal to a, describe an arc intersecting the line AE at C and C′. DrawBC and BC′. Then both the△sABC and ABC′ fulfil the conditions, and hence we have two constructions. This is called the ambiguous case. A B D EH a b A B D E ab Discussion. If the side a is equal to the ⊥BH, the arc described from B will touchAE, and there will be but one construction, the right △ABH. If the given side a is less than the ⊥ from B, the arc described from B will not intersect or touch AE, and hence the problem is impossible. If the ∠A is right or obtuse, the problem is impossible; for the side opposite a right or obtuse angle is the greatest side. § 153

BOOK II. PLANE GEOMETRY. 148 Case 2. If a is equal to b. A B C D E ab If the ∠A is acute, and a =b, the arc described from B as a centre, and with a radius equal to a, will cut the line AE at the points A and C. There is therefore but one solution: the isosceles △ABC. Discussion. If the ∠A is right or obtuse, the problem is impossible; for equal sides of a△ have equal ∠s opposite them, and a△ cannot have two right ∠s or two obtuse ∠s. A B CC′ DE a b a A B CC′ DE a b a A B CC′ DE a b a Case 3. If a is greater than b. If the given ∠A is acute, the arc described from B will cut the line ED on opposite sides of A, at C andC′. The△ABC answers the required conditions, but the△ABC′ does not, for it does not contain the acute ∠A. There is then only one solution; namely, the△ABC. If the ∠A is right, the arc described from B cuts the line ED on opposite sides of A, and we have two equal right△s which fulfil the required conditions. If the ∠A is obtuse, the arc described from B cuts the lineED on opposite sides ofA, at the points C andC′. The△ABC answers the required conditions, but the △ABC′ does not, for it does not contain the obtuse ∠A. There is then only one solution; namely, the△ABC. q.e.f.

PROBLEMS OF CONSTRUCTION. 149 Proposition XXXIII. Problem. 312. To construct a triangle when the three sides of the triangle are given. A B C ab c c a b Let the three sides be c, a, and b. TakeAB equal to c. From A as a centre, with a radius equal to b, describe an arc. From B as a centre, with a radius equal to a, describe an arc, intersecting the other arc at C. DrawCA and CB. △CAB is the△ required. q.e.f. Discussion. The problem is impossible when one side is equal to or greater than the sum of the other two sides.

BOOK II. PLANE GEOMETRY. 150 Proposition XXXIV. Problem. 313. To construct a parallelogram when two sides and the included angle are given. C m o A B EH D m o Let m and o be the two sides, and C the included angle. TakeAB equal to o. AtA construct ∠BAD equal to ∠C. § 305 TakeAH equal to m. From H as a centre, with a radius equal to o, describe an arc, and from B as a centre, with a radius equal to m, describe an arc, intersecting the other arc at E; and draw EH and EB. The quadrilateral ABEH is the / / required. § 182 q.e.f.

PROBLEMS OF CONSTRUCTION. 151 Proposition XXXV. Problem. 314. To circumscribe a circle about a given triangle. A B C D E F G O Let ABC be the given triangle. Bisect AB and BC. § 302 AtE and D, the points of bisection, erect ⊥s. § 301 Since BC is not the prolongation of AB, these⊥s will intersect at some point O. FromO, with a radius equal to OB, describe a circle. The⊙ABC is the⊙ required. Proof. The point O is equidistant from A and B, and also is equidistant from B and C. § 160 ∴ the point O is equidistant from A, B, and C, and a⊙ described from O as a centre, with a radius equal to OB, will pass through the vertices A, B, and C. q.e.f. The same construction serves to describe a circumference which shall pass through three points not in the same straight line; also to find the centre of a given circle or of a given arc. Note. The point O is called the circum-centre of the triangle.

BOOK II. PLANE GEOMETRY. 152 Proposition XXXVI. Problem. 315. To inscribe a circle in a given triangle. A B C K E M H Let ABC be the given triangle. Bisect the ∠sA and C. § 304 FromE, the intersection of the bisectors, drawEH⊥ to the side AC. § 300 FromE as centre, with radius EH , describe the⊙KHM . The⊙KHM is the⊙ required. Proof. SinceE is in the bisector of the ∠A, it is equidistant from the sides AB and AC; and since E is in the bisector of the ∠C, it is equidistant from the sides AC and BC. § 162 ∴ a⊙ described from E as centre, with a radius equal to EH , will touch the sides of the△ and be inscribed in it. q.e.f. Note. The point E is called the in-centre of the triangle.

PROBLEMS OF CONSTRUCTION. 153 A B C DE F O P Q 316. The intersections of the bisectors of the exterior angles of a triangle are the centres of three circles, each of which will touch one side of the triangle, and the two other sides produced. These three circles are calledescribed circles; and their centres are called the ex-centres of the triangle.

BOOK II. PLANE GEOMETRY. 154 Proposition XXXVII. Problem. 317. Through a given point, to draw a tangent to a given circle. Case 1. When the given point is on the circumference. A C M O O M H E Let C be the given point on the circumference whose centre is O. From the centre O draw the radius OC. Through C drawAM⊥ to OC. § 301 Then AM is the tangent required. § 253 Case 2. When the given point is without the circle. Let O be the centre of the given circle, E the given point. DrawOE. On OE as a diameter, describe a circumference intersecting the given circumference at the points M and H. DrawOM and EM . Then EM is the tangent required. Proof. ∠OME is a right angle. § 290 ∴EM is tangent to the circle at M. § 253 In like manner, we may prove EH tangent to the given⊙. q.e.f. Ex. 146. To draw a tangent to a given circle, so that it shall be parallel to a given straight line.

PROBLEMS OF CONSTRUCTION. 155 Proposition XXXVIII. Problem. 318. Upon a given straight line, to describe a segment of a circle in which a given angle may be inscribed. A B E F O M K Let AB be the given line, and M the given angle. Construct the ∠ABE equal to the ∠M. § 305 Bisect the line AB by the⊥OF . § 302 From the point B drawBO⊥ to EB. § 301 FromO, the point of intersection ofFO andBO, as a centre with a radius equal to OB, describe a circumference. The segment AKB is the segment required. Proof. The point O is equidistant from A and B. § 160 ∴ the circumference will pass through A. But BE is⊥ to OB. Const. ∴BE is tangent to the⊙, §

253 (a straight line ⊥ to a radius at its extremity is tangent to the ⊙). ∠ABE is measured by 1 2 arcAB, § 295 (being an ∠ formed by a tangent and a chord). But any ∠ as ∠K inscribed in the segment AKB is measured by 1 2 arcAB.§ 289 the ∠M may be inscribed in the segment AKB. q.e.f.

BOOK II. PLANE GEOMETRY. 156 SOLUTION OF PROBLEMS. 319. If a problem is so simple that the solution is obvious from a known theorem, we have only to make the construction according to the theorem, and then give a synthetic proof, if a proof is necessary, that the construction is correct, as in the examples of the fundamental problems already given. 320. But problems are usually of a more difficult type. The application of known theorems to their solution is not immediate, and often far from obvious. To discover the mode of application is the first and most difficult part of the solution. The best way to attack such problems is by a method resembling the analytic proof of a theorem, called the analysis of the problem. 1. Suppose the construction made , and let the figure represent all parts concerned, both given and required. 2. Study the relations among the parts with the aid of known theorems, and try to find some relation that will suggest the construction. 3. If this attempt fails, introduce new relations by drawing auxiliary lines, and study the new relations. If this attempt fails, make a new trial, and so on till a clue to the right construction is found. 321. A problem is determinate if it has a definite number of solutions, indeterminate if it has an indefinite number of solutions, and impossible if it has no solution. A problem is sometimes determinate for certain relative posi- tions or magnitudes of the given parts, and indeterminate for other positions or magnitudes of the given parts. 322. The discussion of a problem consists in examining the problem with reference to all possible conditions, and in determining the conditions necessary for its solution.

PROBLEMS OF CONSTRUCTION. 157 Ex. 147. Problem. To construct a circle that shall pass through a given point and cut chords of a given length from two parallels. A B C D E F G M N O P Analysis. Suppose the problem solved. Let A be the given point, BC and DE the given parallels, MN the given length, and O the centre of the required circle. Since the circle cuts equal chords from two parallels its centre must be equidistant from them. Therefore, one locus for O isFG‖ toBC and equidis- tant from BC and DE. Draw the⊥ bisector of MN , cutting FG in P . PM is the radius of the circle required. With A as centre and radius PM describe an arc cutting FG at O. Then O is the centre of the required circle. Discussion. The problem is impossible if the distance from A to FG is greater than PM .

BOOK II. PLANE GEOMETRY. 158 A B C DE F G H m n Ex. 148. Problem. To construct a triangle, having given the perimeter, one angle, and the altitude from the vertex of the given angle. Analysis. Suppose the problem solved, and let ABC be the△ required, ACB the given ∠, and CD the given altitude. Produce AB both ways, and take AE =AC, and BF =BC, then EF = the given perimeter. Join CE and CF , forming the isosceles △sCAE and CBF . In the△ECF , ∠E + ∠F + ∠ECF = 180◦ (why?), but ∠ECF = ∠ECA + ∠FCB + ∠ACB. Since ∠E = ∠ECA and ∠F = ∠FCB , we have ∠ECF = ∠E + ∠F + ∠ACB. 2∠E + 2∠F + ∠ACB = 180◦. ∠E + ∠F + 1 2 ∠ACB = 90◦, and ∠E + ∠F = 90◦− 1 2 ∠ACB. By substitution, ∠ECF = 90◦ + 1 2 ∠ACB. ∠ECF is

known. Construction. To find the pointC, construct on EF a segment that will contain the ∠ECF (§ 318), and draw a parallel to EF at the distance CD, the given altitude. To find the pointsA andB, draw the⊥ bisectors of the lines CE andCF , and the points A and B will be vertices of the required △. Why? PROBLEMS OF CONSTRUCTION. Ex. 149. Find the locus of a point at a given distance from a given circumference. Find the locus of the centre of a circle: Ex. 150. Which has a given radiusr and passes through a given point P .

EXERCISES. 159 Ex. 151. Which has a given radius r and touches a given line AB. Ex. 152. Which passes through two given points P and Q. Ex. 153. Which touches a given straight line AB at a given point P . Ex. 154. Which touches each of two given parallels. Ex. 155. Which touches each of two given intersecting lines. Ex. 156. To find in a given line a point X which is equidistant from two given points. The required point is the intersection of the given line with the perpendic- ular bisector of the line joining the two given points (§ 160). Ex. 157. To find a point X equidistant from three given points. P Q R X X′ d Ex. 158. To find a point X equidistant from two given points and at a given distance from a third given point. Ex. 159. To construct a circle which has a given radius and passes through two given points. Ex. 160. To find a point X at given distances from two given points. Ex. 161. To construct a circle which has its centre in a given line and passes through two given points. Ex. 162. To find a point X equidistant from two given points and also equidistant from two given intersecting lines (§§ 160 and 162). Ex. 163. To find a point X equidistant from two given points and also equidistant from two given parallel lines. Ex. 164. To find a point X equidistant from two given intersecting lines and also equidistant from two given parallels.

BOOK II. PLANE GEOMETRY. 160 A BC D E X Y Z P T d Ex. 165. To find a point X equidistant from two given intersecting lines and at a given distance from a given point. Ex. 166. To find a point X which lies in one side of a given triangle and is equidistant from the other two sides. A B CD E X Y Z T A B C D E F Ex. 167. A straight railway passes two miles from a town. A place is four miles from the town and one mile from the railway. To find by construction the places that answer this description. Ex. 168. In a triangleABC, to drawDE parallel to the baseBC, cutting the sides of the triangle inD andE, so thatDE shall equalDB +EC (§ 162). A B C D EF d Ex. 169. To draw through two sides of a triangle a line parallel to the third side so that the part intercepted between the sides shall have a given length. TakeBD =d. Ex. 170. Prove that the locus of the vertex of a right triangle, having a given hypotenuse as base, is the circumference described upon the given hypotenuse as diameter (§ 290). Ex. 171. Prove that the locus of the vertex of a triangle, having a given base and a given angle at the vertex, is the arc which forms with the base a segment capable of containing the given angle (§ 318). Ex. 172. Find the locus of the middle point of a chord of a given length that can be drawn in a given circle.

EXERCISES. 161 Ex. 173. Find the locus of the middle point of a chord drawn from a given point in a given circumference. A B C D E OM A B C D O M A BO M Ex. 174. Find the locus of the middle point of a straight line drawn from a given exterior point to a given circumference. Ex. 175. A straight line moves so that it remains parallel to

a given line, and touches at one end a given circumference. Find the locus of the other end. Ex. 176. A straight rod moves so that its ends constantly touch two fixed rods which are perpendicular to each other. Find the locus of its middle point. Ex. 177. In a given circle let AOB be a diameter, OC any radius, CD the perpendicular from C toAB. Upon OC takeOM equal to CD. Find the locus of the point M as OC turns about O. AB C D E O M A B C D O Ex. 178. To construct an equilateral triangle, having given the radius of the circumscribed circle. To construct on isosceles triangle, having given: Ex. 179. The angle at the vertex and the base (§ 160 and§ 318). Ex. 180. The base and the radius of the circumscribed circle. Ex. 181. The base and the radius of the inscribed circle.

BOOK II. PLANE GEOMETRY. 162 A B C DE F Ex. 182. The perimeter and the altitude. Let ABC be the △ required, EF the given perimeter. The altitude CD passes through the middle of EF , and the△sAEC, BFC are isosceles. To construct a right triangle, having given: Ex. 183. The hypotenuse and one leg. Ex. 184. One leg and the altitude upon the hypotenuse. Ex. 185. The median and the altitude drawn from the vertex of the right angle. Ex. 186. The hypotenuse and the altitude upon the hypotenuse. Ex. 187. The radius of the inscribed circle and one leg. Ex. 188. The radius of the inscribed circle and an acute angle. Ex. 189. An acute angle and the sum of the legs. Ex. 190. An acute angle and the difference of the legs. A B C D E F P Q R O Ex. 191. To construct an equilateral triangle, having given the radius of the inscribed circle. To construct a triangle, having given: Ex. 192. The base, the altitude, and an angle at the base.

EXERCISES. 163 Ex. 193. The base, the altitude, and the ∠ at the vertex. Ex. 194. The base, the corresponding median, and the ∠ at the vertex. Ex. 195. The perimeter and the angles. Ex. 196. One side, an adjacent ∠, and the sum of the other sides. To construct a triangle, having given: Ex. 197. One side, an adjacent ∠, and the difference of the other sides. Ex. 198. The sum of two sides and the angles. Ex. 199. One side, an adjacent ∠, and the radius of the circumscribed circle. Ex. 200. The angles and the radius of the circumscribed circle. Ex. 201. The angles and the radius of the inscribed circle. Ex. 202. An angle, and the bisector and the altitude drawn from the vertex of the given angle. Ex. 203. Two sides and the median corresponding to the other side. Ex. 204. The three medians. To construct a square, having given: Ex. 205. The diagonal. Ex. 206. The sum of the diagonal and one side. LetABCD be the square required,CA the diagonal. Produce CA, making AE =AB.△sABC and ABE are isosceles and ∠BAC = ∠BCA = 45◦.

BOOK II. PLANE GEOMETRY. 164 A B C D E A B C D E F G M N O Ex. 207. Given two perpendiculars, AB and CD, intersecting in O, and a straight line intersecting these perpendiculars in E and F ; to construct a square, one of whose angles shall coincide with one of the right angles atO, and the vertex of the opposite angle of the square shall lie in EF . (Two solutions.) To construct a rectangle, having given: Ex. 208. One side and the angle between the diagonals. Ex. 209. The perimeter and the diagonal. Ex. 210. The perimeter and the angle between the diagonals. Ex. 211. The difference of two adjacent sides and the angle between the diagonals. To construct a rhombus, having given: Ex. 212. The two diagonals. Ex. 213. One side and the radius

of the inscribed circle. Ex. 214. One angle and the radius of the inscribed circle. Ex. 215. One angle and one of the diagonals. To construct a rhomboid, having given: Ex. 216. One side and the two diagonals. Ex. 217. The diagonals and the angle between them.

EXERCISES. 165 Ex. 218. One side, one angle, and one diagonal. Ex. 219. The base, the altitude, and one angle. To construct an isosceles trapezoid, having given: Ex. 220. The bases and one angle. Ex. 221. The bases and the altitude. Ex. 222. The bases and the diagonal. A B CD E F G O Ex. 223. The bases and the radius of the circumscribed circle. LetABCD be the isosceles trapezoid required, O the centre of the circum- scribed⊙. A diameter ⊥ to AB is⊥ to CD, and bisects both AB and CD. DrawCG∥ to FE . Then EG =FC = 1 2DC. To construct a trapezoid, having given: Ex. 224. The four sides. Ex. 225. The two bases and the two diagonals. Ex. 226. The bases, one diagonal, and the ∠ between the diagonals. To construct a circle which has the radius r and which also: Ex. 227. Touches each of two intersecting lines AB and CD. Ex. 228. Touches a given line AB and a given circle K. Ex. 229. Passes through a given point P and touches a given line AB. Ex. 230. Passes through a given point P and touches a given circle K.

BOOK II. PLANE GEOMETRY. 166 To construct a circle which shall: Ex. 231. Touch two given parallels and pass through a given point P . Ex. 232. Touch three given lines two of which are parallel. Ex. 233. Touch a given line AB at P and pass through a given point Q. Ex. 234. Touch a given circle at P and pass through a given point Q. Ex. 235. Touch two given lines and touch one of them at a given pointP . Ex. 236. Touch a given line and touch a given circle at a point P . Ex. 237. Touch a given line AB at P and also touch a given circle. Ex. 238. To inscribe a circle in a given sector. Ex. 239. To construct within a given circle three equal circles, so that each shall touch the other two and also the given circle. Ex. 240. To describe circles about the vertices of a given triangle as centres, so that each shall touch the two others. A B C D E F GHI Ex. 241. To bisect the angle formed by two lines, without producing the lines to their point of intersection. Draw any line EF ∥ to BA. Take EG = EH , and produce GH to meet BA at I. Draw the ⊥ bisector of GI. A B C D E FG P A B C D E P Q A BC D E F P Q Ex. 242. To draw through a given point P between the sides of an angle BAC a line terminated by the sides of the angle and bisected at P .

EXERCISES. 167 Ex. 243. Given two pointsP ,Q, and a line AB; to draw lines from P and Q which shall meet on AB and make equal angles with AB. Make use of the point which forms with P a pair of points symmetrical with respect to AB. Ex. 244. To find the shortest path from P to Q which shall touch a line AB. A B C D E FG H O O ′ P Q R S Ex. 245. To draw a common tangent to two given circles. Let r and r′ denote the radii of the circles, O and O′ their centres. With centre O and radius r−r′ describe a ⊙. From O′ draw the tangents O′M, O′N. Produce OM andON to meet the circumference at A andC. Draw the radii O′B and O′D∥, respectively, to OA and OC. Draw AB and CD. To draw the internal tangents use an auxiliary ⊙ of radius r +r′.

BOOK III. PROPORTION. SIMILAR POLYGONS. THE THEORY OF PROPORTION. 323. A proportion is an expression of equality between two equal ratios; and is written in one of the following forms: a :b =c :d; a :b ::c :d; a b = c d. This proportion

is read, “a is to b asc is to d”; or “the ratio of a tob is equal to the ratio of c to d.” 324. The terms of a proportion are the four quantities compared; the first and third terms are called the antecedents, the second and fourth terms, the consequents; the first and fourth terms, the extremes, the second and third terms, the means. Thus, in the proportion a :b =c :d; a and c are the antecedents, b and d the consequents, a and d the extremes, b and c the means. 325. The fourth proportional to three given quantities is the fourth term of the proportion which has for its first three terms the three given quantities taken in order. Thus,d is the fourth proportional to a, b, and c in the proportion a :b =c :d. 326. The quantitiesa,b,c,d,e, are said to be in continued proportion, if a :b =b :c =c :d =d :e. If three quantities are in continued proportion, the second is called the mean proportional between the other two, and the third is called the third proportional to the other two. Thus, in the proportion a :b =b :c; b is the mean proportional between a and c; and c is the third proportional to a and b.

THEORY OF PROPORTION. 169 Proposition I. Theorem. 327. In every proportion the product of the extremes is equal to the product of the means. Let a : b = c : d. Then a b = c d. § 323 Whence ad =bc. q.e.d. Proposition II. Theorem. 328. The mean proportional between two quantities is equal to the square root of their product. Let a : b = b : c. Then b2 =ac. § 327 Whence, extracting the square root, b =√ac. q.e.d. Proposition III. Theorem. 329. If the product of two quantities is equal to the product of two others, either two may be made the extremes of the proportion in which the other two are made the means. Let ad = bc. To prove that a :b =c :d Divide both members of the given equation by bd. Then a b = c d. Or a :b =c :d. q.e.d.

BOOK III. PLANE GEOMETRY. 170 Proposition IV. Theorem. 330. If four quantities are in proportion, they are in proportion by alter- nation; that is, the first term is to the third as the second is to the fourth. Let a : b = c : d. To prove that a :c =b :d. Now ad =bc. § 327 Divide each member of the equation by cd. Then a c = b d. Or a :c =b :d. q.e.d. Proposition V. Theorem. 331. If four quantities are in proportion, they are in proportion by inver- sion; that is, the second term is to the first as the fourth is to the third. Let a : b = c : d. To prove that b :a =d :c. Now bc =ad. § 327 Divide each member of the equation by ac. Then b a = d c . Or b :a =d :c. q.e.d.

THEORY OF PROPORTION. 171 Proposition VI. Theorem. 332. If four quantities are in proportion, they are in proportion by com- position that is, the sum of the first two terms is to the second term as the sum of the last two terms is to the fourth term. Let a : b = c : d. To prove that a +b :b =c +d :d. Now a b = c d. Then a b + 1 = c d + 1; that is, a +b b = c +d d . Or a +b :b =c +d :d. In like manner a +b :a =c +d :c. q.e.d.

BOOK III. PLANE GEOMETRY. 172 Proposition VII. Theorem. 333. If four quantities are in proportion, they are in proportion by divi- sion; that is, the difference of the first two terms is to the second term as the difference of the last two terms is to the fourth term. Let a : b = c : d. To prove that a−b :b =c−d :d. Now a b = c d. Then a b− 1 = c d− 1; that is, a−b b = c−d d . Or a−b :b =c−d :d. In like manner a−b :a =c−d :c. q.e.d.

THEORY OF PROPORTION. 173 Proposition VIII. Theorem. 334. If four quantities are in proportion, they are in proportion by com- position and division ; that is, the sum of the first two terms is to their difference as the sum of the last two terms is to their difference. Let a : b = c : d Then a +b a = c +d c . § 332 And a−b a = c−d c . § 333 Divide, a +b a−b = c +d c−d. Or a +b :a−b =c +d :c−d. q.e.d.

BOOK III. PLANE GEOMETRY. 174 Proposition IX. Theorem. 335. In a series of equal ratios, the sum of the antecedents is to the sum of the consequents as any antecedent is to its consequent. Let a : b = c : d = e : f = g : h. To prove that a +c +e +g :b +d +f +h =a :b. Let r = a b = c d = e f = g h. Then a =br, c =dr, e =fr, g =hr. And a +c +e +g = (b +d +f +h)r. Divide by (b +d +f +h). Then a +c +e +g b +d +f +h =r = a b . Or a +c +e +g :b +d +f +h =a :b. q.e.d.

THEORY OF PROPORTION. 175 Proposition X. Theorem. 336. The products of the corresponding terms of two or more proportions are in proportion. Let a : b = c : d, e : f = g : h, k : l = m : n. To prove that aek :bfl =cgm :dhn. Now a b = c d, e f = g h, k l = m n . The products of the first members and of the second members of these equations give aek bfl = cgm dhn . Or aek :bfl =cgm :dhn. q.e.d. 337. Cor. If three quantities are in continued proportion, the first is to the third as the square of the first is to the square of the second. Proposition XI. Theorem. 338. Like powers of the terms of a proportion are in proportion. Let a : b = c : d. To prove that an :bn =cn :dn. Now a b = c d. Raise to the nth power, an bn = cn dn . Or an :bn =cn :dn. q.e.d.

BOOK III. PLANE GEOMETRY. 176 339. Def. Equimultiples of two quantities are the products obtained by multiplying each of them by the same number. Thus, ma and mb are equimultiples of a and b. Proposition XII. Theorem. 340. Equimultiples of two quantities are in the same ratio as the quantities themselves. Let a and b be any two quantities. To prove that ma :mb =a :b. Now a b = a b . Multiply both terms of the first fraction by m. Then ma mb = a b . Or ma :mb =a :b. q.e.d. 341. Scholium. In the treatment of proportion, it is assumed that the quantities involved are expressed by their numerical measures. It is evident that the ratio of two quantities of the same kind may be represented by a frac- tion, if the two quantities are expressed in integers in terms of a common unit. If there is no unit in terms of which both quantities can be expressed in inte- gers, it is still possible by taking the unit of measure sufficiently small to find a fraction that will represent the ratio to any required degree of accuracy.§ 269 If we speak of the product of two quantities, it must be understood that we mean simply the product of the numbers which represent them when they are expressed in terms of a common unit. In order that four quantities,a,b,c,d, may form a proportion,a andb must be quantities of the same kind; and c and d must be quantities of the same kind; though c andd need not be of the same kind as a andb. In alternation, however, the four quantities must be of the same kind.

THEORY OF PROPORTION. 177 Proposition XIII. Theorem. 342. If a line is drawn through two sides of a triangle parallel to the third side, it divides those sides proportionally. A B C E F M A B C E F K H In the triangle ABC, let EF be drawn parallel to BC. To prove that EB :AE =FC :AF . Case 1. When AE and EB (Fig. 1) are commensurable. Proof. Find a common measure of AE and EB, as MB . Let MB be

contained m times in EB, and n times in AE. Then EB :AE =m :n. At the points of division on BE and AE draw lines‖ to BC. These lines will divideAC intom+n equal parts, of whichFC will containm, andAF will contain n. § 187 □FC :AF =m :n. □EB :AE =FC :AF . Ax. 1 Case 2. When AE and EB (Fig. 2) are incommensurable. Proof. DivideAE into any number of equal parts, and apply one of these parts to EB as many times as EB will contain it. Since AE and EB are incommensurable, a certain number of these parts will extend from E to some point K, leaving a remainder KB less than one of these parts. Draw KH‖BC. Then EK :AE =FH :AF Case 1 By increasing the number of equal parts into which AE is divided, we can make the length of each part less than any assigned value, however small, but not zero. Hence,KB, which is less than one of these equal parts, has zero for a limit.§ 275 And the corresponding segment HC has zero for a limit.

BOOK III. PLANE GEOMETRY. 178 Therefore, EK approachesEB as a limit, § 271 and FH approachesFC as a limit. □ the variable EK AE approaches EB AE as a limit, § 280 and the variable FH AF approaches FC AF as a limit. But EK AE is constantly equal to FH AF Case 1 □ EB AE = FC AF . § 284 q.e.d. 343. Cor. 1. One side of a triangle is to either part cut off by a straight line parallel to the base as the other side is to the corresponding part. For AE :EB =AF :FC . By composition, AE +EB :AE =AF +FC :AF . § 332 Or AB :AE =AC :AF . 344. Cor. 2. If two lines are cut by any number of parallels the corre- sponding intercepts are proportional. A B C D F G H K L M N DrawAN‖ to CD. Then AL =CG, LM =GK, MN =KD. § 180 Now AH :AM =AF :AL =FH :LM =HB :MN . § 343 Or AF :CG =FH :GK =HB :KD.

THEORY OF PROPORTION. 179 Proposition XIV. Theorem. 345. If a straight line divides two sides of a triangle proportionally, it is parallel to the third side. A B C E F H In the triangle ABC, let EF be drawn so that AB AE = AC AF . To prove that EF is‖ to BC. Proof. FromE drawEH‖ to BC. Then AB :AE =AC :AH, § 343 (one side of a triangle is to either part cut off by a line parallel to the base as the other side to the corresponding part). But AB :AE =AC :AF . Hyp. □AC :AF =AC :AH. Ax. 1 □AF =AH. □EF and EH coincide. § 47 But EH is‖ to BC. Const. □EF , which coincides with EH , is‖ to BC. q.e.d. Ex. 246. Find the fourth proportional to 91, 65, and 133. Ex. 247. Find the mean proportional between 39 and 351. Ex. 248. Find the third proportional to 54 and 3.

BOOK III. PLANE GEOMETRY. 180 346. If a given line AB is divided at M, a point between the extremities A andB, it is said to be divided internally into the segments MA andMB ; and if it is divided at M′, a point in the prolongation of AB, it is said to be divided externally into the segments M′A and M′B. A B M M ′ In either case the segments are the distances from the point of division to the extremities of the line. If the line is divided internally, the sum of the segments is equal to the line; and if the line is divided externally, thedifference of the segments is equal to the line. Suppose it is required to divide the given line AB internally and ex- ternally in the same ratio ; as, for example, the ratio of the two numbers 3 and 5. xM ′ A M B y We divide AB into 5 + 3, or 8, equal parts, and take 3 parts from A; we then have the point M, such that MA :MB = 3 : 5. (1) Secondly, we divide AB into 5− 3, or 2, equal parts, and lay off on the prolongation of AB, to the left of A, three of these equal parts; we then have the

point M′, such that M′A :M′B = 3 : 5. (2) Comparing (1) and (2), MA :MB =M′A :M′B. 347. Def. If a given straight line is divided internally and externally into segments having the same ratio, the line is said to be divided harmonically.

THEORY OF PROPORTION. 181 Proposition XV. Theorem. 348. The bisector of an angle of a triangle divides the opposite side into segments which are proportional to the adjacent sides. A B C E M Let CM bisect the angle C of the triangle CAB. To prove that MA :MB =CA :CB. Proof. DrawAE∥ to MC , meeting BC produced at E. Then MA :MB =CE :CB, § 342 (if a line is drawn through two sides of a △ parallel to the third side, it divides those sides proportionally). Also, ∠ACM = ∠CAE, § 110 (being alt.-int. ∠s of ∥ lines); and ∠BCM = ∠CEA, § 112 (being ext.-int. ∠s of ∥ lines). But ∠ACM = ∠BCM . Hyp. □ ∠CAE = ∠CEA. Ax. 1 □CE =CA. § 147 Put CA for its equal, CE, in the first proportion. Then MA :MB =CA :CB. q.e.d. Ex. 249. In a triangle ABC, AB = 12, AC = 14, BC = 13. Find the segments of BC made by the bisector of the angle A. Ex. 250. In a triangle CAB, CA = 6, CB = 12, AB = 15. Find the segments of AB made by the bisector of the angle C.

BOOK III. PLANE GEOMETRY. 182 Proposition XVI. Theorem. 349. The bisector of an exterior angle of a triangle divides the opposite side externally into segments which are proportional to the adjacent sides. A B C E F M ′ Let CM ′ bisect the exterior angle ACE of the triangle CAB, and meet BA produced at M ′. To prove that M′A :M′B =CA :CB. Proof. DrawAF∥ to M′C, meeting BC at F . Then M′A :M′B CF :CB. § 343 Now ∠M′CE = ∠AFC , § 112 and ∠M′CA = ∠CAF , § 110 (being alt.-int. ∠s of ∥ lines). But ∠M′CE = ∠M′CA. Hyp. □ ∠AFC = ∠CAF . Ax. 1 □CA =CF . § 147 Put CA for its equal, CF , in the first proportion. Then M′A :M′B =CA :CB. q.e.d. Question. To what case does this theorem not apply? (See Ex. 41, page 79.) 350. Cor. The bisectors of an interior angle and an exterior angle at one vertex of a triangle meeting the opposite side divide that side harmoni- cally. § 347

SIMILAR POLYGONS. 183 SIMILAR POLYGONS. 351. Def. Similar polygons are polygons that have their homologous angles equal, and their homologous sides proportional. A B C DE A′ B′ C′ D′E′ Thus, the polygons ABCDE and A′B′C′D′E′ are similar, if the ∠sA, B, C, etc., are equal, respectively, to the ∠sA′, B′, C′, etc., and if AB :A′B′ =BC :B′C′ =CD :C′D′, etc. 352. Def. Homologous lines in similar polygons are lines similarly sit- uated. 353. Def. The ratio of any two homologous lines in similar polygons, is called the ratio of similitude of the polygons. The primary idea of similarity is likeness of form . The two conditions necessary to similarity are: 1. For every angle in one of the figures there must be an equal angle in the other. 2. The homologous sides must be proportional. Thus, Q and Q′ are not similar; the homologous sides are proportional, but the homologous angles are not equal. Also R and R′ are not similar; the homologous angles are equal, but the sides are not proportional. Q Q ′ R R′ In the case of triangles, either condition involves the other (see § 354 and § 358).

BOOK III. PLANE GEOMETRY. 184 Proposition XVII. Theorem. 354. Two mutually equiangular triangles are similar. A B C E H A′ B′ C′ In the triangles ABC and A′B′C′, let the angles A, B, C be equal to the angles A′, B′, C′, respectively. To prove that the △sABC and A′B′C′ are similar. Since the△s are mutually equiangular, we have only to

prove that AB :A′B′ =AC :A′C′ =BC :B′C′. § 351 Proof. Place the△A′B′C′ on the△ABC so that ∠A′ shall coincide with its equal, the ∠A; and B′C′ take the position EH . Then ∠AEH = ∠B Hyp. □EH is‖ to BC. § 114 □AB :AE =AC :AH. § 343 That is, AB :A′B′ =AC :A′C′. Similarly, by placing △A′B′C′ on△ABC, so that ∠B′ shall coincide with its equal, the ∠B, we may prove that AB :A′B′ =BC :B′C′ q.e.d. 355. Cor. 1. Two triangles are similar if two angles of the one are equal, respectively, to two angles of the other. 356. Cor. 2. Two right triangles are similar if an acute angle of the one is equal to an acute angle of the other.

SIMILAR POLYGONS. 185 Proposition XVIII. Theorem. 357. If two triangles have an angle of the one equal to an angle of the other, and the including sides proportional, they are similar . A B C E H A′ B′ C′ In the triangles ABC and A′B′C′, let ∠A = ∠A′, and let AB : A′B′ = AC : A′C′. To prove that the △sABC and A′B′C′ are similar. In this case we prove the △s similar by proving them mutually equiangular. Proof. Place the△A′B′C′ on the△ABC, so that the ∠A′ shall coincide with its equal, the ∠A. Then the△A′B′C′ will take the position of △AEH . Now AB A′B′ = AC A′C′ . Hyp. That is, AB AE = AC AH . □EH is‖ to BC, § 345 (if a line divides two sides of a △ proportionally, it is ‖ to the third side). □ ∠AEH = ∠B, and ∠AHE = ∠C. § 112 □△AEH is similar to△ABC. § 354 □△A′B′C′ is similar to△ABC. q.e.d.

BOOK III. PLANE GEOMETRY. 186 Proposition XIX. Theorem. 358. If two triangles have their sides respectively proportional, they are similar. A B C E H A′ B′ C′ In the triangles ABC and A′B′C′, let AB : A′B′ = AC : A′C′ = BC : B′C′. To prove that the △sABC and A′B′C′ are similar. Proof. UponAB takeAE equal toA′B′, and upon AC takeAH equal toA′C′; and draw EH . Now AB :A′B′ =AC :A′C′. Hyp. Or, since AE =A′B′ and AH =A′C′, AB :AE =AC :AH. □△sABC and AEH are similar. § 357 □AB :AE =BC :EH ; § 351 that is, AB :A′B′ =BC :EH . But AB :A′B′ =BC :B′C′. Hyp. □BC :EH =BC :B′C′. Ax. 1 □EH =B′C′. Hence, the△sAEH and A′B′C′ are equal. § 150 But △AEH is similar to△ABC. □△A′B′C′ is similar to△ABC. q.e.d.

SIMILAR POLYGONS. 187 Proposition XX. Theorem. 359. Two triangles which have their sides respectively parallel, or respec- tively perpendicular, are similar. A B C A′B′ C′ x y D E F D′ E′ F ′ H K O Let ABC and A′B′C′ have their sides respectively parallel; and DEF and D′E′F ′ have their sides respectively perpendicular. To prove that the △sABC and A′B′C′ are similar; and that the △sDEF and D′E′F′ are similar. Proof. 1. Prolong BC and AC to B′A′, forming ∠sx and y. Then ∠B′ = ∠x (§ 112), and ∠B = ∠x. § 110 Therefore, ∠B′ = ∠B Ax. 1 In like manner, ∠A′ = ∠A. Therefore,△A′B′C′ is similar to△ABC. § 355 2. Prolong DE and FD to meet D′E′ at H and D′F′ at K. The quadrilateral EHE′O has ∠sEHE′ and E′OE right angles, by hypothesis. Therefore, ∠E′ and ∠OEH are supplementary. § 206 But ∠DEF and ∠OEH are supplementary. § 86 Therefore, ∠DEF = ∠E′. § 85 In like manner, ∠EDF = ∠D′. Therefore,△DEF is similar to△D′E′F′. § 355 q.e.d. 360. Cor. The parallel sides and the perpendicular sides are homologous sides of the triangles.

BOOK III. PLANE GEOMETRY. 188 Proposition XXI. Theorem. 361. The homologous altitudes of two similar triangles have the same ratio as any two homologous sides. A B C O A ′ B′ C′ O′ In the two similar triangles ABC and A′B′C′, let CO and C′O′ be homologous altitudes. To prove that CO C′O′ = AC A′C′ = AB A′B′ = BC

B′C′ . Proof. In the rt.△sCOA and C′O′A′, ∠A = ∠A′, § 351 (being homologous △s of the similar △sABC and A′B′C′). ∴△sCOA and C′O′A′ are similar, § 356 (two rt. △s having an acute ∠ of the one equal to an acute ∠ of the other are similar). ∴ CO C′O′ = AC A′C′ . § 351 In the similar△sABC and A′B′C′, AC A′C′ = AB A′B′ = BC B′C′ . § 351 Therefore, CO C′O′ = AC A′C′ = AB A′B′ = BC B′C′ . q.e.d. Ex. 251. The base and altitude of a triangle are 7 feet 6 inches and 5 feet 6 inches, respectively. If the homologous base of a similar triangle is 5 feet 6 inches, find its homologous altitude.

SIMILAR POLYGONS. 189 Proposition XXII. Theorem. 362. If two parallels are cut by three or more transversals that pass through the same point, the corresponding segments are proportional . A B C D E A′ B′ C′ D′ E′ A B C D E A′B′C′D′E′ O Let the two parallels AE and A′E′ be cut by the transversals OA, OB, OC, OD, OE in A, A′, B, B′, etc. To prove that AB A′B′ = BC B′C′ = CD C′D′ = DE D′E′ . Proof. Since A′E′ is‖ to AE, the pairs of △sOAB and OA′B′, OBC and OB′C′, etc., are similar. § 354 ∴ AB A′B′ = OB OB′ and BC B′C′ = OB OB′ . § 351 (homologous sides of similar △s are proportional). ∴ AB A′B′ = BC B′C′ . Ax. 1 In a similar way it may be shown that BC B′C′ = CD C′D′ and CD C′D′ = DE D′E′ . q.e.d. Note. A condensed form of writing the above is AB A′B′ = (OB OB′) = BC B′C′ = (OC OC′) = CD C′D′ = (OD OD′) = DE D′E′ . A parenthesis about a ratio signifies that this ratio is used to prove the equality of the ratios immediately preceding and following it.

BOOK III. PLANE GEOMETRY. 190 Proposition XXIII. Theorem. 363. Conversely: If three or more non-parallel straight lines intercept proportional segments upon two parallels, they pass through a common point. A B C D E F O Let AB, CD, EF cut the parallels AE and BF so that AC : BD = CE : DF. To prove that AB, CD, EF prolonged meet in a point. Proof. Prolong AB and CD until they meet in O. DrawOE. Designate by F′ the point where OE cuts BF . Then AC :BD =CE :DF′. § 362 But AC :BD =CE :DF . Hyp. ∴CE :DF′ =CE :DF . Ax. 1 ∴DF′ =DF . ∴F′ coincides with F . ∴EF coincides with EF′. § 47 ∴EF prolonged passes through O. ∴AB, CD, and EF prolonged meet in the point O. q.e.d.

SIMILAR POLYGONS. 191 Proposition XXIV. Theorem. 364. The perimeters of two similar polygons have the same ratio as any two homologous sides. A B C D E A B C D E Let the two similar polygons be ABCDE and A′B′C′D′E′, and let P and P ′ represent their perimeters. To prove that P :P′ =AB :A′B′. Proof. AB :A′B′ =BC :B′C′ =CD :C′D′, etc. § 351 ∴AB +BC + etc. : A′B′ +B′C′ + etc. = AB :A′B′, § 335 (in a series of equal ratios the sum of the antecedents is to the sum of the consequents as any antecedent is to its consequent). That is, P :P′ =AB :A′B′. q.e.d. Ex. 252. If the line joining the middle points of the bases of a trapezoid is produced, and the two legs are also produced, the three lines will meet in the same point. Ex. 253. AB and AC are chords drawn from any point A in the cir- cumference of a circle, and AD is a diameter. The tangent to the circle at D intersects AB and AC at E and F , respectively. Show that the triangles ABC and AEF are similar. Ex. 254. AD and BE are two altitudes of the triangle CAB. Show that the triangles CED and CAB are similar.

BOOK III. PLANE GEOMETRY. 192 Ex. 255. If two circles are tangent to each other, the chords formed by a straight line drawn through the point of contact have the same ratio as the diameters of the circles.

SIMILAR POLYGONS. 193 Proposition XXV. Theorem. 365. If two polygons are similar, they are composed of the same number of triangles, similar each to each, and similarly placed. A B C D E A B C D E Let the polygons ABCDE and A′B′C′D′E′ be similar. From two homologous vertices, asE andE′, draw diagonalsEB,EC, andE′B′, E′C′. To prove that the △sEAB, EBC , ECD are similar, respectively, to the △sE′A′B′, E′B′C′, E′C′D′. Proof. The△sEAB and E′A′B′ are similar. § 357 For ∠A = ∠A′, § 351 and AE :A′E′ =AB :A′B′. § 351 Now ∠ABC = ∠A′B′C′, § 351 and ∠ABE = ∠A′B′E′. § 351 By subtracting, ∠EBC = ∠E′B′C′. Ax. 3 Now EB :E′B′ =AB :A′B′ § 351 and BC :B′C′ =AB :A′B′ § 351 ∴EB :E′B′ =BC :B′C′. Ax. 1 ∴△sEBC and E′B′C′ are similar. § 357 In like manner△sECD and E′C′D′ are similar. q.e.d.

BOOK III. PLANE GEOMETRY. 194 Proposition XXVI. Theorem. 366. Conversely: If two polygons are composed of the same number of triangles, similar each to each, and similarly placed, the polygons are similar. A B C D E A B C D E In the two polygons ABCDE and A′B′C′D′E′, let the triangles AEB, BEC , CED be similar, respectively, to the triangles A′E′B′, B′E′C′, C′E′D′; and similarly placed. To prove that ABCDE is similar to A′B′C′D′E′. Proof. ∠A = ∠A′ § 351 Also, ∠ABE = ∠A′B′E′, and ∠EBC = ∠E′B′C′. § 351 By adding, ∠ABC = ∠A′B′C′. Ax. 2 In like manner, ∠BCD = ∠B′C′D′, ∠CDE = ∠C′D′E′, etc. Hence, the polygons are mutually equiangular. Also, AB A′B′ = (EB E′B′) = BC B′C′ = (EC E′C′) = CD C′D′ , etc. § 351 Hence, the polygons have their homologous sides proportional. Therefore, the polygons are similar. § 351 q.e.d.

EXERCISES. 195 THEOREMS. Ex. 256. If two circles are tangent to each other externally, the correspond- ing segments of two lines drawn through the point of contact and terminated by the circumferences are proportional. Ex. 257. In a parallelogram ABCD , a line DE is drawn, meeting the diagonal AC in F , the side BC in G, and the side AB produced in E. Prove that DF 2 =FG×FE . Ex. 258. Two altitudes of a triangle are inversely proportional to the corresponding bases. Ex. 259. Two circles touch at P . Through P three lines are drawn, meeting one circle in A,B,C, and the other in A′,B′,C′, respectively. Prove that the triangles ABC, A′B′C′ are similar. Ex. 260. Two chords AB, CD intersect at M, and A is the middle point of the arc CD. Prove that the product AB×AM is constant if the chord AB is made to turn about the fixed point A. Draw the diameter AE, and draw BE. Ex. 261. If two circles touch each other, their common external tangent is the mean proportional between their diameters. Let AB be the common tangent. Draw the diameters AC, BD. Join the point of contactP toA,B,C, andD. Show that APD andBPC are straight lines⊥ to each other, and that △sCAB, ABD are similar. Ex. 262. If two circles are tangent internally, all chords of the greater circle drawn from the point of contact are divided proportionally by the cir- cumference of the smaller circle. Draw any two of the chords, and join the points where they meet the circumferences. The△s thus formed are similar (Ex. 120).

BOOK III. PLANE GEOMETRY. 196 A B CD E Ex. 263. In an inscribed quadrilateral, the product of the diagonals is equal to the sum of the products of the opposite sides. DrawDE, making ∠CDE = ∠ADB. The△sABD andECD are similar; and the△sBCD and AED are similar. Ex. 264. Two isosceles triangles with equal vertical

angles are similar. Ex. 265. The bisector of the vertical angleA of the triangleABC intersects the base at D and the circumference of the circumscribed circle at E. Show that AB×AC =AD×AE.

NUMERICAL PROPERTIES OF FIGURES. 197 NUMERICAL PROPERTIES OF FIGURES. Proposition XXVII. Theorem. 367. If in a right triangle a perpendicular is drawn from the vertex of the right angle to the hypotenuse: 1. The triangles thus formed are similar to the given triangle, and to each other. 2. The perpendicular is the mean proportional between the segments of the hypotenuse. 3. Each leg of the right triangle is the mean proportional between the hy- potenuse and its adjacent segment. A B C F a ba′ b′ In the right triangle ABC, let CF be drawn from the vertex of the right angle C, perpendicular to AB. 1. To prove that△'s BCA, CFA , BFC are similar. Proof. The rt.△sCFA and BCA are similar, § 356 since the ∠a′ is common. The rt.△sBFC and BCA are similar, § 356 since the ∠b is common. Since the△sCFA andBFC are each similar to△BCA, they are similar to each other. § 354 2. To prove that AF :CF =CF :FB . Proof. In the similar△sCFA and BFC , AF :CF =CF :FB . § 351

BOOK III. PLANE GEOMETRY. 198 3. To prove that AB :AC =AC :AF , and AB :BC =BC :BF . Proof. In the similar△sBCA and CFA , AB :AC =AC :AF § 351 In the similar△sBCA and BFC , AB :BC =BC :BF . § 351 q.e.d. 368. Cor. 1. The squares of the two legs of a right triangle are proportional to the adjacent segments of the hypotenuse. From the proportions in§ 367,3, AC 2 =AB×AF , and BC 2 =AB×BF . § 327 Hence, AC 2 BC 2 = AB×AF AB×BF = AF BF . 369. Cor. 2. The squares of the hypotenuse and either leg are proportional to the hypotenuse and the adjacent segment. For AB 2 AC 2 = AB×AB AB×AF = AB AF . A B C D 370. Cor. 3. The perpendicular from any point in the circumference to the diameter of a circle is the mean proportional between the segments of the diameter. The chord drawn from any point in the circumference to either extremity of the diameter is the mean proportional between the diameter and the adjacent segment. For the ∠ACB is a rt. ∠. § 290

NUMERICAL PROPERTIES OF FIGURES. 199 Proposition XXVIII. Theorem. 371. The sum of the squares of the two legs of a right triangle is equal to the square of the hypotenuse. A B C F Let ABC be a right triangle with its right angle at C. To prove that AC 2 +CB 2 =AB 2 . Proof. DrawCF ⊥ to AB. Then AC 2 =AB×AF , and CB 2 =AB×BF . § 367 By adding, AC 2 +CB 2 =AB(AF +BF) =AB 2 q.e.d. 372. Cor. 1. The square of either leg of a right triangle is equal to the difference of the square of the hypotenuse and the square of the other leg. A B CD A B C D P R 373. Cor. 2. The diagonal and a side of a square are incommensurable. For AC 2 =AB 2 +BC 2 = 2AB 2 . □AC =AB √ 2. 374. Def. The projection of any line upon a second line is the segment of the second line included between the perpendiculars drawn to it from the extremities of the first line. Thus, PR is the projection of CD upon AB.

BOOK III. PLANE GEOMETRY. 200 Proposition XXIX. Theorem. 375. In any triangle, the square of the side opposite an acute angle is equal to the sum of the squares of the other two sides diminished by twice the product of one of those sides by the projection of the other upon that side. A B C D A B CD Let C be an acute angle of the triangle ABC, and DC the projection of AC upon BC. To prove that AB 2 =BC 2

+AC 2 − 2BC×DC. Proof. If D falls upon the base (Fig. 1), DB =BC−DC. If D falls upon the base produced (Fig. 2), DB =DC−BC. In either case, DB 2 =BC 2 +DC 2 − 2BC×DC. Add AD 2 to both sides of this equality, and we have AD 2 +DB 2 =BC 2 +AD 2 +DC 2 − 2BC×DC. But AD 2 +DB 2 =AB 2 and AD 2 +DC 2 =AC 2 § 371 Put AB 2 and AC 2 for their equals in the above equality. Then AB 2 =BC 2 +AC 2 − 2BC×DC. q.e.d.

NUMERICAL PROPERTIES OF FIGURES. 201 Proposition XXX. Theorem. 376. In any obtuse triangle, the square of the side opposite the obtuse angle is equal to the sum of the squares of the other two sides increased by twice the product of one of those sides by the projection of the other upon that side. A C BD Let C be the obtuse angle of the triangle ABC, and CD be the pro- jection of AC upon BC produced. To prove that AB 2 =BC 2 +AC 2 + 2BC×DC. Proof. DB =BC +DC. Squaring, DB 2 =BC 2 +DC 2 + 2BC×DC. Add AD 2 to both sides, and we have AD 2 +DB 2 =BC 2 +AD 2 +DC 2 + 2BC×DC. But AD 2 +DB 2 =AB 2 , and AD 2 +DC 2 =AC 2 . § 371 Put AB 2 and AC 2 for their equals in the above equality. Then AB 2 =BC 2 +AC 2 + 2BC×DC. q.e.d. Note 1. By the Principle of Continuity the last three theorems may be included in one theorem. Let the student explain. Note 2. The last three theorems enable us to compute the lengths of the alti- tudes of a triangle if the lengths of the three sides are known.

BOOK III. PLANE GEOMETRY. 202 Proposition XXXI. Theorem. 377. 1. The sum of the squares of two sides of a triangle is equal to twice the square of half the third side increased by twice the square of the median upon that side. 2. The difference of the squares of two sides of a triangle is equal to twice the product of the third side by the projection of the median upon that side. A B CDM In the triangle ABC, let AM be the median and M D the projection of AM upon the side BC. Also, let AB be greater than AC. To prove that 1. AB 2 +AC 2 = 2BM 2 + 2AM 2 . 2. AB 2 −AC 2 = 2BC×MD. Proof. Since AB > AC, the ∠AMB will be obtuse, and the ∠AMC will be acute. § 155 Then AB 2 =BM 2 +AM 2 + 2BM×MD, § 376 and AC 2 =MC 2 +AM 2 − 2MC×MD. § 375 Add these two equalities, and observe that BM =MC . Then AB 2 +AC 2 = 2BM 2 + 2AM 2 . Subtract the second equality from the first. Then AB 2 −AC 2 = 2BC×MD. q.e.d. Note. This theorem enables us to compute the lengths of the medians of a triangle if the lengths of the three sides are known.

NUMERICAL PROPERTIES OF FIGURES. 203 Proposition XXXII. Theorem. 378. If two chords intersect in a circle, the product of the segments of one is equal to the product of the segments of the other. M N O P Q a c a′ c′ Let any two chords M N and P Q intersect at O. To prove that OM×ON =OQ×OP . Proof. DrawMP and NQ. ∠a = ∠a′, § 289 (each being measured by 1 2 arcPN). And ∠c = ∠c′, § 289 (each being measured by 1 2 arcMQ). ∴ the△sNOQ and POM are similar. § 355 ∴OQ :OM =ON :OP . § 351 ∴OM×ON =OQ×OP . § 327 q.e.d. 379. Scholium. This proportion may be written OM OQ = OP ON , or OM OQ = 1 ON OP ; that is, the ratio of two corresponding segments is equal to the reciprocal of the ratio of the other two segments. Hence, these segments are said to be reciprocally proportional.

BOOK III. PLANE GEOMETRY. 204 380. Def. A secant from a point to a circle is understood to mean the segment of the secant lying between the point and the

second point of intersection of the secant and circumference. Proposition XXXIII. Theorem. 381. If from a point without a circle a secant and a tangent are drawn, the tangent is the mean proportional between the whole secant and its external segment. A B C D a b a′ Let AD be a tangent and AC a secant drawn from the point A to the circle BCD. To prove that AC :AD =AD :AB. Proof. DrawDC and DB. The△sADC and ABD are similar. § 355 For ∠b is common; and ∠a′ = ∠a, §§ 289, 295 (each being measured by 1 2 arcBD). ∴AC :AD =AD :AB. § 351 q.e.d. 382. Cor. If from a fixed point without a circle a secant is drawn, the product of the secant and its external segment is constant in whatever direction the secant is drawn. For AC×AB =AD 2 . § 327

NUMERICAL PROPERTIES OF FIGURES. 205 Proposition XXXIV. Theorem. 383. The square of the bisector of an angle of a triangle is equal to the product of the sides of this angle diminished by the product of the segments made by the bisector upon the third side of the triangle. M N O P Q a b a′ b′ Let N O bisect the angle M N P of the triangle M N P. To prove that NO 2 =NM×NP−OM×OP . Proof. Circumscribe the⊙MNP about the△MNP . § 314 Produce NO to meet the circumference in Q, and draw PQ . The△sNQP and NMO are similar. § 355 For ∠b = ∠b′ Hyp. and ∠a = ∠a′ § 289 Whence NQ :NM =NP :NO. § 351 ∴NM×NP =NQ×NO = (NO +OQ)NO =NO 2 +NO×OQ. But NO×OQ =MO×OP . § 378 ∴MN×NP =NO 2 +MO×OP . Whence NO 2 =NM×NP =MO×OP . Ax. 3 q.e.d.

BOOK III. PLANE GEOMETRY. 206 Note. This theorem enables us to compute the lengths of the bisectors of the angles of a triangle if the lengths of the sides are known. Proposition XXXV. Theorem. 384. In any triangle the product of two sides is equal to the product of the diameter of the circumscribed circle by the altitude upon the third side. M N O Q P a a′ Let NMQ be a triangle, NO the altitude, and QNMP the circle cir- cumscribed about the triangle NMQ. Draw the diameter NP , and draw PQ . To prove that NM×NQ =NP×NO . Proof. In the△sNOM and NQP , ∠NOM is a rt. ∠, Hyp. ∠NQP is a rt. ∠, § 290 and ∠a = ∠a′, § 289 (each being measured by 1 2 arcNQ). ∴△sNOM and NQP are similar. § 356 Whence NM :NP =NO :NQ. § 351 ∴NM×NQ =NP×NO. § 327 q.e.d. Note. This theorem enables us to compute the length of the radius of a circle circumscribed about a triangle, if the lengths of the three sides of the triangle are known.

EXERCISES. 207 Ex. 266. IfOE,OF ,OG are the perpendiculars from any point O within the triangleABC upon the sidesAB,BC,CA, respectively, show thatAE 2 + BF 2 +CG 2 =EB 2 +FC 2 +GA 2 . THEOREMS. Ex. 267. The sum of the squares of the segments of two perpendicular chords is equal to the square of the diameter of the circle. If AB, CD are the chords, draw the diameter BE, draw AC, ED, BD. Prove that AC =ED, and apply§ 371. Ex. 268. The tangents to two intersecting circles drawn from any point in their common chord produced, are equal. (§ 381.) Ex. 269. The common chord of two intersecting circles, if produced, will bisect their common tangents. (§ 381.) A B C D E P O Ex. 270. If three circles intersect one another, the common chords all pass through the same point. Let two of the chordsAB andCD meet atO. Join the point of intersection E to O, and suppose that EO produced meets the same two circles at two different pointsP andQ. Then prove that OP =OQ (§ 378);

hence, that the points P and Q coincide. Ex. 271. If two circles are tangent to each other, the common internal tangent bisects the two common external tangents. Ex. 272. If the perpendiculars from the vertices of the triangleABC upon the opposite sides intersect at D, show that AB 2 −AC 2 =BD 2 −CD 2 .

BOOK III. PLANE GEOMETRY. 208 Ex. 273. In an isosceles triangle, the square of a leg is equal to the square of any line drawn from the vertex to the base, increased by the product of the segments of the base. Ex. 274. The squares of two chords drawn from the same point in a circumference have the same ratio as the projections of the chords on the diameter drawn from the same point. Ex. 275. The difference of the squares of two sides of a triangle is equal to the difference of the squares of the segments of the third side, made by the perpendicular on the third side from the opposite vertex. Ex. 276. E is the middle point of BC, one of the parallel sides of the trapezoid ABCD ; AE and DE produced meet DC and AB produced at F and G, respectively. Show that FG is parallel to DA. △sAGD and BGE are similar; and△sAFD and EFC are similar. Ex. 277. If two tangents are drawn to a circle at the extremities of a diameter, the portion of a third tangent intercepted between them is divided at its point of contact into segments whose product is equal to the square of the radius. Ex. 278. If two exterior angles of a triangle are bisected, the line drawn from the point of intersection of the bisectors to the opposite angle of the triangle bisects that angle. Ex. 279. The sum of the squares of the diagonals of a quadrilateral is equal to twice the sum of the squares of the lines that join the middle points of the opposite sides. A B CD EF A B C D F H Ex. 280. The sum of the squares of the four sides of any quadrilateral is equal to the sum of the squares of the diagonals, increased by four times the square of the line joining the middle points of the diagonals. Apply§ 377 to the △s ABC and ADC, add the results, and eliminate BE 2 +DE 2 by applying§ 377 to the△BDE .

EXERCISES. 209 Ex. 281. The square of the bisector of an exterior angle of a triangle is equal to the product of the external segments determined by the bisector upon one of the sides, diminished by the product of the other two sides. LetCD bisect the exterior ∠BCH of the△ABC.△s ACD andFCB are similar (§ 355). Apply§ 382. Ex. 282. If a point O is joined to the vertices of a triangle ABC; through any pointA′ inOA a line parallel toAB is drawn, meetingOB atB′; through B′ a line parallel toBC, meetingOC atC′; andC′ is joined toA′; the triangle A′B′C′ is similar to the triangle ABC. Ex. 283. If the line of centres of two circles meets the circumferences at the consecutive points A, B, C, D, and meets the common external tangent at P , then PA×PD =PB×PC . Ex. 284. The line of centres of two circles meets the common external tan- gent atP , and a secant is drawn fromP , cutting the circles at the consecutive points E, F , G, H. Prove that PE×PH =PF×PG . Draw radii to the points of contact, and to E,F ,G,H. Let fall⊥s onPH from the centres of the ⊙s. The various pairs of △s are similar. Ex. 285. If a line drawn from a vertex of a triangle divides the opposite side into segments proportional to the adjacent sides, the line bisects the angle at the vertex.

BOOK III. PLANE GEOMETRY. 210 PROBLEMS OF CONSTRUCTION. Proposition XXXVI. Problem. 385. To divide a given straight line into parts proportional to any

number of given lines. A B C E F H K X m n pm n p Let AB, m, n, and p be given straight lines. To divide AB into parts proportional to m, n, and p. DrawAX, making any convenient ∠ with AB. On AX takeAC equal to m, CE to n, EF to p. DrawBF . FromE and C drawEK and CH‖ to FB . Through A draw a line‖ to BF . K and H are the division points required. Proof. AH AC = HK CE = KB EF , § 344 (if two lines are cut by any number of parallels, the corresponding intercepts are proportional). Substitute m, n, and p for their equals AC, CE, and EF . Then AH m = HK n = KB p . q.e.f. Ex. 286. Divide a line 12 inches long into three parts proportional to the numbers 3, 5, 7.

PROBLEMS OF CONSTRUCTION. 211 Proposition XXXVII. Problem. 386. To find the fourth proportional to three given straight lines. A B C D F x y m n p m n p Let the three given lines be m, n, and p. To find the fourth proportional to m, n, and p. DrawAx and Ay containing any convenient angle. On Ax takeAB equal to m, BC to n. On Ay takeAD equal to p. DrawBD. FromC drawCF‖ to BD, meeting Ay at F . DF is the fourth proportional required. Proof. AB :BC =AD :DF , § 342 (a line drawn through two sides of a △‖ to the third side divides those sides proportionally). Substitute m, n, and p for their equals AB, BC, and AD. Then m :n =p :DF q.e.f. Ex. 287. The square of the altitude of an equilateral triangle is equal to three fourths of the square of one side of the triangle.

BOOK III. PLANE GEOMETRY. 212 Proposition XXXVIII. Problem. 387. To find the third proportional to two given straight lines. A B C D E m n Let m and n be the two given straight lines. To find the third proportional to m and n. Construct any convenient angle A, and take AB equal to m, AC equal to n. Produce AB to D, making BD equal to AC. DrawBC. Through D drawDE‖ to BC, meeting AC produced at E. CE is the third proportional required. Proof. AB :BD =AC :CE, § 342 (a line drawn through two sides of a △ parallel to the third side divides those sides proportionally). Substitute, in the above proportion, AC for its equal BD. Then AB :AC =AC :CE, that is, m :n =n :CE. q.e.f. Ex. 288. Construct x, if (1) x = ab c , (2) x = a2 c . Special cases: (1) a = 2, b = 8, c = 4; (2) a = 3, b = 7, c = 11; (3) a = 2, c = 3; (4) a = 3, c = 5; (5) a = 2c.

PROBLEMS OF CONSTRUCTION. 213 Proposition XXXIX. Problem. 388. To find the mean proportional between two given straight lines. A BC H Em n m n Let the two given lines be m and n. To find the mean proportional between m and n. On the straight line AE takeAC equal to m, and CB equal to n. On AB as a diameter describe a semicircumference. AtC erect the⊥ CH meeting the circumference at H. CH is the mean proportional between m and n. Proof. AC :CH =CH :CB § 370 (the ⊥ let fall from a point in a circumference to the diameter of a circle is the mean proportional between the segments of the diameter). Substitute for AC and CB their equals m and n. Then m :CH =CH :n. q.e.f. 389. Def. A straight line is divided in extreme and mean ratio , when one of the segments is the mean proportional between the whole line and the other segment. Ex. 289. Construct x, if x = √ ab. Special cases: (1) a = 2, b = 3; (2) a = 1, b = 6; (3) a = 3, b = 7.

BOOK III. PLANE GEOMETRY. 214 Proposition XL. Problem. 390. To divide a given line in extreme and mean ratio. A B C E F G C′ Let AB be the given line. To divide AB

in extreme and mean ratio. AtB erect a⊥BE equal to half of AB. FromE as a centre, with a radius equal to EB, describe a⊙. DrawAE, meeting the circumference in F and G. On AB takeAC equal to AF . On BA produced take AC′ equal to AG. Then AB is divided internally at C and externally at C′ in extreme and mean ratio. AG :AB =AB :AF . § 381 AB 2 =AF×AG AB 2 =AG×AF =AC(AF +AG) =C′A(AG−AF) =AC(AC +AB) =C′A(C′A−AB) =AC 2 +AB×AC. =C′A 2 −AB×C′A. □AB 2 −AB×AC =AC 2 . □AB 2 +AB×C′A =C′A 2 . □AB(AB−AC) = AC 2 . □AB(AB +C′A) = C′A 2 . □AB×CB =AC 2 . □AB×C′B =C′A 2 . q.e.f.

PROBLEMS OF CONSTRUCTION. 215 Proposition XLI. Problem. 391. Upon a given line homologous to a given side of a given polygon, to construct a polygon similar to the given polygon. A B C D E A B C D E Let A′E′ be the given line homologous to AE of the given polygon ABCDE . To construct on A′E′ a polygon similar to the given polygon. FromE draw the diagonals EB and EC. FromE′ drawE′B′, E′C′, and E′D′, making△'s A′E′B′, B′E′C′, and C′E′D′ equal, respectively, to △sAEB, BEC , and CED . FromA′ drawA′B′, making ∠E′A′B′ equal to ∠EAB, and meeting E′B′ at B′. FromB′ drawB′C′, making ∠E′B′C′ equal to ∠EBC , and meeting E′C′ at C′. FromC′ drawC′D′, making ∠E′C′D′ equal to ∠ECD , and meeting E′D′ at D′. Then A′B′C′D′E′ is the required polygon. Proof. The△sABE, A′B′E′, etc., are similar. § 354 Therefore, the two polygons are similar. § 366 q.e.f.

BOOK III. PLANE GEOMETRY. 216 PROBLEMS OF CONSTRUCTION. Ex. 290. To divide one side of a given triangle into segments proportional to the adjacent sides (§ 348). A B C D E F G H O A B C D OP A B C O P A BC DE F O P Ex. 291. To find in one side of a given triangle a point whose distances from the other sides shall be to each other in the given ratio m :n. TakeAG =m⊥ to AC, GH =n⊥ to BC. Draw CD∥ to OG. Ex. 292. Given an obtuse triangle; to draw a line from the vertex of the obtuse angle to the opposite side which shall be the mean proportional between the segments of that side. Ex. 293. Through a given point P within a given circle to draw a chord AB so that the ratio AP :BP shall equal the given ratio m :n. Draw OPC so that OP : PC = n : m. Draw CA equal to the fourth proportional to n, m, and the radius of the circle. Ex. 294. To draw through a given pointP in the arc subtended by a chord AB a chord which shall be bisected by AB. On radius OP takeCD equal to CP . Draw DE∥ to BA. A B C D P A B C DP A B C DE O P G D P E F O A O′ Ex. 295. To draw through a given external point P a secant PAB to a given circle so that the ratio PA :AB shall equal the given ratio m :n. PD :DC =m :n. PD :PA =PA :PC.

EXERCISES. 217 Ex. 296. To draw through a given external point P a secant PAB to a given circle so that AB 2 =PA×PB . PC :CD =CD :PD. PA =CD. Ex. 297. To find a point P in the arc subtended by a given chord AB so that the ratio PA :PB shall equal the given ratio m :n. Ex. 298. To draw through one of the points of intersection of two circles a secant so that the two chords that are formed shall be in the given ratio m :n. Ex. 299. Having given the greater segment of a line divided in extreme and mean ratio, to construct the line. Ex. 300. To construct a circle which shall pass through two given points and touch a given straight line. Ex. 301. To construct a circle which shall pass through a given point and touch two given straight lines. Ex. 302. To inscribe a square in a semicircle. A B C D E FG H M N Ex. 303. To inscribe a square

in a given triangle. LetDEFG be the required inscribed square. Draw CM‖ toAB, meeting AF produced in M. Draw CH and MN ⊥ to AB, and produce AB to meet MN at N. The △sACM , AGF are similar; also, the △sAMN , AFE are similar. By these triangles show that the figure CMNH is a square. By constructing this square, the point F can be found. Ex. 304. To inscribe in a given triangle a rectangle similar to a given rectangle. Ex. 305. To inscribe in a circle a triangle similar to a given triangle.

BOOK III. PLANE GEOMETRY. 218 Ex. 306. To inscribe in a given semicircle a rectangle similar to a given rectangle. Ex. 307. To circumscribe about a circle a triangle similar to a given triangle. Ex. 308. To construct the expression, x = 2abc de ; that is, 2ab d × c e. Ex. 309. To construct two straight lines, having given their sum and their ratio. Ex. 310. To construct two straight lines, having given their difference and their ratio. Ex. 311. Given two circles, with centresO andO′, and a point A in their plane, to draw through the pointA a straight line, meeting the circumferences at B and C, so that AB :AC =m :n.

EXERCISES. 219 PROBLEMS OF COMPUTATION. Ex. 312. To compute the altitudes of a triangle in terms of its sides. AB C D a b c h At least one of the angles A or B is acute. Suppose B is acute. In the△ CDB , h2 =a2−BD 2 , § 372 In the△ ABC, b2 =a2 +c2− 2c× BD. § 376 Whence BD = a2 +c2−b2 2c . Hence, h2 =a2− (a2 +c2−b2)2 4c2 = 4a2c2− (a2 +c2−b2)2 4c2 = (2ac +a2 +c2−b2)(2ac−a2−c2 +b2) 4c2 ={(a +c)2−b2}{b2− (a−c)2} 4c2 = (a +b +c)(a +c−b)(b +a−c)(b−a +c) 4c2 . Let a +b +c = 2s. Then a +c−b = 2(s−b), b +a−c = 2(s−c), b−a +c = 2(s−a). Hence, h2 = 2s× 2(s−a)× 2(s−b)× 2(2−c) 4c2 . By simplifying, and extracting the square root,

BOOK III. PLANE GEOMETRY. 220 h = 2 c √ s(s−a)(s−b)(s−c). A B C DF ab c m h Ex. 313. To compute the medians of a triangle in terms of its sides. By§ 377, a2 +b2 = 2m2 + 2 (c 2)2 . Whence 4m2 = 2(a2 +b2)−c2. ∴m = 1 2 √ 2(a2 +b2)−c2.

EXERCISES. 221 A B C D E ab c t Ex. 314. To compute the bisectors of a triangle in terms of the sides. By§ 383, t2 =ab−AD×BD. By§348, AD B = BD a = AD +BD a +b = c a +b. ∴AD = bc a +b, and BD = ac a +b. Whence t2 =ab− abc2 (a +b)2 =ab [1− c2 (a +b)2] = ab{(a +b)2−c2} (a +b)2 = ab(a +b +c)(a +b−c) (a +b)2 = ab× 2s× 2(s−c) (a +b2) . Whence t = 2 a +b √ abs(s−c).

BOOK III. PLANE GEOMETRY. 222 B C A D E bc a Ex. 315. To compute the radius of the circle circumscribed about a triangle in terms of the sides of the triangle. By§384, AC×AB =AE×AD, or, bc = 2B×AD. But AD = 2 a √ s(s−a)(s−b)(s−c). Ex. 312 ∴R = abc 4 √ s(s−a)(s−b)(s−c) . Ex. 316. If the sides of a triangle are 3, 4, and 5, is the angle opposite 5 right, acute, or obtuse? Ex. 317. If the sides of a triangle are 7, 9, and 12, is the angle opposite 12 right, acute, or obtuse? Ex. 318. If the sides of a triangle are 7, 9, and 11, is the angle opposite 11 right, acute, or obtuse? Ex. 319. The legs of a right triangle are 8 inches and 12 inches; find the lengths of the projections of these legs upon the hypotenuse, and the distance of the vertex of the right angle from the hypotenuse. Ex. 320. If the sides of a triangle are 6 inches, 9 inches, and 12 inches, find the lengths (1) of the altitudes; (2) of the medians; (3) of the bisectors; (4) of the radius of the circumscribed circle. Ex. 321. A line is drawn

parallel to a side AB of a triangleABC, cutting AC in D, BC in E. If AD :DC = 2 : 3, and AB = 20 inches, find DE.

EXERCISES. 223 Ex. 322. The sides of a triangle are 9, 12, 15. Find the segments of the sides made by bisecting the angles. Ex. 323. A tree casts a shadow 90 feet long, when a post 6 feet high casts a shadow 4 feet long. How high is the tree? Ex. 324. The lower and upper bases of a trapezoid are a,b, respectively; and the altitude is h. Find the altitudes of the two triangles formed by pro- ducing the legs until they meet. Ex. 325. The sides of a triangle are 6, 7, 8, respectively. In a similar triangle the side homologous to 8 is 40. Find the other two sides. Ex. 326. The perimeters of two similar polygons are 200 feet and 300 feet. If a side of the first is 24 feet, find the homologous side of the second. Ex. 327. How long a ladder is required to reach a window 24 feet high, if the lower end of the ladder is 10 feet from the side of the house? Ex. 328. If the side of an equilateral triangle is a, find the altitude. Ex. 329. If the altitude of an equilateral triangle is h, find the side. Ex. 330. Find the length of the longest chord and of the shortest chord that can be drawn through a point 6 inches from the centre of a circle whose radius is 10 inches. Ex. 331. The distance from the centre of a circle to a chord 10 feet long is 12 feet. Find the distance from the centre to a chord 24 feet long. Ex. 332. The radius of a circle is 5 inches. Through a point 3 inches from the centre a diameter is drawn, and also a chord perpendicular to the diameter. Find the length of this chord, and the distance from one end of the chord to the ends of the diameter. Ex. 333. The radius of a circle is 6 inches. Find the lengths of the tangents drawn from a point 10 inches from the centre, and also the length of the chord joining the points of contact. Ex. 334. The sides of a triangle are 407 feet, 368 feet, and 351 feet. Find the three bisectors and the three altitudes.

BOOK III. PLANE GEOMETRY. 224 Ex. 335. If a chord 8 inches long is 8 inches distant from the centre of the circle, find the radius, and the chords drawn from the end of the chord to the ends of the diameter which bisects the chord. Ex. 336. From the end of a tangent 20 inches long a secant is drawn through the centre of the circle. If the external segment of this secant is 8 inches, find the radius of the circle. Ex. 337. The radius of a circle is 13 inches. Through a point 5 inches from the centre any chord is drawn. What is the product of the two segments of the chord? What is the length of the shortest chord that can be drawn through the point? Ex. 338. The radius of a circle is 9 inches and the length of a tangent 12 inches. Find the length of a line drawn from the extremity of the tangent to the centre of the circle. Ex. 339. Two circles have radii of 8 inches and 3 inches, respectively, and the distance between their centres is 15 inches. Find the lengths of their common tangents. Ex. 340. Find the segments of a line 10 inches long divided in extreme and mean ratio. Ex. 341. The sides of a triangle are 4, 5, 5. Is the largest angle acute, right, or obtuse? Ex. 342. Find the third proportional to two lines whose lengths are 28 feet and 42 feet. Ex. 343. If the sides of a triangle are a,b,c, respectively, find the lengths of the three altitudes. Ex. 344. The diameter of a circle is 30 feet and is divided into five equal parts. Find the lengths of the chords drawn through the points of division perpendicular to the diameter. Ex. 345. The radius of a circle is 2 inches. From a point 4 inches from the centre a secant is drawn so that the internal segment is 1 inch. Find the length of the secant.

EXERCISES. 225 Ex. 346. The sides of a triangular pasture are 1551 yards, 2068 yards, 2585 yards. Find the median to the longest side. Ex. 347. The diagonal of a rectangle is d, and the perimeter is p. Find the sides. Ex. 348. The radius of a circle is r. Find the length of a chord whose distance from the centre is 1 2r.

BOOK IV. AREAS OF POLYGONS. 392. Def. The unit of surface is a square whose side is a unit of length . 393. Def. The area of a surface is the number of units of surface it contains. 394. Def. Plane figures that have equal areas but cannot be made to co- incide are called equivalent. Note. In propositions relating to areas, the words “rectangle,” “triangle,” etc., are often used for “area of rectangle,” “area of triangle,” etc. Proposition I. Theorem. 395. Two rectangles having equal altitudes are to each other as their bases. A B CD O A E FD O Let the rectangles AC and AF have the same altitude AD. To prove that rect.AC : rect.AF = baseAB : baseAE. Case 1. When AB and AE are commensurable. Proof. SupposeAB andAE have a common measure, asAO, which is contained m times in AB and n times in AE. Then AB :AE =m :n.

AREAS OF POLYGONS. 227 Apply AO as a unit of measure to AB and AE, and at the several points of division erect⊥s. The rect.AC is divided into m rectangles, and the rect.AF is divided into n rectangles. § 107 These rectangles are all equal. § 186 Hence, rect.AC : rect.AF =m :n. Therefore, rect.AC : rect.AF =AB :AE. Ax. 1 Case 2. When AB and AE are incommensurable. A B CD A K E FHD Proof. Divide AB into any number of equal parts, and apply one of them to AE as many times as AE will contain it. Since AB and AE are incommensurable, a certain number of these parts will extend from A to some point K, leaving a remainder KE less than one of the equal parts of AB. DrawKH‖ to EF . Then AB and AK are commensurable by construction. Therefore, rect.AH rect.AC = AK AB. Case 1 If the number of equal parts into which AB is divided is indefinitely increased, the varying values of these ratios will continue equal, and approach for their respec- tive limits the ratios rect.AF rect.AC and AE AB . (See § 287.) □ rect.AF rect.AC = AE AB. § 284 q.e.d. 396. Cor. Two rectangles having equal bases are to each other as their altitudes.

BOOK IV. PLANE GEOMETRY. 228 Proposition II. Theorem. 397. Two rectangles are to each other as the products of their bases by their altitudes. R R′ S a b a′ b′ a′ b Let R and R′ be two rectangles, having for their bases b and b′, and for their altitudes a and a′, respectively. To prove that R R′ = a×b a′×b′ . Proof. Construct the rectangle S, with its base equal to that of R, and its altitude equal to that of R′. Then R S = a a′ , § 396 and S R′ = b b′ . § 395 The products of the corresponding members of these equations give R R′ = a×b a′×b′ . q.e.d. Ex. 349. Find the ratio of a rectangular lawn 72 yards by 49 yards to a grass turf 18 inches by 14 inches. Ex. 350. Find the ratio of a rectangular courtyard 181 2 yards by 151 2 yards to a flagstone 31 inches by 18 inches. Ex. 351. A square and a rectangle have the same perimeter, 100 yards. The length of the rectangle is 4 times its breadth. Compare their areas. Ex. 352. On a certain map the linear scale is 1 inch to 5 miles. How many acres are represented on this map by a square the perimeter of which is 1 inch?

AREAS OF POLYGONS. 229 Proposition III. Theorem. 398. The area of a rectangle is equal to the product of its base by its altitude. Ra b U1 1 Let R be a rectangle, b its base, and a its altitude. To prove that the area of R =a×b. Proof. Let U be the unit of

surface. R U = a×b 1× 1 =a×b, (two rectangles are to each other as the products of their bases and altitudes). But R U = the number of units of surface in R. § 393 ∴ the area of R =a×b. q.e.d. 399. Scholium. When the base and altitude each contain the linear unit an integral number of times, this proposition is rendered evident by dividing the figure into squares, each equal to the unit of surface. Thus, if the base contains seven linear units, and the altitude four, the figure may be divided into twenty-eight squares, each equal to the unit of surface.

BOOK IV. PLANE GEOMETRY. 230 Proposition IV. Theorem. 400. The area of a parallelogram is equal to the product of its base by its altitude. A B C D E F a b A B C D E F a b Let AEF D be a parallelogram, b its base, and a its altitude. To prove that the area of the / / AEFD =a×b. Proof. FromA drawAB∥ to DC to meet FE produced. Then the figureABCD is a rectangle, with the same base and the same altitude as the / / AEFD . The rt.△sABE and DCF are equal. § 151 ForAB =CD, and AE =DF . § 178 FromABFD take the△DCF ; the rect.ABCD is left. FromABFD take the△ABE; the / / AEFD is left. ∴ rect.ABCD ∴ / / AEFD Ax. 3 But the area of the rect.ABCD =a×b. § 398 ∴ the area of the / / AEFD =a×b. Ax. 1 q.e.d. 401. Cor. 1. Parallelograms having equal bases and equal altitudes are equivalent. 402. Cor. 2. Parallelograms having equal bases are to each other as their altitudes; parallelograms having equal altitudes are to each other as their bases; any two parallelograms are to each other as the products of their bases by their altitudes.

AREAS OF POLYGONS. 231 Proposition V. Theorem. 403. The area of a triangle is equal to half the product of its base by its altitude. A B C D H a b Let a be the altitude and b the base of the triangle ABC. To prove that the area of the △ABC = 1 2a×b. Proof. Construct on AB and BC the parallelogram ABCH . Then △ABC = 1 2 / / ABCH . § 179 The area of the / / ABCH =a×b. § 400 Therefore, the area of △ABC = 1 2a×b. Ax. 7 q.e.d. 404. Cor. 1. Triangles having equal bases and equal altitudes are equivalent. 405. Cor. 2. Triangles having equal bases are to each other as their alti- tudes; triangles having equal altitudes are to each other as their bases; any two triangles are to each other as the products of their bases by their altitudes. 406. Cor. 3. The product of the legs of a right triangle is equal to the product of the hypotenuse by the altitude from the vertex of the right angle. Ex. 353. The lines which join the middle point of either diagonal of a quadrilateral to the opposite vertices divide the quadrilateral into two equiv- alent parts.

BOOK IV. PLANE GEOMETRY. 232 Proposition VI. Theorem. 407. The area of a trapezoid is equal to half the sum of its bases multiplied by the altitude. A B CE F H O P a b b′ Let b and b′ be the bases and a the altitude of the trapezoid ABCH . To prove that the area of the ABCH = 1 2a(b +b′). Proof. Draw the diagonal AC. Then the area of the △ABC = 1 2a×b, and the area of the △AHC = 1 2a×b′. § 403 ∴ the area of ABCH = 1 2a(b +b′). Ax. 2 q.e.d. 408. Cor. The area of a trapezoid is equal to the product of the median by the altitude. § 190

AREAS OF POLYGONS. 233 A B C D E FG 409. Scholium. The area of an irregular polygon may be found by divid- ing the polygon into triangles, and by finding the area of each of these triangles separately. Or, we may draw the longest diagonal, and let fall perpendiculars upon this diagonal from the other vertices of the polygon. The

sum of the areas of the right triangles, rectangles, and trapezoids thus formed is the area of the polygon.

BOOK IV. PLANE GEOMETRY. 234 Proposition VII. Theorem. 410. The areas of two triangles which have an angle of the one equal to an angle of the other are to each other as the products of the sides including the equal angles. A B C D E Let the triangles ABC and ADE have the common angle A. To prove that △ABC △ADE = AB×AC AD×AE . Proof. DrawBE. Now △ABC △ABE = AC AE , and △ABE △ADE = AB AD. § 405 The products of the first members and of the second members of these equalities give △ABC △ADE = AB×AC AD×AE . q.e.d. Ex. 354. The areas of two triangles which have an angle of the one supplementary to an angle of the other are to each other as the products of the sides including the supplementary angles.

COMPARISON OF POLYGONS. 235 COMPARISON OF POLYGONS. Proposition VIII. Theorem. 411. The areas of two similar triangles are to each as the squares of any two homologous sides. A O C B A′ O′ C′ B′ Let the two similar triangles be ACB and A′C′B′. To prove that △ACB △A′C′B′ = AB 2 A′B′2 . Proof. Draw the altitudes CO and C′O′. Then △ACB △A′C′B′ = AB×CO A′B′×C′O′ = AB A′B′× CO C′O′ , § 405 (two △s are to each other as the products of their bases by their altitudes). But AB A′B′ = CO C′O′ . § 361 (the homologous altitudes of two similar △s have the same ratio as any two homologous sides). Substitute, in the above equality, for CO C′O′ its equal AB A′B′ ; then △ACB △A′C′B′ = AB A′B′× AB A′B′ = AB 2 A′B′2 . q.e.d. Ex. 355. Prove this proposition by§ 410.

BOOK IV. PLANE GEOMETRY. 236 Proposition IX. Theorem. 412. The areas of two similar polygons are to each other as the squares of any two homologous sides. A B C D E A′ B′ C′ D′ E′ Let S and S′ denote the areas of the two similar polygons ABC etc. and A′B′C′ etc. To prove that S :S′ =AB 2 :A′B′2. Proof. By drawing all the diagonals from any homologous vertices E and E′, the two similar polygons are divided into similar triangles. § 365 □ AB 2 A′B′2 = △ABE △A′B′E′ = (BE 2 B′E′2) = △BCE △B′C′E′ = etc. § 411 That is, △ABE △A′B′E′ = △BCE △B′C′E′ = △CDE △C′D′E′ . □ △ABE +△BCE +△CDE △A′B′E′ +△B′C′E′ +△C′D′E′ = △ABE △A′B′E′ = AB 2 A′B′2 . § 335 □S :S′ =AB 2 :A′B′2 q.e.d.

COMPARISON OF POLYGONS. 237 413. Cor. 1. The areas of two similar polygons are to each other as the squares of any two homologous lines. 414. Cor. 2. The homologous sides of two similar polygons have the same ratio as the square roots of their areas.

BOOK IV. PLANE GEOMETRY. 238 Proposition X. Theorem. 415. The square on the hypotenuse of a right triangle is equivalent to the sum of the squares on the two legs. A B C D E F G H K L Let BE, CH, AF , be squares on the three sides of the right triangle ABC. To prove that BE □CH +AF . Proof. Through A drawAL‖ to CE, and draw AD and CF . Since ∠sBAC, BAG, and CAH are rt. ∠s, CAG and BAH are straight lines. § 90 The △ABD =△FBC . § 143 For BD =BC, BA =BF , § 168 and ∠ABD = ∠FBC , Ax. 2 (each being the sum of a rt. ∠ and the ∠ABC). Now the rectangle BL is double the△ABD, (having the same base BD, and the same altitude, the distance between the ‖sAL and BD), and the square AF is double the△FBC , (having the same base FB , and

the same altitude AB).

EXERCISES. 239 □ the rectangle BL is equivalent to the square AF . Ax. 6 In like manner, by drawing AE and BK, it may be proved that the rectangle CL is equivalent to the square CH . Hence, the square BE, the sum of the rectangles BL and CL, is equivalent to the sum of the squares CH and AF . q.e.d. 416. Cor. The square on either leg of a right triangle is equivalent to the difference of the square on the hypotenuse and the square on the other leg. THEOREMS. A B C D E F K H G A C B L E G F H K D E H D G F A C B Ex. 356. The square constructed upon the sum of two straight lines is equivalent to the sum of the squares constructed upon these two lines, in- creased by twice the rectangle of these lines: Let AB and BC be the two straight lines, and AC their sum. Construct the squares ACGK andABED upon AC andAB, respectively. Prolong BE and DE until they meet KG and CG, respectively. Then we have the square EFGH , with sides each equal to BC. Hence, the square ACGK is the sum of the squares ABED and EFGH , and the rectangles DEHK and BCFE , the dimensions of which are equal to AB and BC.

BOOK IV. PLANE GEOMETRY. 240 Ex. 357. The square constructed upon the difference of two straight lines is equivalent to the sum of the squares constructed upon these two lines, di- minished by twice the rectangle of these lines. Let AB and AC be the two straight lines, and BC their difference. Con- struct the square ABFG upon AB, the square ACKH upon AC, and the square BEDC upon BC (as shown in the figure). Prolong ED to meet AG in L. The dimensions of the rectanglesLEFG andHKDL areAB andAC, and the squareBCDE is evidently the difference between the whole figure and the sum of these rectangles; that is, the square constructed upon BC is equivalent to the sum of the squares constructed upon AB andAC, diminished by twice the rectangle of AB and AC. Ex. 358. The difference between the squares constructed upon two straight lines is equivalent to the rectangle of the sum and difference of these lines. Let ABDE and BCFG be the squares constructed upon the two straight lines AB and BC. The difference between these squares is the polygon ACGFDE , which is composed of the rectanglesACHE andGFDH . Prolong AE andCH toI andK, respectively, makingEI andHK each equal toBC, and draw IK . The rectangles GFDH and EHKI are equal. The difference between the squares ABDE and BCGF is then equivalent to the rectangle ACKI , which has for dimensions AI, equal to AB +BC, and EH , equal to AB−BC. Ex. 359. The area of a rhombus is equal to half the product of its diago- nals. Ex. 360. Two isosceles triangles are equivalent if their legs are equal each to each, and the altitude of one is equal to half the base of the other. Ex. 361. The area of a circumscribed polygon is equal to half the product of its perimeter by the radius of the inscribed circle. Ex. 362. Two parallelograms are equal if two adjacent sides of the one are equal, respectively, to two adjacent sides of the other, and the included angles are supplementary.

EXERCISES. 241 Ex. 363. If ABC is a right triangle, C the vertex of the right angle, BD a line cutting AC in D, then BD 2 +AC 2 =AB 2 +DC 2 . Ex. 364. Upon the sides of a right triangle as homologous sides three similar polygons are constructed. Prove that the polygon upon the hypotenuse is equivalent to the sum of the polygons upon the legs. Ex. 365. If the middle points of two adjacent sides of a parallelogram are

joined, a triangle is formed which is equivalent to one eighth of the parallelo- gram. Ex. 366. If any point within a parallelogram is joined to the four vertices, the sum of either pair of triangles having parallel bases is equivalent to half the parallelogram. Ex. 367. Every straight line drawn through the intersection of the diag- onals of a parallelogram divides the parallelogram into two equal parts. Ex. 368. The line which joins the middle points of the bases of a trapezoid divides the trapezoid into two equivalent parts. Ex. 369. Every straight line drawn through the middle point of the median of a trapezoid cutting both bases divides the trapezoid into two equivalent parts. Ex. 370. If two straight lines are drawn from the middle point of either leg of a trapezoid to the opposite vertices, the triangle thus formed is equivalent to half the trapezoid. Ex. 371. The area of a trapezoid is equal to the product of one of the legs by the distance from this leg to the middle point of the other leg. Ex. 372. The figure whose vertices are the middle points of the sides of any quadrilateral is equivalent to half the quadrilateral.

BOOK IV. PLANE GEOMETRY. 242 PROBLEMS OF CONSTRUCTION. Proposition XI. Problem. 417. To construct a square equivalent to the sum of two given squares. R R′ B A C S Let R and R′ be two given squares. To construct a square equivalent to R′ +R. Construct the rt. ∠A. TakeAC equal to a side of R′, and AB equal to a side of R; and draw BC. Construct the square S, having each of its sides equal to BC. Then S is the square required. Proof. BC 2 □AC 2 +AB 2 , § 415 (the square on the hypotenuse of a rt. △ is equivalent to the sum of the squares on the two legs). □S □R′ +R. q.e.f. Ex. 373. If the perimeter of a rectangle is 72 feet, and the length is equal to twice the width, find the area. Ex. 374. How many tiles 9 inches long and 4 inches wide will be required to pave a path 8 feet wide surrounding a rectangular court 120 feet long and 36 feet wide? Ex. 375. The bases of a trapezoid are 16 feet and 10 feet; each leg is equal to 5 feet. Find the area of the trapezoid.

PROBLEMS OF CONSTRUCTION. 243 Proposition XII. Problem. 418. To construct a square equivalent to the difference of two given squares. R R′ B A C X S Let R be the smaller square and R′ the larger. To construct a square equivalent to R′−R. Construct the rt. ∠A. TakeAB equal to a side of R. FromB as a centre, with a radius equal to a side of R′, describe an arc cutting the line AX at C. Construct the square S, having each of its sides equal to AC. Then S is the square required. Proof. AC 2 □BC 2 −AB 2 , § 416 (the square on either leg of a rt. △ is equivalent to the difference of the square on the hypotenuse and the square on the other leg). □S □R′−R. q.e.f. Ex. 376. Construct a square equivalent to the sum of two squares whose sides are 3 inches and 4 inches. Ex. 377. Construct a square equivalent to the difference of two squares whose sides are 2 1 2 inches and 2 inches. Ex. 378. Find the side of a square equivalent to the sum of two squares whose sides are 24 feet and 32 feet. Ex. 379. Find the side of a square equivalent to the difference of two squares whose sides are 24 feet and 40 feet.

BOOK IV. PLANE GEOMETRY. 244 Ex. 380. A rhombus contains 100 square feet, and the length of one diagonal is 10 feet. Find the length of the other diagonal. Proposition XIII. Problem. 419. To construct a polygon similar to two given similar polygons and equivalent to their sum. R A B R′ A′ B′ R″ A″ B″ O P H Let R and R′ be

two similar polygons, and AB and A′B′ two homol- ogous sides. To construct a similar polygon equivalent to R +R′. Construct the rt. ∠P . TakePH equal to A′B′, and PO equal to AB. DrawOH, and take A″B″ equal to OH. Upon A″B″, homologous to AB, construct R″ similar to R. Then R″ is the polygon required. Proof. PO 2 +PH 2 =OH 2 . § 415 Put for PO , PH , and OH their equals AB, A′B′, and A″B″. Then AB 2 +A′B′2 =A″B″2. Now R R″ = AB 2 A″B″2 , and R′ R″ = A′B′2 A″B″2 . § 412 By addition, R +R′ R″ = AB 2 +A′B′2 A″B″2 = 1. Ax. 2 □R″ □R +R′. q.e.f.

PROBLEMS OF CONSTRUCTION. 245 Proposition XIV. Problem. 420. To construct a triangle equivalent to a given polygon. A B C D E F H I K Let ABCDHE be the given polygon. To construct a triangle equivalent to the given polygon. Let D, H, and E be any three consecutive vertices of the polygon. Draw the diagonal DE. FromH drawHF∥ to DE. Produce AE to meet HF at F , and draw DF . Again, drawCF , and draw DK∥ toCF to meet AF produced at K, and draw CK . In like manner continue to reduce the number of sides of the polygon until we obtain the△CIK . Then△CIK is the triangle required. Proof. The polygon ABCDF has one side less than the polygon ACBDHE , but the two polygons are equivalent. For the part ACBDE is common, and the△DEF □△DEH , § 404 (for the base DE is common, and their vertices F and H are in the line FH∥ to the base). In like manner it may be proved that ABCK □ABCDF , and CIK □ABCK . q.e.f.

BOOK IV. PLANE GEOMETRY. 246 Proposition XV. Problem. 421. To construct a square equivalent to a given parallelogram. A B C D a b R M N O P X Let ABCD be the parallelogram, b its base, and a its altitude. To construct a square equivalent to the / / ABCD . Upon a line MX takeMN equal to a, NO equal to b. Upon MO as a diameter, describe a semicircle. AtN erect NP ⊥ to MO, meeting the circumference at P . Then the square R, constructed upon a line equal to NP , is equivalent to the / / ACBD . Proof. MN :NP =NP :NO, § 370 (a ⊥ let fall from any point of a circumference to the diameter is the mean proportional between the segments of the diameter). □NP 2 =MN×NO =a×b. § 327 Therefore, R □ / / ABCD . q.e.f. 422. Cor. 1. A square may be constructed equivalent to a given triangle, by taking for its side the mean proportional between the base and half the altitude of the triangle. 423. Cor. 2. A square may be constructed equivalent to a given polygon, by first reducing the polygon to an equivalent triangle, and then constructing a square equivalent to the triangle.

PROBLEMS OF CONSTRUCTION. 247 Proposition XVI. Problem. 424. To construct a parallelogram equivalent to a given square, and having the sum of its base and altitude equal to a given line. R M C N P S Q Let R be the given square, and let the sum of the base and altitude of the required parallelogram be equal to the given line M N. To construct a / / equivalent to R, with the sum of its base and altitude equal to MN . Upon MN as a diameter, describe a semicircle. AtM erect MP , a⊥ to MN , equal to a side of the given square R. DrawPQ∥ to MN , cutting the circumference at S. DrawSC⊥ to MN . Any / / havingCM for its altitude and CN for its base is equivalent to R. Proof. SC =PM . §§ 104, 180 □SC 2 =PM 2 =R. MC :SC =SC :CN , § 370 (a ⊥ let fall from any point of a circumference to the diameter is the mean proportional between the segments of the diameter). Then SC 2 □MC×CN . § 327 q.e.f. Note. This

problem may be stated as follows: To construct two straight lines the sum and product of which are known.

BOOK IV. PLANE GEOMETRY. 248 Proposition XVII. Problem. 425. To construct a parallelogram equivalent to a given square, and having the difference of its base and altitude equal to a given line. R M C N B S R′ Let R be the given square, and let the difference of the base and altitude of the required parallelogram be equal to the given line M N. To construct a / / equivalent to R, with the difference of its base and altitude equal to MN . Upon the given line MN as a diameter, describe a circle. FromM drawMS , tangent to the⊙, and equal to a side of the given square R. Through the centre of the ⊙ draw SB intersecting the circumference at C and B. Then any / / , asR′, havingSB for its base andSC for its altitude, is equivalent to R. Proof. SB :SM =SM :SC, § 381 (if from a point without a ⊙ a secant and a tangent are drawn, the tangent is the mean proportional between the whole secant and the external segment). Then SM 2 □SB×SC, § 327 and the difference between SB and SC is the diameter of the ⊙, that is, MN . q.e.f. Note. This problem may be stated: To construct two straight lines the difference and product of which are known.

PROBLEMS OF CONSTRUCTION. 249 Proposition XVIII. Problem. 426. To construct a polygon similar to a given polygon P and equivalent to a given polygon Q. P A B Q P ′ A′ B′ Let P and Q be the two given polygons, and AB a side of P . To construct a polygon similar to P and equivalent to Q. Find squares equivalent to P and Q, § 423 and let m and n respectively denote their sides. Find A′B′, the fourth proportional to m, n, and AB. § 386 Upon A′B′, homologous to AB, construct P′ similar to P . Then P′ □Q. Proof. m :n =AB :A′B′. Const. □m2 :n2 =AB 2 :A′B′2 . § 338 But P □m2, and Q □n2. Const. □P :Q =m2 :n2 =AB 2 :A′B′2 . But P :P′ =AB 2 :A′B′2 . § 412 □P :Q =P :P′. Ax. 1 □P′ □Q. q.e.f. Ex. 381. To construct a square equivalent to the sum of any number of given squares. Ex. 382. To construct a polygon similar to two given similar polygons and equivalent to their difference.

BOOK IV. PLANE GEOMETRY. 250 Proposition XIX. Problem. 427. To construct a square which shall have a given ratio to a given square. R m n A B C D E F a b m n x y Let R be the given square, and n m the given ratio. To construct a square which shall be to R as n is to m. TakeAB equal to a side of R, and draw Ay, making any convenient angle with AB. On Ay takeAE equal to m, EF equal to n, and draw EB. DrawFC∥ to EB meeting AB produced at C. On AC as a diameter, describe a semicircle. AtB erect the⊥BD, meeting the semicircumference at D. Then BD is a side of the square required. Proof. Denote AB bya, BC byb, and BD byx. Now a :x =x :b. § 370 Therefore, a2 :x2 =a :b. § 337 But a :b =m :n. § 342 Therefore, a2 :x2 =m :n. Ax. 1 By inversion, x2 :a2 =n :m. § 331 Hence, the square on BD will have the same ratio to R as n has to m. q.e.f.

PROBLEMS OF CONSTRUCTION. 251 Proposition XX. Problem. 428. To construct a polygon similar to a given polygon and having a given ratio to it. R A B m n S A′ B′ Let R be the given polygon, and n m the given ratio. To construct a polygon similar to R, which shall be to R as n is to m. Construct a line A′B′, such that the square on A′B′ shall be to the square on AB as n is to m. § 427 Upon A′B′, as a side homologous to AB, construct the polygon S similar to R. § 391 Then S is the polygon required. Proof. S :R

=A′B′2 :AB 2 . § 412 But A′B′2 :AB 2 =n :m. Const. Therefore, S :R =n :m. Ax. 1 q.e.f. Ex. 383. To construct a triangle equivalent to a given triangle, and having one side equal to a given length l. Ex. 384. To transform a triangle into an equivalent right triangle. Ex. 385. To transform a given triangle into an equivalent right triangle, having one leg equal to a given length. Ex. 386. To transform a given triangle into an equivalent right triangle, having the hypotenuse equal to a given length.

BOOK IV. PLANE GEOMETRY. 252 PROBLEMS OF CONSTRUCTION. Ex. 387. To transform a triangleABC into an equivalent triangle, having a side equal to a given length l, and an angle equal to angle BAC. Upon AB (produced if necessary), take AD equal to l, drawBE∥ toCD, meeting AC (produced if necessary) at E. △BED □△BEC . Ex. 388. To transform a given triangle into an equivalent isosceles triangle, having the base equal to a given length. To construct a triangle equivalent to: Ex. 389. The sum of two given triangles. Ex. 390. The difference of two given triangles. Ex. 391. To transform a given triangle into an equivalent equilateral triangle. To transform a parallelogram into an equivalent: Ex. 392. Parallelogram having one side equal to a given length. Ex. 393. Parallelogram having one angle equal to a given angle. Ex. 394. Rectangle having a given altitude. To transform a square into an equivalent: Ex. 395. Equilateral triangle. Ex. 396. Right triangle having one leg equal to a given length. Ex. 397. Rectangle having one side equal to a given length. To construct a square equivalent to: Ex. 398. Five eighths of a given square. Ex. 399. Three fifths of a given pentagon.

EXERCISES. 253 Ex. 400. To divide a given triangle into two equivalent parts by a line through a given point P in one of the sides. Ex. 401. To find a point within a triangle, such that the lines joining this point to the vertices shall divide the triangle into three equivalent parts. Ex. 402. To divide a given triangle into two equivalent parts by a line parallel to one of the sides. Ex. 403. To divide a given triangle into two equivalent parts by a line perpendicular to one of the sides. PROBLEMS OF COMPUTATION. A B CD a a/2 h A B CD a b c h A B C D E a b c h Ex. 404. To find the area of an equilateral triangle in terms of its side. Denote the side by a, the altitude by h, and the area by S. Then h2 a2− a2 4 = 3a2 4 = a2 4 × 3. § 372 □h = a 2 √ 3. But S = a×h 2 . § 403 □S = a 2× a √ 3 2 = a2√ 3 4 .

BOOK IV. PLANE GEOMETRY. 254 Ex. 405. To find the area of a triangle in terms of its sides. By Ex. 312, h = 2 b √ s(s−a)(s−b)(s−c). Hence, S = b 2× 2 b √ s(s−a)(s−b)(s−c) § 403 = √ s(s−a)(s−b)(s−c). Ex. 406. To find the area of a triangle in terms of the radius of the circumscribed circle. If R denotes the radius of the circumscribed circle, and h the altitude of the triangle, we have, by§ 384, b×c = 2R×h. Multiply by a, and we have, a×b×c = 2R×a×h. But a×h = 2S. § 403 □a×b×c = 4R×S. □S = abc 4R . Show that the radius of the circumscribed circle is equal to abc 4S . Ex. 407. Find the area of a right triangle, if the length of the hypotenuse is 17 feet and the length of one leg is 8 feet. Ex. 408. Find the ratio of the altitudes of two equivalent triangles, if the base of one is three times that of the other. Ex. 409. The bases of a trapezoid are 8 feet and 10 feet, and the altitude is 6 feet. Find the base of the equivalent rectangle that has an equal altitude. Ex. 410. Find the area of a rhombus, if the sum of its diagonals is 12 feet, and their ratio is 3 : 5. Ex. 411. Find

the area of an isosceles right triangle, if the hypotenuse is 20 feet.

EXERCISES. 255 Ex. 412. In a right triangle the hypotenuse is 13 feet, one leg is 5 feet. Find the area. Ex. 413. Find the area of an isosceles triangle, if base = b, and leg = c. Ex. 414. Find the area of an equilateral triangle, if one side = 8 feet. Ex. 415. Find the area of an equilateral triangle, if the altitude = h. Ex. 416. A house is 40 feet long, 30 feet wide, 25 feet high to the roof, and 35 feet high to the ridge-pole. Find the number of square feet in its entire exterior surface. Ex. 417. The sides of a right triangle are as 3 : 4 : 5. The altitude upon the hypotenuse is 12 feet. Find the area. Ex. 418. Find the area of a right triangle, if one leg = a, and the altitude upon the hypotenuse = h. Ex. 419. Find the area of a triangle, if the lengths of the sides are 104 feet, 111 feet, and 175 feet. Ex. 420. The area of a trapezoid is 700 square feet. The bases are 30 feet and 40 feet, respectively. Find the altitude. Ex. 421. ABCD is a trapezium; AB = 87 feet, BC = 119 feet, CD = 41 feet, DA = 169 feet, AC = 200 feet. Find the area. Ex. 422. What is the area of a quadrilateral circumscribed about a circle whose radius is 25 feet, if the perimeter of the quadrilateral is 400 feet? What is the area of a hexagon that has a perimeter of 400 feet and is circumscribed about the same circle of 25 feet radius (Ex. 361)? Ex. 423. The base of a triangle is 15 feet, and its altitude is 8 feet. Find the perimeter of an equivalent rhombus, if the altitude is 6 feet. Ex. 424. Upon the diagonal of a rectangle 24 feet by 10 feet a triangle equivalent to the rectangle is constructed. What is its altitude? Ex. 425. Find the side of a square equivalent to a trapezoid whose bases are 56 feet and 44 feet, and each leg is 10 feet.

BOOK IV. PLANE GEOMETRY. 256 Ex. 426. Through a point P in the side AB of a triangle ABC, a line is drawn parallel to BC so as to divide the triangle into two equivalent parts. Find the value of AP in terms of AB. Ex. 427. What part of a parallelogram is the triangle cut off by a line from one vertex to the middle point of one of the opposite sides? Ex. 428. In two similar polygons, two homologous sides are 15 feet and 25 feet. The area of the first polygon is 450 square feet. Find the area of the second polygon. Ex. 429. The base of a triangle is 32 feet, its altitude 20 feet. What is the area of the triangle cut off by a line parallel to the base at a distance of 15 feet from the base? Ex. 430. The sides of two equilateral triangles are 3 feet and 4 feet. Find the side of an equilateral triangle equivalent to their sum. Ex. 431. If the side of one equilateral triangle is equal to the altitude of another, what is the ratio of their areas? Ex. 432. The sides of a triangle are 10 feet, 17 feet, and 21 feet. Find the areas of the parts into which the triangle is divided by the bisector of the angle formed by the first two sides. Ex. 433. In a trapezoid, one base is 10 feet, the altitude is 4 feet, the area is 32 square feet. Find the length of a line drawn between the legs parallel to the bases and distant 1 foot from the lower base. Ex. 434. The diagonals of a rhombus are 90 yards and 120 yards, respec- tively. Find the area, the length of one side, and the perpendicular distance between two parallel sides. Ex. 435. Find the number of square feet of carpet that are required to cover a triangular floor whose sides are, respectively, 26 feet, 35 feet, and 51 feet. Ex. 436. If the altitude h of a triangle is increased by a length m, how much must be taken from the base a that the area may remain the same?

EXERCISES. 257 Ex. 437. Find the area of a right triangle, having given the segments p, q, into which the hypotenuse is divided by a perpendicular drawn to the hypotenuse from the vertex of the right angle.

BOOK V. REGULAR POLYGONS AND CIRCLES. 429. Def. A regular polygon is a polygon which is both equilateral and equiangular. The equilateral triangle and the square are examples. Proposition I. Theorem. 430. An equilateral polygon inscribed in a circle is a regular polygon. A B C D F Let ABC etc. be an equilateral polygon inscribed in a circle. To prove that the polygon ABC etc. is a regular polygon. Proof. The arcs AB, BC, CD, etc., are equal. § 243 Hence, arcs ABC, BCD , etc., are equal. Ax. 2 Therefore, arcs CFA , DFB , etc., are equal. Ax. 3 Therefore, ∠sA, B, C, etc., are equal. § 289 Therefore, the polygon ABC etc. is a regular polygon, being equilateral and equiangular. § 429 q.e.d.

REGULAR POLYGONS AND CIRCLES. 259 Proposition II. Theorem. 431. A circle may be circumscribed about, and a circle may be inscribed in, any regular polygon. A B C D E F O Let ABCDE be a regular polygon. 1. To prove that a circle may be circumscribed about ABCDE . Proof. Let O be the centre of the circle which may be passed through A, B, and C. § 258 DrawOA, OB, OC, and OD. Then ∠ABC = ∠BCD , § 429 and ∠OBC = ∠OCB . § 145 By subtraction, ∠OBA = ∠OCD. Ax. 3 The△sOBA and OCD are equal. § 143 For ∠OBA = ∠OCD, OB =OC, § 217 and AB =CD. § 429 □OA =OD. § 128 □ the circle passing through A, B, C, passes through D. In like manner it may be proved that the circle passing through B, C, and D also passes through E; and so on. Therefore, the circle described from O as a centre, with a radius OA, will be circumscribed about the polygon. § 231 2. To prove that a circle may be inscribed in ABCDE .

BOOK V. PLANE GEOMETRY. 260 Proof. Since the sides of the regular polygon are equal chords of the circum- scribed circle, they are equally distant from the centre. § 249 Therefore, the circle described from O as a centre, with the distance from O to a side of the polygon as a radius, will be inscribed in the polygon (§ 232). q.e.d. 432. Def. The radius of the circumscribed circle,OA, is called the radius of the polygon. 433. Def. The radius of the inscribed circle, OF , is called the apothem of the polygon. 434. Def. The common centre, O, of the circumscribed and inscribed circles is called the centre of the polygon. 435. Def. The angle between radii drawn to the extremities of any side is called the angle at the centre of the polygon. By joining the centre to the vertices of a regular polygon, the polygon can be decomposed into as many equal isosceles triangles as it has sides. 436. Cor. 1. The angle at the centre of a regular polygon is equal to four right angles divided by the number of sides of the polygon. Hence, the angles at the centre of any regular polygon are all equal. 437. Cor. 2. The radius drawn to any vertex of a regular polygon bisects the angle at the vertex. 438. Cor. 3. The angle at the centre of a regular polygon and an interior angle of the polygon are supplementary. For ∠sFOB and FBO are complementary. § 135 □ their doubles AOB and FBC are supplementary. Ax. 6

REGULAR POLYGONS AND CIRCLES. 261 Proposition III. Theorem. 439. If the circumference of a circle is divided into any number of equal arcs, the chords joining the successive points of division form a regular in- scribed polygon; and the tangents

drawn at the points of division form a regular circumscribed polygon. A B C D E F G HJ K Suppose the circumference divided into equal arcs AB, BC, etc. Let AB, BC, etc., be the chords, F BG, GCH, etc., the tangents. 1. To prove that ABCDE is a regular polygon. Proof. The sides AB, BC, CD, etc., are equal. § 241 Therefore, the polygon is regular. § 430 2. To prove that To prove that FGHIK is a regular polygon. Proof. The△sAFB , BGC, CHD , etc., are all equal isosceles triangles. §§ 295,139 □ ∠ sF , G, H, etc., are equal, and FB , BG, GC, etc., are equal. □FG =GH =HI , etc. Ax. 6 □FGHIK is a regular polygon. § 429 q.e.d. 440. Cor. 1. Tangents to a circle at the vertices of a regular inscribed polygon form a regular circumscribed polygon of the same number of sides as the inscribed polygon.

BOOK V. PLANE GEOMETRY. 262 A′ B′ C′ D′ E′ O A B C D E M N PQ R A B CD E F H K D K I C L H M G A E F B O 441. Cor. 2. Tangents to a circle at the middle points of the arcs subtended by the sides of a regular inscribed polygon form a circumscribed regular polygon, whose sides are parallel to the sides of the inscribed polygon and whose vertices lie on the radii (prolonged) of the inscribed polygon. For two corresponding sides, AB and A′B′, are perpendicular to OM (§§ 248, 254), and are parallel (§ 104); and the tangents MB′ andNB′, inter- secting at a point equidistant from OM and ON (§ 261), intersect upon the bisector of the ∠MON (§ 162); that is, upon the radius OB. 442. Cor. 3. If the vertices of a regular inscribed polygon are joined to the middle points of the arcs subtended by the sides of the polygon, the joining lines form a regular inscribed polygon of double the number of sides. 443. Cor. 4. Tangents at the middle points the arcs between adjacent points of contact of the sides of a regular circumscribed polygon form a regular circumscribed polygon of double the number of sides. 444. Cor. 5. The perimeter of an inscribed polygon is less than the perime- ter of an inscribed polygon of double the number of sides; and the perimeter of a circumscribed polygon is greater than the perimeter of a circumscribed polygon of double the number of sides. For two sides of a triangle are together greater than the third side. § 138

REGULAR POLYGONS AND CIRCLES. 263 Proposition IV. Theorem. 445. Two regular polygons of the same number of sides are similar. A B C D E Q A′ B′ C′ D′ E′ Q′ Let Q and Q′ be two regular polygons, each having n sides. To prove that Q and Q′ are similar. Proof. The sum of the interior ∠s of each polygon is equal to (n− 2)2 rt. ∠s, § 205 (the sum of the interior ∠s of a polygon is equal to 2 rt. ∠s taken as many times less two as the polygon has sides). Each angle of either polygon = (n− 2)2 rt. ∠s n , § 206 (for the ∠s of a regular polygon are all equal, and hence each ∠ is equal to the sum of the ∠s divided by their number). Hence, the two polygons Q and Q′ are mutually equiangular. Since AB =BC, etc., and A′B′ =B′C′, etc., § 429 AB :A′B′ =BC :B′C′, etc. Hence, the two polygons have their homologous sides proportional. Therefore the two polygons are similar. § 351 q.e.d. 446. Cor. The areas of two regular polygons of the same number of sides are to each other as the squares of any two homologous sides. § 412

BOOK V. PLANE GEOMETRY. 264 Proposition V. Theorem. 447. The perimeters of two regular polygons of the same number of sides are to each other as the radii of their circumscribed circles, and also as the radii of their inscribed circles. A B C D E

O M A′ B′ C′ D′ E′ O′ M ′ Let P and P ′ denote the perimeters, O and O′ the centres, of the two regular polygons. FromO, O′ drawOA, O′A′, OB, O′B′, and the⊥sOM, O′M′. To prove that P :P′ =OA :O′A′ =OM :O′M′. Proof. Since the polygons are similar, § 445 P :P′ =AB :A′B′. § 364 The△sOAB and O′A′B′ are isosceles. § 431 Now ∠O = ∠O′, § 436 and OA :OB =O′A′ :O′B′. ∴ the△sOAB and O′A′B′ are similar. § 357 ∴AB :A′B′ =OA :O′A′. § 351 Also, AB :A′B′ =OM :O′M′. § 361 ∴P :P′ =OA :O′A′ =OM :O′M′. Ax. 1 q.e.d. 448. Cor. The areas of two regular polygons of the same number of sides are to each other as the squares of the radii of the circumscribed circles, and of the inscribed circles. § 413

REGULAR POLYGONS AND CIRCLES. 265 Proposition VI. Theorem. 449. If the number of sides of a regular inscribed polygon is indefinitely increased, the apothem of the polygon approaches the radius of the circle as its limit. O A B P Let AB be a side and OP the apothem of a regular polygon of n sides inscribed in the circle whose radius is OA. To prove that OP approachesOA as a limit, when n increases indefinitely. Proof. OP <OA , § 97 and OA−OP <AP . § 138 ∴OA−OP <AB , which is twice AP . § 245 Now, if n is taken sufficiently great, AB, and consequently OA−OP , can be made less than any assigned value, however small, but cannot be made zero. Since OA−OP can be made less than any assigned value by increasing n, but cannot be made zero, OA is the limit of OP by the test for a limit. § 275 q.e.d. 450. Cor. If the number of sides of a regular inscribed polygon is indefi- nitely increased, the square of the apothem approaches the square of the radius of the circle as a limit. For OA 2 −OP 2 =AP 2 . § 372 But by taking n sufficiently great, AB and consequently AP , the half of AB, can be made less than any assigned value.

BOOK V. PLANE GEOMETRY. 266 Therefore, AP 2 , the product of AP by AP , can be made less than any assigned value; for the product of two finite factors approaches zero as a limit, if either factor approaches zero as a limit (§ 276); and for a still stronger reason, the product approaches zero as a limit, if each of the factors approaches zero as a limit. Proposition VII. Theorem. 451. An arc of a circle is less than any line which envelops it and has the same extremities. A B C D E F Let ACB be an arc of a circle, and AB its chord. To prove that the arc ACB is less than any other line which envelops this arc and terminates at A and B. Proof. Of all the lines that can be drawn, each to include the areaACB between itself and the chord AB, there must be at least one shortest line; for all the lines are not equal. Now the enveloping lineADB cannot be the shortest; for drawingECF tangent to the arc ACB at C, the line AECFB <AEDFB , since ECF <EDF . § 49 In like manner it can be shown that no other enveloping line can be the shortest. Therefore, ACB is the shortest. q.e.d. 452. Cor. 1. The circumference of a circle is less than the perimeter of any polygon circumscribed about it. 453. Cor. 2. Any convex curve is less than the perimeter of a polygon circumscribed about it.

REGULAR POLYGONS AND CIRCLES. 267 Proposition VIII. Theorem. 454. The circumference of a circle is the limit which the perimeters of regular inscribed polygons and of similar circumscribed polygons approach, if the number of sides of the polygons is indefinitely increased; and the area of a circle is the limit which the areas of these polygons approach. O A B A′ B′C D Let P and P ′ denote the lengths of

the perimeters, AB and A′B′ two homologous sides, R and R′ the radii, of the polygons, and C the circumference of the circle. To prove thatC is the limit of P and of P′, if the number of sides of the polygons is indefinitely increased. Proof. Since the polygons are similar by hypothesis, P′ :P =R′ :R. § 447 Therefore, P′−P :P =R′−R :R. § 333 Whence, R(P′−P) = P (R′−R). § 327 Therefore, P′−P = P R(R′−R). NowP is always less than C. § 273 □P′−P < C R(R′−R). ButR′−R, which is less than A′C (§ 138), can be made less than any assigned quantity by increasing the number of sides of the polygons; and therefore C R(R′−R) can be made less than any assigned quantity. § 276 Hence, P′−P can be made less than any assigned quantity.

BOOK V. PLANE GEOMETRY. 268 Since P′ is always greater than C (§ 452), and P is always less than C (§ 273), the difference between C and either P′ or P is less than the difference P′−P , and consequently can be made less than any assigned quantity, but cannot be made zero. Therefore, C is the common limit of P′ and P . § 275 Let K denote the area of the circle, S the area of the inscribed poly- gon, and S′ the area of the circumscribed polygon. 2. To prove that K is the limit of S and S′. Proof. S′ :S =R′2 :R2. § 448 By division, S′−S :S =R′2−R2 :R2. § 333 Whence S′−S = S R2 (R′2−R2). NowK is always greater than S. Ax. 8 Therefore, S′−S <K R2 (R′2−R2). But R′2−R2, which is equal to (R′ +R)(R′−R), can be made less than any assigned quantity; and therefore K R2 (R′2−R2) can be made less than any assigned quantity. § 276 Hence, S′−S can be made less than any assigned quantity. Since S′>K always, and S <K always (Ax. 8), the difference between K and either S′ or S is less than the difference S′−S, and consequently can be made less than any assigned quantity, but cannot be made zero. Therefore, K is the common limit of S′ and S. § 275 q.e.d.

REGULAR POLYGONS AND CIRCLES. 269 Proposition IX. Theorem. 455. Two circumferences have the same ratio as their radii. Q Q ′ Let C and C′ be the circumferences, R and R′ the radii, of the two circles Q and Q′. To prove that C :C′ =R :R′. Proof. Inscribe in the⊙s two similar regular polygons, and denote their perime- ters by P and P′. Then P :P′ =R :R′. § 447 Conceive the number of sides of these regular polygons to be indefinitely in- creased, the polygons continuing similar. Then P and P′ will have C and C′ as limits. § 454 But P :P′ will always be equal to R :R′. § 447 □C :C′ =R :R′. § 285 q.e.d. 456. Cor. The ratio of the circumference of a circle to its diameter is constant. For C :C′ =R :R′. § 455 □C :C′ = 2R : 2R′. § 340 By alternation, C : 2R =C′ : 2R′. § 330 457. Def. The constant ratio of the circumference of a circle to its diameter is represented by the Greek letter π.

BOOK V. PLANE GEOMETRY. 270 458. Cor. π = C 2R. □C = 2πR. Proposition X. Theorem. 459. The area of a regular polygon is equal to half the product of its apothem by its perimeter. A B C DE F O M Let P represent the perimeter, R the apothem, and S the area of the regular polygon ABC etc. To prove that S = 1 2R×P . Proof. Draw the radii OA, OB, OC, etc. The polygon is divided into as many △s as it has sides. The apothem is the common altitude of these △s, and the area of each △ = 1 2R multiplied by the base. § 403 Hence, the area of all the △s is equal to 1 2R multiplied by the sum of all the bases. But the sum of the areas of all the △s is equal to the area of the polygon. Ax. 9 And the sum of all the bases of the △s is equal to the perimeter of the polygon. Ax. 9 □S = 1 2R×P . q.e.d. 460. Def. In different circles

similar arcs, similar sectors, and similar segments are such as correspond to equal angles at the centre .

REGULAR POLYGONS AND CIRCLES. 271 Proposition XI. Theorem. 461. The area of a circle is equal to half the product of its radius by its circumference. O M A B C D E Let R represent the radius, C the circumference, and S the area, of the circle whose centre is O. To prove that S = 1 2R×C. Proof. Circumscribe any regular polygon about the circle, and denote its perimeter by P , and its area by S′. Then S′ = 1 2R×P . § 459 Conceive the number of sides of the polygon to be indefinitely increased. Then P approachesC as its limit, § 454 1 2R×P approaches 1 2R×C as its limit, § 279 and S′ approachesS as its limit. § 454 But S′ = 1 2R×P , always. § 459 □S = 1 2R×C. § 284 q.e.d. 462. Cor. 1. The area of a sector is equal to half the product of its radius by its arc. For the sector and its arc are like parts of the circle and its circumference, respectively.

BOOK V. PLANE GEOMETRY. 272 463. Cor. 2. The area of a circle is equal to π times the square of its radius. For the area of the⊙ = 1 2R×C = 1 2R× 2πR =πR2. 464. Cor. 3. The areas of two circles are to each other as the squares of their radii. For, ifS and S′ denote the areas, and R and R′ the radii, S :S′ =πR2 :πR′2 =R2 :R′2. 465. Cor. 4. Similar arcs are to each other as their radii; similar sectors are to each other as the squares of their radii.

REGULAR POLYGONS AND CIRCLES. 273 Proposition XII. Theorem. 466. The areas of two similar segments are to each other as the squares of their radii. A B C P A′ B′ C′ P ′ Let AC and A′C′ be the radii of the two similar sectors ACB and A′C′B′, and let ABP and A′B′P ′ be the corresponding segments. To prove that ABP :A′B′P′ =AC 2 :A′C′2 . Proof. Sector ACB : Sector A′C′B′ =AC 2 :A′C′2 . § 465 The△sACB and A′C′B′ are similar. § 357 □△ACB :△A′C′B′ =AC 2 :A′C′2 . § 411 □ sector ACB : sector A′C′B′ =△ACB :△A′C′B′. Ax. 1 □ sector ACB :△ACB = sector A′C′B′ :△A′C′B′. § 330 □ sector ACB−△ACB sector A′C′B′−△A′C′B′ = △ACB △A′C′B′ = AC 2 A′C′2 . § 333 That is, ABP :A′B′P′ =AC 2 :A′C′2 . q.e.d.

BOOK V. PLANE GEOMETRY. 274 PROBLEMS OF CONSTRUCTION. Proposition XIII. Problem. 467. To inscribe a square in a given circle. O A B C D Let O be the centre of the given circle. To inscribe a square in the given circle. Draw two diameters AC and BD⊥ to each other. DrawAB, BC, CD, and DA. Then ABCD is the square required. Proof. The ∠sABC, BCD , etc., are rt. ∠s, § 290 (each being inscribed in a semicircle), and the sides AB, BC, etc., are equal, § 241 (in the same ⊙ equal arcs are subtended by equal chords). Hence the quadrilateral ABCD is a square. § 168 q.e.f. 468. Cor. By bisecting the arcs AB, BC, etc., a regular polygon of eight sides may be inscribed in the circle; and, by continuing the process, regular polygons of sixteen, thirty-two, sixty-four, etc., sides may be inscribed. Ex. 438. The area of a circumscribed square is equal to twice the area of the inscribed square. Ex. 439. The area of a circular ring is equal to that of a circle whose diameter is a chord of the outer circle tangent to the inner circle.

PROBLEMS OF CONSTRUCTION. 275 Proposition XIV. Problem. 469. To inscribe a regular hexagon in a given circle. O A B C D E F Let O be the centre of the given

circle. To inscribe a regular hexagon in the given circle. FromO draw any radius, as OC. FromC as a centre, with a radius equal to OC, describe an arc intersecting the circumference at F . DrawOF and CF . Then CF is a side of the regular hexagon required. Proof. The△OFC is equiangular, § 146 (since it is equilateral by construction). Hence, the ∠FOC is 1 3 of 2 rt. ∠s, or 1 6 of 4 rt. ∠s. § 136 ∴ the arc FC is 1 6 of the circumference, and the chord FC is a side of a regular inscribed hexagon. Hence, to inscribe a regular hexagon apply the radius six times as a chord.q.e.f. 470. Cor. 1. By joining the alternate vertices A, C, D, an equilateral triangle is inscribed in the circle. 471. Cor. 2. By bisecting the arcs AB, BC, etc., a regular polygon of twelve sides may be inscribed in the circle; and, by continuing the process, regular polygons of twenty-four, forty-eight, etc., sides may be inscribed.

BOOK V. PLANE GEOMETRY. 276 Proposition XV. Problem. 472. To inscribe a regular decagon in a given circle. O B C F S Let O be the centre of the given circle. To inscribe a regular decagon in the given circle. Draw any radius OC, and divide it in extreme and mean ratio, so that OC shall be to OS as OS is to SC. § 389 FromC as a centre, with a radius equal to OS, describe an arc intersecting the circumference at B. DrawBC. Then BC is a side of the regular decagon required. Proof. DrawBS and BO. Now OC :OS =OS :SC, Const. and BC =OS. Const. ∴OC :BC =BC :SC. Moreover, ∠OCB = ∠SCB . Iden. Hence, the△sOCB and BCS are similar. § 357 But the△OCB is isosceles. § 217 ∴△BCS , which is similar to the △OCB , is isosceles, and CB =BS =SO. § 120 ∴ the△SOB is isosceles, and the ∠O = ∠SBO. § 145

PROBLEMS OF CONSTRUCTION. 277 But the ext. ∠CSB = ∠O + ∠SBO = 2∠O. § 137 Hence, ∠SCB = 2∠O, and ∠OBC = 2∠O. ∴ the sum of the ∠s of the△OCB = 5∠O = 2rt. ∠s, and ∠O = 1 5 of 2 rt. ∠s, or 1 10 of 4 rt. ∠s. Therefore, the arc BC is 1 10 of the circumference, and the chord BC is a side of a regular inscribed decagon. Therefore, to inscribe a regular decagon, divide the radius internally in extreme and mean ratio, and apply the greater segment ten times as a chord. q.e.f. 473. Cor. 1. By joining the alternate vertices of a regular inscribed decagon, a regular pentagon is inscribed. 474. Cor. 2. By bisecting the arcs BC, CF , etc., a regular polygon of twenty sides may be inscribed in the circle; and, by continuing the process, regular polygons of forty, eighty, etc., sides may be inscribed. If R denotes the radius of a regular inscribed polygon, r the apothem, a one side, A an interior angle, and C the angle at the centre, show that Ex. 440. In a regular inscribed triangle a = R √ 3, r = 1 2R, A = 60◦, C = 120◦. Ex. 441. In an inscribed square a =R √ 2,r = 1 2R √ 2,A = 90◦,C = 90◦. Ex. 442. In a regular inscribed hexagon a = R, r = 1 2R √ 3, A = 120◦, C = 60◦. Ex. 443. In a regular inscribed decagon a = R(√ 5− 1) 2 , r = 1 4R √ 10 + 2 √ 5, A= 144◦, C = 36◦.

BOOK V. PLANE GEOMETRY. 278 Proposition XVI. Problem. 475. To inscribe in a given circle a regular pentedecagon, or polygon of fifteen sides. Q A B C D E F H Let Q be the given circle. To inscribe in Q a regular pentedecagon. DrawEH equal to the radius of the circle, and EF equal to a side of the regular inscribed decagon. § 472 DrawFH . Then FH is a side of the regular pentedecagon required. Proof. The arc EH is 1 6 of the circumference, § 469 and the arc EF is 1 10 of the circumference. Const. Hence, the arc FH is 1 6− 1 10, or 1 15, of the circumference. And the chord FH is a

side of a regular inscribed pentedecagon. By applying FH fifteen times as a chord, we have the polygon required. q.e.f. 476. Cor. By bisecting the arcs FH , HA, etc., a regular polygon of thirty sides may be inscribed; and, by continuing the process, regular polygons of sixty, one hundred twenty, etc., sides may be inscribed.

PROBLEMS OF CONSTRUCTION. 279 Proposition XVII. Problem. 477. To inscribe in a given circle a regular polygon similar to a given regular polygon. O′ A′ B′ C′ D′ E′ F ′ O A B C D E F Let ABC etc. be the given regular polygon, and O′ the centre of the given circle. To inscribe in the circle a regular polygon similar to ABC etc. FromO, the centre of the given polygon, drawOD and OC. FromO′, the centre of the given circle, drawO′C′ and O′D′, making the ∠O′ equal to the ∠O. DrawC′D′. Then C′D′ is a side of the regular polygon required. Proof. Each polygon has as many sides as the ∠O, or ∠O′, is contained times in 4 rt. ∠s. Therefore, the polygon C′D′E′ etc. is similar to the polygon CDE etc., § 445 (two regular polygons of the same number of sides are similar). q.e.f. Ex. 444. The area of an inscribed regular octagon is equal to that of the rectangle whose sides are equal to the sides of the inscribed and the circumscribed squares.

BOOK V. PLANE GEOMETRY. 280 Proposition XVIII. Problem. 478. Given the side and the radius of a regular inscribed polygon, to find the side of the regular inscribed polygon of double the number of sides. O A B C D H Let AB be a side of the regular inscribed polygon. To findAD, a side of the regular inscribed polygon of double the number of sides. Denote the radius by R, and AB bya. FromD drawDH through the centre O, and draw OA, AH. DH is⊥ to AB at its middle point C. § 161 In the rt.△OCA, OC 2 =R2− 1 4a2. § 372 Therefore, OC = √ R2− 1 4a2, and DC =R− √ R2− 1 4a2. The ∠DAH is a rt. ∠. § 290 In the rt.△DAH, AD 2 =DH×DC. § 367 But DH = 2R, and DC =R− √ R2− 1 4a2. □AD = √ 2R(R− √ R2− 1 4a2) = √ R(2R− √ 4R2−a2). q.e.f. 479. Cor. If R = 1, AD = √ 2− √ 4−a2.

PROBLEMS OF CONSTRUCTION. 281 Proposition XIX. Problem. 480. To find the numerical value of the ratio of the circumference of a circle to its diameter. Let C be the circumference, when the radius is unity. To find the numerical value of π. By§ 458, 2πR =C. □π = 1 2C when R = 1. Let S6 be the length of a side of a regular polygon of 6 sides, S12 of 12 sides, and so on. If R = 1, by§ 469, S6 = 1 and by§ 479 we have Form of Computation. Length of Side. Length of Perimeter. S12 = √ 2− √ 4− 12 0.51763809 6 .21165708 S24 = √ 2− √ 4− (0.51763809)2 0.26105238 6 .26525722 S48 = √ 2− √ 4− (0.26105238)2 0.13080626 6 .27870041 S96 = √ 2− √ 4− (0.13080626)2 0.06543817 6 .28206396 S192 = √ 2− √ 4− (0.06543817)2 0.03272346 6 .28290510 S384 = √ 2− √ 4− (0.03272346)2 0.01636228 6 .28311544 S768 = √ 2− √ 4− (0.01636228)2 0.00818121 6 .28316941 □C = 6.28317 approximately; that is, π = 3.14159 nearly. q.e.f. 481. Scholium. π is incommensurable. We generally take π = 3.1416, and 1 π = 0.31831.

BOOK V. PLANE GEOMETRY. 282 MAXIMA AND MINIMA. 482. Def. Among geometrical magnitudes which satisfy given conditions, the greatest is called the maximum; and the smallest is called the minimum. Thus, the diameter of a circle is the maximum among all chords; and the perpendicular is the minimum among all lines drawn to a given line from a given external point. 483. Def. Isoperimetric

polygons are polygons which have equal perimeters. Proposition XX. Theorem. 484. Of all triangles having two given sides, that in which these sides in- clude a right angle is the maximum. A B CD D E E Let the triangles ABC and EBC have the sides AB and BC equal to EB and BC, respectively; and let the angle ABC be a right angle. To prove that △ABC >△EBC . Proof. FromE draw the altitude ED. The△sABC and EBC , having the same base, BC, are to each other as their altitudes AB and ED. § 405 Now EB >ED . § 97 But EB =AB. Hyp. □AB >ED. □△ABC >△EBC . § 405 q.e.d.

MAXIMA AND MINIMA. 283 Proposition XXI. Theorem. 485. Of all isoperimetric triangles having the same base the isosceles tri- angle is the maximum. A B C D E F H K M P Let the △sACB and ADB have equal perimeters, and let AC and CB be equal, and AD and DB be unequal. To prove that△ACB >△ADB. Proof. Produce AC to H, making CH =AC; and draw HB. Produce HB, take DP equal to DB, and draw AP . DrawCE and DF⊥ to AB, and CK and DM∥ to AB. The ∠ABH is a right ∠, for it may be inscribed in the semicircle whose centre is C and radius CA. § 290 ADP is not a straight line, for then the ∠sDBA and DAB would be equal, being complements of the equal ∠sDBM andDPM , respectively; and DA andDB would be equal (§ 147), which is contrary to the hypothesis. Hence, AP <AD +DP , □<AD +DB, □<AC +CB, □<AH . □BH >BP . § 102 □CE(= 1 2BH) >DF (= 1 2BP). Ax. 7 Therefore, △ACB >△ADB. § 405 q.e.d.

BOOK V. PLANE GEOMETRY. 284 Proposition XXII. Theorem. 486. Of all polygons with sides all given but one, the maximum can be inscribed in a semicircle which has the undetermined side for its diameter. A B C D EM N Let ABCDE be the maximum of polygons with sides AB, BC, CD, DE, and the extremities A and E on the straight line M N. To prove that ABCDE can be inscribed in a semicircle. Proof. From any vertex, as C, draw CA and CE. The△ACE must be the maximum of all△s having the sides CA andCE, and the third side on MN ; otherwise by increasing or diminishing the ∠ACE, keeping the lengths of the sides CA and CE unchanged, but sliding the extremities A and E along the line MN , we could increase the △ACE, while the rest of the polygon would remain unchanged; and therefore increase the polygon. But this is contrary to the hypothesis that the polygon is the maximum polygon. Hence, the△ACE is the maximum of△s that have the sides CA and CE. Therefore, the ∠ACE is a right angle. § 484 Therefore, C lies on the semicircumference. § 290 Hence, every vertex lies on the circumference; that is, the maximum polygon can be inscribed in a semicircle having the undetermined side for a diameter. q.e.d.

MAXIMA AND MINIMA. 285 Proposition XXIII. Theorem. 487. Of all polygons with given sides, that which can be inscribed in a circle is the maximum. A B C D E H A′ B′ C′ D′ E′ H ′ Let ABCDE be a polygon inscribed in a circle, and A′B′C′D′E′ be a polygon, equilateral with respect to ABCDE , which cannot be inscribed in a circle. To prove that that ABCDE >A′B′C′D′E′. Proof. Draw the diameter AH, and draw CH and DH. Upon C′D′ construct the△C′H′D′ =△CHD , and draw A′H′. Since, by hypothesis, a ⊙ cannot pass through all the vertices of A′B′C′D′E′, one or both of the parts ABCH , AEDH must be greater than the corresponding part of A′B′C′H′D′E′. § 486 If either of these parts is not greater than its corresponding part, it is equal to it, § 486 (for ABCH and AEDH are the maxima of polygons that have sides equal to AB, BC, CH ,

and AE, ED, DH, respectively, and the remaining side undetermined). □ABCHDE >A′B′C′H′D′E′. Ax. 4 Take away from the two figures the equal△sCHD and C′H′D′. Then ABCDE >A′B′C′D′E′. Ax. 5 q.e.d.

BOOK V. PLANE GEOMETRY. 286 Proposition XXIV. Theorem. 488. Of isoperimetric polygons of the same number of sides, the maximum is equilateral. A B C D K Let ABCD etc. be the maximum of isoperimetric polygons of any given number of sides. To prove that AB, BC, CD, etc., are equal. Proof. DrawAC. The△ABC must be the maximum of all the △s which are formed upon AC with a perimeter equal to that of △ABC. Otherwise a greater△AKC could be substituted for△ABC, without changing the perimeter of the polygon. But this is inconsistent with the hypothesis that the polygon ABCD etc. is the maximum polygon. □ the△ABC is isosceles. § 485 □AB =BC. In like manner it may be proved that BC =CD, etc. q.e.d. 489. Cor. The maximum of isoperimetric polygons of the same number of sides is a regular polygon. For the maximum polygon is equilateral (§ 488), and can be inscribed in a circle (§ 487), and is, therefore, regular. § 430

MAXIMA AND MINIMA. 287 Proposition XXV. Theorem. 490. Of isoperimetric regular polygons, that which has the greatest number of sides is the maximum. A B C D E Q Q′ Let Q be a regular polygon of three sides, and Q′ a regular polygon of four sides, and let the two polygons have equal perimeters. To prove that Q′ is greater than Q. Proof. DrawCD from C to any point in AB. Invert the△CDA and place it in the position DCE , letting D fall at C, C at D, and A at E. The polygon DBCE is an irregular polygon of four sides, which by construction has the same perimeter as Q′, and the same area as Q. Then the irregular polygon DBCE of four sides is less than the isoperimetric regular polygon Q′ of four sides. § 489 In like manner it may be shown that Q′ is less than an isoperimetric regular polygon of five sides, and so on. q.e.d. Ex. 445. Of all equivalent parallelograms that have equal bases, the rectangle has the minimum perimeter. Ex. 446. Of all equivalent rectangles, the square has the minimum perime- ter. Ex. 447. Of all triangles that have the same base and the same altitude, the isosceles has the minimum perimeter. Ex. 448. Of all triangles that can be inscribed in a given circle, the equilateral is the maximum and has the maximum perimeter.

BOOK V. PLANE GEOMETRY. 288 Proposition XXVI. Theorem. 491. Of regular polygons having a given area, that which has the greatest number of sides has the least perimeter. Q Q′ Q″ Let Q and Q′ be regular polygons having the same area, and let Q′ have the greater number of sides. To prove the perimeter of Q >the perimeter of Q′. Proof. Let Q″ be a regular polygon having the same perimeter as Q′, and the same number of sides as Q. Then Q′ >Q″ § 490 (of isoperimetric regular polygons, that which has the greatest number of sides is the maximum). But Q □Q′. Hyp. □Q >Q″. □ the perimeter of Q >the perimeter of Q″. But the perimeter of Q′ = the perimeter of Q″. Hyp. □ the perimeter of Q >the perimeter of Q′. q.e.d. Ex. 449. To inscribe in a semicircle the maximum rectangle. Ex. 450. Of all polygons of a given number of sides which may be inscribed in a given circle, that which is regular has the maximum area and the maximum perimeter. Ex. 451. Of all polygons of a given number of sides which may be cir- cumscribed about a given circle, that which is

regular has the minimum area and the minimum perimeter.

EXERCISES. 289 THEOREMS. Ex. 452. Every equilateral polygon circumscribed about a circle is regular if it has an odd number of sides. Ex. 453. Every equiangular polygon inscribed in a circle is regular if it has an odd number of sides. Ex. 454. Every equiangular polygon circumscribed about a circle is reg- ular. Ex. 455. The side of a circumscribed equilateral triangle is equal to twice the side of the similar inscribed triangle. Ex. 456. The apothem of an inscribed regular hexagon is equal to half the side of the inscribed equilateral triangle. Ex. 457. The area of an inscribed regular hexagon is three fourths of the area of the circumscribed regular hexagon. Ex. 458. The area of an inscribed regular hexagon is the mean propor- tional between the areas of the inscribed and the circumscribed equilateral triangles. Ex. 459. The square of the side of an inscribed equilateral triangle is equal to three times the square of a side of the inscribed regular hexagon. Ex. 460. The area of an inscribed equilateral triangle is equal to half the area of the inscribed regular hexagon. Ex. 461. The square of the side of an inscribed equilateral triangle is equal to the sum of the squares of the sides of the inscribed square and of the inscribed regular hexagon. Ex. 462. The square of the side of an inscribed regular pentagon is equal to the sum of the squares of the radius of the circle and the side of the inscribed regular decagon.

BOOK V. PLANE GEOMETRY. 290 If R denotes the radius of a circle, and a one side of an inscribed regular polygon, show that: Ex. 463. In a regular pentagon, a = 1 2R √ 10− 2 √ 5. Ex. 464. In a regular octagon, a =R √ 2− √ 2. Ex. 465. In a regular dodecagon, a =R √ 2− √ 3. Ex. 466. If two diagonals of a regular pentagon intersect, the longer segment of each is equal to a side of the pentagon. Ex. 467. The apothem of an inscribed regular pentagon is equal to half the sum of the radius of the circle and the side of the inscribed regular decagon. Ex. 468. The side of an inscribed regular pentagon is equal to the hy- potenuse of the right triangle which has for legs the radius of the circle and the side of the inscribed regular decagon. Ex. 469. The radius of an inscribed regular polygon is the mean propor- tional between its apothem and the radius of the similar circumscribed regular polygon. Ex. 470. If squares are constructed outwardly upon the six sides of a regular hexagon, the exterior vertices of these squares are the vertices of a regular dodecagon. Ex. 471. If the alternate vertices of a regular hexagon are joined by straight lines, show that another regular hexagon is thereby formed. Find the ratio of the areas of these two hexagons. Ex. 472. If on the legs of a right triangle as diameters semicircles are described external to the triangle, and from the whole figure a semicircle on the hypotenuse is subtracted, the remaining figure is equivalent to the given right triangle. Ex. 473. The star-shaped polygon, formed by producing the sides of a regular hexagon, is equivalent to twice the given hexagon.

EXERCISES. 291 Ex. 474. The sum of the perpendiculars drawn to the sides of a regular polygon from any point within the polygon is equal to the apothem multiplied by the number of sides. Ex. 475. If two chords of a circle are perpendicular to each other, the sum of the four circles described on the four segments as diameters is equivalent to the given circle. Ex. 476. If the diameter of a circle is divided into any

two segments, and upon these segments as diameters semicircumferences are described upon opposite sides of the diameter, these semicircumferences divide the circle into two parts which have the same ratio as the two segments of the diameter. Ex. 477. The diagonals that join any vertex of a regular polygon to all the vertices not adjacent divide the angle at that vertex into as many equal parts less two as the polygon has sides. PROBLEMS OF CONSTRUCTION. Ex. 478. To circumscribe an equilateral triangle about a given circle. Ex. 479. To circumscribe a square about a given circle. Ex. 480. To circumscribe a regular hexagon about a given circle. Ex. 481. To circumscribe a regular octagon about a given circle. Ex. 482. To circumscribe a regular pentagon about a given circle. Ex. 483. To draw through a given point a line so as to divide a given circumference into two parts having the ratio 3 : 7. Ex. 484. To construct a circumference equal to the sum of two given circumferences. Ex. 485. To construct a circumference equal to the difference of two given circumferences. Ex. 486. To construct a circle equivalent to the sum of two given circles.

BOOK V. PLANE GEOMETRY. 292 Ex. 487. To construct a circle equivalent to the difference of two given circles. Ex. 488. To construct a circle equivalent to three times a given circle. Ex. 489. To construct a circle equivalent to three fourths of a given circle. Ex. 490. To construct a circle whose ratio to a given circle shall be equal to the given ratio m :n. Ex. 491. To divide a given circle by a concentric circumference into two equivalent parts. Ex. 492. To divide a given circle by concentric circumferences into five equivalent parts. Ex. 493. To construct an angle of 18◦; of 36◦; of 9◦. Ex. 494. To construct an angle of12◦; of 24◦; of 6◦. To construct with a side of a given length: Ex. 495. An equilateral triangle. Ex. 496. A square. Ex. 497. A regular hexagon. Ex. 498. A regular octagon. Ex. 499. A regular pentagon. Ex. 500. A regular decagon. Ex. 501. A regular dodecagon. Ex. 502. A regular pentedecagon. PROBLEMS OF COMPUTATION. Ex. 503. Find the area of a circle whose radius is 12 inches.

EXERCISES. 293 Ex. 504. Find the circumference and the area of a circle whose diameter is 8 feet. Ex. 505. A regular pentagon is inscribed in a circle whose radius is R. If the length of a side is a, find the apothem. Ex. 506. A regular polygon is inscribed in a circle whose radius is R. If the length of a side is a, show that the apothem is 1 2 √ R2−a2. Ex. 507. Find the area of a regular decagon inscribed in a circle whose radius is 16 inches. Ex. 508. Find the side of a regular dodecagon inscribed in a circle whose radius is 20 inches. Ex. 509. Find the perimeter of a regular pentagon inscribed in a circle whose radius is 25 feet. Ex. 510. The length of each side of a park in the shape of a regular decagon is 100 yards. Find the area of the park. Ex. 511. Find the cost, at $2 per yard, of building a wall around a cemetery in the shape of a regular hexagon, that contains 16 , 627.84 square yards. Ex. 512. The side of an inscribed regular polygon of n sides is 16 feet. Find the side of an inscribed regular polygon of 2 n sides. Ex. 513. If the radius of a circle is R, and the side of an inscribed regular polygon is a, show that the side of the similar circumscribed regular polygon is 2aR√ 4R2−a2 . Ex. 514. What is the width of the circular ring between two concentric circumferences whose lengths are 650 feet and 425 feet? Ex. 515. Find the angle subtended at the centre by an arc 5 feet 10 inches long, if the radius of the circle is 9 feet 4 inches. Ex. 516. The chord of a segment is

10 feet, and the radius of the circle is 16 feet. Find the area of the segment.

BOOK V. PLANE GEOMETRY. 294 Ex. 517. Find the area of a sector, if the angle at the centre is 20 ◦, and the radius of the circle is 20 inches. Ex. 518. The chord of half an arc is 12 feet, and the radius of the circle is 18 feet. Find the height of the segment subtended by the whole arc. Ex. 519. Find the side of a square which is equivalent to a circle whose diameter is 35 feet. Ex. 520. The diameter of a circle is 15 feet. Find the diameter of a circle twice as large. Three times as large. Ex. 521. Find the radii of the concentric circumferences that divide a circle 11 inches in diameter into five equivalent parts. Ex. 522. The perimeter of a regular hexagon is 840 feet, and that of a regular octagon is the same. By how many square feet is the octagon larger than the hexagon? Ex. 523. The diameter of a bicycle wheel is 28 inches. How many revolutions does the wheel make in going 10 miles? Ex. 524. Find the diameter of a carriage wheel that makes 264 revolutions in going half a mile. Ex. 525. The sides of three regular octagons are 6 feet, 7 feet, 8 feet, respectively. Find the side of a regular octagon equivalent to the sum of the three given octagons. Ex. 526. A circular pond 100 yards in diameter is surrounded by a walk 10 feet wide. Find the area of the walk. Ex. 527. The span (chord) of a bridge in the form of a circular arc is 120 feet, and the highest point of the arch is 15 feet above the piers. Find the radius of the arc. Ex. 528. Three equal circles are described each tangent to the other two. If the common radius is R, find the area contained between the circles.

EXERCISES. 295 Ex. 529. Given p, P , the perimeters of regular polygons of n sides in- scribed in and circumscribed about a given circle. Find p′, P′, the perimeters of regular polygons of 2n sides inscribed in and circumscribed about the given circle. Ex. 530. Given the radius R, and the apothem r of an inscribed regular polygon ofn sides. Find the radius R′ and the apothemr′ of an isoperimetrical regular polygon of 2n sides. MISCELLANEOUS EXERCISES. THEOREMS. Ex. 531. If two adjacent angles of a quadrilateral are right angles, the bisectors of the other two angles are perpendicular. Ex. 532. If two opposite angles of a quadrilateral are right angles, the bisectors of the other two angles are parallel. Ex. 533. The two lines that join the middle points of the opposite sides of a quadrilateral bisect each other. Ex. 534. The line that joins the feet of the perpendiculars dropped from the extremities of the base of an isosceles triangle to the opposite sides is parallel to the base. Ex. 535. If AD bisects the angle A of a triangle ABC, and BD bisects the exterior angle CBF , then angle ADB equals one half angle ACB. Ex. 536. The sum of the acute angles at the vertices of a pentagram (five-pointed star) is equal to two right angles. Ex. 537. The altitudesAD,BE,CF of the triangleABC bisect the angles of the triangle DEF . Circles with AB,BC,AC as diameters will pass through E andD,E and F , D and F , respectively. Ex. 538. The segments of any straight line intercepted between the circumferences of two concentric circles are equal.

BOOK V. PLANE GEOMETRY. 296 Ex. 539. If a circle is circumscribed about any triangle, the feet of the perpendiculars dropped from any point in the circumference to the sides of the triangle lie in one straight line. Ex. 540. Two circles are tangent internally at P , and a chord AB of the larger circle touches the smaller circle at C. Prove that PC bisects the angle APB . Ex. 541. The diagonals of a trapezoid divide

each other into segments which are proportional. Ex. 542. If through a point P in the circumference of a circle two chords are drawn, the chords and the segments betweenP and a chord parallel to the tangent at P are reciprocally proportional. Ex. 543. The perpendiculars from two vertices of a triangle upon the opposite sides divide each other into segments reciprocally proportional. Ex. 544. The perpendicular from any point of a circumference upon a chord is the mean proportional between the perpendiculars from the same point upon the tangents drawn at the extremities of the chord. Ex. 545. In an isosceles right triangle either leg is the mean proportional between the hypotenuse and the perpendicular upon it from the vertex of the right angle. Ex. 546. If two circles intersect in the pointsA andB, and throughA any secantCAD is drawn limited by the circumferences at C and D, the straight lines BC, BD are to each other as the diameters of the circles. Ex. 547. The area of a triangle is equal to half the product of its perimeter by the radius of the inscribed circle. Ex. 548. The perimeter of a triangle is to one side as the perpendicular from the opposite vertex is to the radius of the inscribed circle.

EXERCISES. 297 Ex. 549. If three straight lines AA′,BB′,CC′, drawn from the vertices of a triangle ABC to the opposite sides, pass through a common point O within the triangle, then OA′ AA′ + OB′ BB′ + OC′ CC′ = 1. Ex. 550. ABC is a triangle, M the middle point of AB, P any point in AB between A and M. If MD is drawn parallel to PC , meeting BC at D, the triangle BPD is equivalent to half the triangle ABC. Ex. 551. Two diagonals of a regular pentagon, not drawn from a common vertex, divide each other in extreme and mean ratio. Ex. 552. If all the diagonals of a regular pentagon are drawn, another regular pentagon is thereby formed. Ex. 553. The area of an inscribed regular dodecagon is equal to three times the square of the radius. Ex. 554. The area of a square inscribed in a semicircle is equal to two fifths the area of the square inscribed in the circle. Ex. 555. The area of a circle is greater than the area of any polygon of equal perimeter. Ex. 556. The circumference of a circle is less than the perimeter of any polygon of equal area. PROBLEMS OF LOCI. Ex. 557. Find the locus of the centre of the circle inscribed in a triangle that has a given base and a given angle at the vertex. Ex. 558. Find the locus of the intersection of the altitudes of a triangle that has a given base and a given angle at the vertex. Ex. 559. Find the locus of the extremity of a tangent to a given circle, if the length of the tangent is equal to a given line.

BOOK V. PLANE GEOMETRY. 298 Ex. 560. Find the locus of a point, tangents drawn from which to a given circle form a given angle. Ex. 561. Find the locus of the middle point of a line drawn from a given point to a given straight line. Ex. 562. Find the locus of the vertex of a triangle that has a given base and a given altitude. Ex. 563. Find the locus of a point the sum of whose distances from two given parallel lines is equal to a given length. Ex. 564. Find the locus of a point the difference of whose distances from two given parallel lines is equal to a given length. Ex. 565. Find the locus of a point the sum of whose distances from two given intersecting lines is equal to a given length. Ex. 566. Find the locus of a point the difference of whose distances from two given intersecting lines is equal to a given length. Ex. 567. Find the locus of a point whose distances from two given points are in the given ratio m :n. Ex. 568. Find the locus of a point whose distances from two given parallel lines are in

the given ratio m :n. Ex. 569. Find the locus of a point whose distances from two given inter- secting lines are in the given ratio m :n. Ex. 570. Find the locus of a point the sum of the squares of whose distances from two given points is constant. Ex. 571. Find the locus of a point the difference of the squares of whose distances from two given points is constant. Ex. 572. Find the locus of the vertex of a triangle that has a given base and the other two sides in the given ratio m :n.

EXERCISES. 299 PROBLEMS OF CONSTRUCTION. Ex. 573. To divide a given trapezoid into two equivalent parts by a line parallel to the bases. Ex. 574. To divide a given trapezoid into two equivalent parts by a line through a given point in one of the bases. Ex. 575. To construct a regular pentagon, given one of the diagonals. Ex. 576. To divide a given straight line into two segments such that their product shall be the maximum. Ex. 577. To find a point in a semicircumference such that the sum of its distances from the extremities of the diameter shall be the maximum. Ex. 578. To draw a common secant to two given circles exterior to each other such that the intercepted chords shall have the given lengths a, b. Ex. 579. To draw through one of the points of intersection of two inter- secting circles a common secant which shall have a given length. Ex. 580. To construct an isosceles triangle, given the altitude and one of the equal base angles. Ex. 581. To construct an equilateral triangle, given the altitude. Ex. 582. To construct a right triangle, given the radius of the inscribed circle and the difference of the acute angles. Ex. 583. To construct an equilateral triangle so that its vertices shall lie in three given parallel lines. Ex. 584. To draw a line from a given point to a given straight line which shall be to the perpendicular from the given point as m :n. Ex. 585. To find a point within a given triangle such that the perpendic- ulars from the point to the three sides shall be as the numbers m, n, p. Ex. 586. To draw a straight line equidistant from three given points.

BOOK V. PLANE GEOMETRY. 300 Ex. 587. To draw a tangent to a given circle such that the segment intercepted between the point of contact and a given straight line shall have a given length. Ex. 588. To inscribe a straight line of a given length between two given circumferences and parallel to a given straight line. Ex. 589. To draw through a given point a straight line so that its distances from two other given points shall be in a given ratio. Ex. 590. To construct a square equivalent to the sum of a given triangle and a given parallelogram. Ex. 591. To construct a rectangle having the difference of its base and altitude equal to a given line, and its area equivalent to the sum of a given triangle and a given pentagon. Ex. 592. To construct a pentagon similar to a given pentagon and equiv- alent to a given trapezoid. Ex. 593. To find a point whose distances from three given straight lines shall be as the numbers m, n, p. Ex. 594. Given an angle and two points P and P′ between the sides of the angle. To find the shortest path from P to P′ that shall touch both sides of the angle. Ex. 595. To construct a triangle, given its angles and its area. Ex. 596. To transform a given triangle into a triangle similar to another given triangle. Ex. 597. Given three points A,B,C. To find a fourth point P such that the areas of the triangles APB , APC , BPC shall be equal. Ex. 598. To construct a triangle, given its base, the ratio of the other sides, and the angle included by them. Ex. 599. To divide a given circle into n equivalent parts by concentric circumferences.

EXERCISES. 301 Ex. 600. In a given equilateral triangle to inscribe three equal circles tangent to each other, each circle tangent to two sides of the triangle. Ex. 601. Given an angle and a point P between the sides of the angle. To draw through P a straight line that shall form with the sides of the angle a triangle with the perimeter equal to a given length a. Ex. 602. In a given square to inscribe four equal circles, so that each circle shall be tangent to two of the others and also tangent to two sides of the square. Ex. 603. In a given square to inscribe four equal circles, so that each circle shall be tangent to two of the others and also tangent to one side of the square.

TABLE OF FORMULAS. PLANE FIGURES. NOTATION. P = perimeter. h = altitude. b = lower base. b′ = upper base. R = radius of circle. D = diameter of circle. C = circumference of circle. r = apothem of regular polygon. a, b, c = sides of triangle. s = 1 2(a +b +c). p = perpendicular of triangle. m,n = segments of third side of triangle adjacent to sides b and a, respectively. S = area. π = 3.1416.

TABLE OF FORMULAS. 303 FORMULAS. Line Values. P AGE Right triangle, b2 = c×m; a2 =c×n 197 p2 = m×n 197 b2 :a2 = m :n 198 b2 :c2 :: m :c 198 a2 +b2 = c2 199 Any triangle, a2 = b2 +c2± 2c×m 200,201 Altitude of triangle on side a, h = 2 a √ s(s−a)(s−b)(s−c) 219 Median of triangle on side a, m = 1 2 √ 2(b2 +c2)−a2 220 Bisector of triangle on side a, t = 2 b +c √ bcs(s−a) 221 Radius of circumscribed circle, R = abc 4 √ s(s−a)(s−b)(s−c) 222 Circumference of circle, C = 2πR 270 “ “ C = πD 270 Areas. Rectangle, S = b×h 229 Square, S = b2 229 Parallelogram, S = b×h 230 Triangle, S = 1 2b×h 231 “ S = √ s(s−a)(s−b)(s−c) 254 “ S = abc 4R 254 Equilateral triangle, S = a2 4 √ 3 253 Trapezoid, S = 1 2h(b +b′) 232 Regular polygon, S = 1 2r×P 270 Circle, S = 1 2R×C 272 “ S = πR2 272 Sector, S = 1 2R× arc 271

PLANE GEOMETRY. 304

INDEX. P AGE Abbreviations 6 Alternation 170 Altitude of parallelogram 52 “ of trapezoid 52 “ of triangle 34 Analysis 4 Angle 10 “ acute 12 “ at centre of regular polygon 260 “ central 90 “ exterior of triangle 33 “ inscribed in circle 90 “ inscribed in segment 90 “ oblique 12 “ obtuse 12 “ reflex 12 “ right 11 “ salient 61 “ straight 11 “ vertical 34 Angles, adjacent 11,33 “ alternate-exterior 28 “ alternate-interior 28 “ complementary 14 “ conjugate 13 “ exterior 28 “ exterior-interior 28 “ interior 28 “ supplementary 14 P AGE “ supplementary-adjacent 18 “ vertical 14 Antecedents 168 Apothem 260 Arc 90 Area 226 Axiom 4 “ of parallel lines 26 Axioms of straight lines 8 “ general 6 Axis of symmetry 65 Base of isosceles triangle 34 “ of parallelogram 51 “ of triangle 34 Bases of trapezoid 52 Bisector 11 Centre of circle 89 “ of regular polygon 260 “ of symmetry 65 Chord 90 Circle 89 “ circumscribed 91 “ inscribed 91 Circles, concentric 91 “ escribed 153 Circum-centre of triangle 151 Circumference 89 Commensurable 109 Complement 14

INDEX. 306 P AGE Composition 171 Conclusion 4 Concurrent lines 77 Congruent figures 7 Consequents 168 Constant 111 Construction 4 Continued proportion 168 Continuity, Principle of 127 Contradictory of a theorem 4 Converse of a theorem 5,94 Convex curve 266 Curved surface 7 Decagon 62 Diagonal 52,61 Diameter 89

Dimensions 1 Distance 8,24 Division 172 Dodecagon 62 Duality, Principle of 39 Equal figures 7 Equimultiples 176 Equivalent figures 7,226 Ex-centres of triangle 153 Extreme and mean ratio 213 Extremes 168 Figure, curvilinear 7 “ geometrical 3 “ plane 7 “ rectilinear 7 Foot of perpendicular 11 P AGE Fourth proportional 168 Geometrical solid 2,2 Geometry 3 Geometry, Plane 3 “ Solid 3 Harmonic division 180 Heptagon 62 Hexagon 62 Homologous angles 35,62 “ lines 183 “ sides 35,62 Hypotenuse 34 Hypothesis 4 In-centre of triangle 152 Incommensurable ratio 110 Intersection 1 Inversion 170 Isoperimetric figures 282 Legs of right triangle 34 “ of trapezoid 52 Limit 111 Line 1,2,3 “ curved 7 “ of centres 106 “ straight 7 Lines, oblique 12 “ parallel 26 “ perpendicular 11 Major arc 90 Maximum 282

INDEX. 307 P AGE Mean proportional 168 Means 168 Median of trapezoid 52 Minimum 282 Minor arc 90 Negative quantities 127 Numerical measure 109 Octagon 62 Opposite of a theorem 5 Origin 8 Parallel lines 26 Parallelogram 51 Pentagon 62 Pentagram 295 Perigon 13 Perimeter 33,61 Perpendicular bisector 49 Perpendicular lines 11 Pi (π) 269 Plane 1,7 Point 1,2 “ of contact 89 “ of tangency 89 Polygon 61 “ angles of 61 “ circumscribed 91 “ concave 61 “ convex 61 “ equiangular 61 “ equilateral 61 “ inscribed 91 “ regular 258 P AGE Polygons mut. equiangular 62 “ mutually equilateral 62 Positive quantities 127 Postulate 4 Projection 201 Proof 3 Proportion 168 Proposition 4 Quadrant 90 Quadrilateral 51,62 Radius of regular polygon 260 Ratio 109 Ratio of similitude 183 Reciprocity, Principle of 39 Rectangle 51 Rhomboid 51 Rhombus 51 Scholium 4 Secant 89,204 Sector 90 Segment of circle 90 “ of line 8 Semicircle 90 Semicircumference 90 Sides of an angle 10 “ of polygon 61 “ of triangle 33 Similar arcs 270 “ figures 7 “ polygons 183 “ sectors 270 “ segments 270 “ triangles 184

INDEX. 308 P AGE Square 51 Superposition 10 Supplement 14 Surface 1,2,2 Symbols 6 Symmetry 65 Tangent 89,106 “ common external 106 “ common internal 106 Terms of a proportion 168 Theorem 4 Third proportional 168 Transversal 27 Trapezium 51 Trapezoid 51 “ isosceles 52 Triangle 33,62 “ equiangular 34 “ equilateral 34 “ isosceles 34 “ obtuse 34 “ right 34 “ scalene 34 “ altitudes of 34 “ angles of 33 “ bisectors of 35 “ medians of 35 “ vertices of 33 Variable 111 Vertex of angle 10 “ of triangle 34 Vertices of polygon 61

LICENSE. 309 End of Project Gutenberg’s Plane Geometry, by George Albert Wentworth